

**LEARNING MANAGEMENT SYSTEM FOR LSPU EMPLOYEES: CIVIL
SERVICE EXAMINATION PREPARATION**

A Capstone Project

Presented to the

Faculty of College of Computer Studies

Laguna State Polytechnic University

Santa Cruz Campus

In Partial Fulfillment of the requirements for the Degree

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

By:

ALOPA, JON CHRISCHAN A.

ASUNCION, JULIUS ROY D.

BANGKIL, FRANCIS ANSELME V.

Under the supervision of:

MIA V. VILLARICA, DIT

MAY 2026

LSPU VISION

An ASEAN Polytechnic University by 2023

LSPU MISSION

A premier ASEAN Polytechnic University that LEADS in multidisciplinary education and applied research, equipping future ready servant LEADers to drive innovation for sustainable regional development

LSPU QUALITY POLICY

LSPU delivers quality education through responsive instruction, distinctive research, sustainable extension, and production services. Thus, we commit to continually improve quality, efficient and effective services to the university stakeholder's highest level of satisfaction through an excellent management system imbued with utmost integrity, professionalism, and innovation

COLLEGE OF COMPUTER STUDIE'S GOAL

The College of Computer Studies graduates are expected to become globally competitive and innovative computing professionals imbued with utmost integrity, contributing to the country's national development goals.

PROGRAM EDUCATIONAL OBJECTIVES

The Bachelor of Science in Information Technology graduates are professionals that can adapt with the fast-paced computing trends responsive to global IT demands. It is designed to enable students to achieve the following by the time they graduate:

1. Apply knowledge for solving computing problems employing design and development solutions for business-driven application, installation, processes, operation, maintenance and administration of IT hardware and software.
2. Utilize modern computing tools and techniques in research and development projects.
3. Communicate effectively as a member or leader of the computing society with social, moral and legal responsibilities to accomplish a common goal.
4. Engage in life-long learning as foundation for continuing professional advancement.

APPROVAL SHEET

The thesis entitled “**LEARNING MANAGEMENT SYSTEM FOR LSPU EMPLOYEES: CIVIL SERVICE EXAMINATION PREPARATION**” prepared and submitted by **CHRISCHAN A. ALOBA, JULIUS ROY D. ASUNCION**, and **FRANCIS ANSELME V. BANGKIL** in partial fulfillment of the requirements for the degree of **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY, MAJOR IN SERVICE MANAGEMENT PROGRAM** is hereby recommended for approval and acceptance.

MIA V. VILLARICA, DIT
Thesis Adviser

Approved by the Committee on Oral Examination with a grade of _____.

MARK P. BERNARDINO
Member

MARIA LAUREEN B. MIRANDA, MIT, LPT
Member

RINA J. ARCIGAL, Ed.D.
Member

MARIA CEZANNE D. DIMACULANGAN
Member

MARIA LAUREEN B. MIRANDA, MIT, LPT
Research Implementing Unit Head

Accepted and approved in partial fulfillment of the requirement for the degree of **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**, major in **SERVICE MANAGEMENT PROGRAM**.

MIA V. VILLARICA, DIT
Associate Dean, CCS

Date Signed

ROSARIO G. CATAPANG, Ph.D., Ffp
Chairperson

RESEARCH CONTRIBUTION NO.	
----------------------------------	--

ACKNOWLEDGEMENT

The researchers would like to express the utmost gratitude and heartfelt thanks to the following individuals for the completion of the study.

Hon. MARIO R. BRIONES, Ed.D., the President of Laguna State Polytechnic University, for making sure that the students and faculty gain the support and quality education.

Engr. MANUEL LUIS R. ALVAREZ, the Director of the University Campus, for his kindness and encouragement to all the students at the university.

MIA V. VILLARICA, DIT, Associate Dean of The College of Computer Studies, by being a guiding hand and paying close attention to the complaints of the students in the College of Computer Studies, as well as being the Technical Editor of the researchers.

MIA V. VILLARICA, DIT, for being the Thesis Adviser and guiding the researchers to become responsible, dedicated students during the time together.

MARK P. BERNARDINO, in appreciation for his role as the Subject Matter Expert and his generous sharing of his knowledge, suggestions specially in the system.

MARIA LAUREEN B. MIRANDA, MIT, LPT, being the Technical Editor of the study, for assisting the researchers in making the necessary corrections to the grammar in documentation and spending her precious time for checking the manuscript.

MARIA CEZANNE D. DIMACULANGAN, as the Language Critic, for her willingness to share her expertise, and her guidance to the researcher in improving the study endeavors.

RINA J. ARCIGAL, ED. D., for being the Statistician of the research, for sharing her statistical knowledge and guidance to maintain the data quality of the study.

TO THE FAMILY, for giving financial, moral support and encouragement to the proponents.

ALMIGHTY GOD, for giving the researchers strength and determination to accomplish the study.

Abstract

The adoption of digital technologies in organizational training has become essential in enhancing employee competence and professional development. This study, entitled “Learning Management System for LSPU Employees: Civil Service Examination Preparation,” aimed to design and develop a web-based platform to support Laguna State Polytechnic University (LSPU) employees in preparing for the Civil Service Examination. The system provides centralized access to review materials, enables self-paced learning, tracks user progress, and generates performance reports for both learners and administrators. The study utilized the Rapid Application Development (RAD) methodology to ensure efficient system design and implementation. System functionality was evaluated through black box testing conducted by IT experts, yielding a 100% success rate across all tested features. Furthermore, the Technology Acceptance Model (TAM) was employed to assess user acceptance, with sixty (60) respondents participating in the evaluation. The results indicated a “Highly Acceptable” rating in terms of perceived usefulness, ease of use, and overall system performance. The system was also evaluated based on ISO/IEC 25010 standards, confirming its quality and reliability.

Keywords: *Learning Management System; Civil Service Examination Preparation LSPU Employee; Rapid Application Development; Black Box Testing; Technology Acceptance Model; ISO/IEC 25010 Software Metrics*

TABLE OF CONTENTS

Preliminaries	Page
Title Page	i
Vision, Mission, Quality Policy, Goals, and Objectives	ii
Approval Sheet.....	iii
Acknowledgement.....	iv
Abstract	v
Table of Contents	vi
List of Table.....	ix
List of Figure.....	xi
List of Appendices	xiii
Definition of Terms.....	xiv
Operational Terms.....	xiv
Technical Terms.....	xvi
 CHAPTER	
I. INTRODUCTION	
Project Context.....	2
Research Objective	3
Conceptual Framework.....	6
Project Purpose	7
Scope and Limitation of the Study.....	7
 CHAPTER	
II. REVIEW OF RELATED LITERATURE	
Learning Management System (LMS).....	9
Black Box Testing.....	18
PHP for Web Development	20

CHAPTER

III. RESEARCH METHODOLOGY

Research Design.....	25
Locale of the Study	26
Population of the Study.....	26
Sampling Design.....	27
Tester Profiles	28
Data Collection Instruments	29
Interview	29
Library Research.....	29
Online Research and Technology Gap Analysis	29
Brain storming	30
Observation.....	30
Interview	30
Internet research.....	30
Consultation	31
<i>Validation of Questionnaire</i>	31
Statistical Treatment	35
Project Design.....	36
System Architecture.....	37
Software Development Model	38
Phase 1. Project requirement and planning.....	38
Phase 2. Prototype.....	39
Phase 3. Construction and feedback.	39
Design and Development of Testing Tool	40

Testing Tool	41
Testing Method	44
Software Evaluation Phase	45
Distribution of Questionnaires	47

CHAPTER

IV. RESULT AND DISCUSSION

Conceptualization	49
Research Objective	50
The Design	52
The Development	54
Demographics of the Respondents	61
Technology Acceptance Evaluation Survey Results and Interpretation ..	62

CHAPTER

V. SUMMARY, CONCLUSION, AND RECOMMENDATION

Summary	68
Conclusions	69
Recommendations	70
LITERATURE CITED	71

List of Table

Table 1. Respondents for the User Evaluation.....	26
Table 2. Testers Profile	28
Table 3. Criteria for Quality Factors	31
Table 4. Criteria for Perceived Ease of Use	32
Table 5. Criteria for Perceived Usefulness.....	32
Table 6. Criteria for Experience.....	33
Table 7. Criteria for Attitude Towards Using	33
Table 8. Criteria for Behavioral Intention to Use.....	34
Table 9. Likert Scale	34
Table 10. <i>Test Case Description: Test the Functionality of Learning Modules.</i>	41
Table 11. <i>Test Case Description: Test the Functionality of Assessments.</i>	42
Table 12. Test Case Description: Test the Functionality of Analytics.....	43
Table 13. Black Box Testing Results for System Administrator	59
Table 14. Black Box Testing Results for Users Section	60
Table 15. Evaluation of TAM in terms of Quality Factors	62
Table 16. Evaluation of TAM in terms of Perceived Ease to Use	63
Table 17. Evaluation of TAM in terms of Perceived Usefulness.....	64
Table 18. Evaluation of TAM in terms of Attitude Towards Using	65
Table 19. Evaluation of TAM in terms of Behavioral Intense to Use	66
Table 20. Evaluation of TAM in terms of Experience.....	67
Table 21. Show the data dictionary of calendar_events.....	101
Table 22. Show the data dictionary of classrooms.....	102
Table 23. Show the data dictionary of classwork.....	103
Table 24. Show the data dictionary of course_progress.....	104

Table 25. Show the data dictionary of course_resources.	105
Table 26. Show the data dictionary of department.	106
Table 27. Show the data dictionary of discussion_replies.	106
Table 28. Show the data dictionary of discussion_threads.	107
Table 29. Show the data dictionary of employee.	108
Table 30. Show the data dictionary of final_exam_submissions.	109
Table 31. Show the data dictionary of final_exam_user_answers.	110
Table 32. Show the data dictionary of learner_achievements.	111
Table 33. Show the data dictionary of learner_analytics.	112
Table 34. Show the data dictionary of learner_notes.	113
Table 35. Show the data dictionary of learner_progress.	114
Table 36. Show the data dictionary of learning_paths.	115
Table 37. Show the data dictionary of mastery_checks.	116
Table 38. Show the data dictionary of mock_exam_settings.	116
Table 39. Show the data dictionary of mock_exam_submissions.	117
Table 40. Show the data dictionary of mock_exam_user_answers.	118
Table 41. Show the data dictionary of notifications.	119
Table 42. Show the data dictionary of password_reset_tokens.	119
Table 43. Show the data dictionary of path_courses.	120
Table 44. Show the data dictionary of test_answer_options.	120
Table 45. Show the data dictionary of test_questions.	121
Table 46. Show the data dictionary of test_submissions.	122
Table 47. Show the data dictionary of test_user_answers.	123
Table 48. Show the data dictionary of users.	124
Table 49. Show the data dictionary of user_preferences.	125

List of Figure

Figure 1. Conceptual Framework of the study.....	6
Figure 2. System Architecture of the study.....	37
Figure 3. Rapid Application Development Methodology.....	38
Figure 4. Black Box Testing. ReQtest (2019).....	40
Figure 5. Technology Acceptance Model adapted from Revyathi & Tselios (2019)...	45
Figure 6. ISO/IEC 25010 Software Quality Metrics.....	46
Figure 7: The study and proof that the researcher conducted an interview.	53
Figure 8: This Figure shows the log in section.....	55
Figure 9: System Admin Account	55
Figure 10. External Client Account	56
Figure 11. Face-to-face testing with Mr. Yasoña and Ms. Besande	57
Figure 12. Screenshots of face-to-face testing with Mr. Ramos.....	58
Figure 13. Screenshots of face-to-face testing with Mr. Domingo.....	58
Figure 14. Screenshots of face-to-face testing with Ms. Etron.....	59
Figure 15. Demographic of the respondents based on role.....	61
Figure 16. Demographics of the respondents based on Level of Proficiency.....	62
Figure 17. The image above shows the current process.	91
Figure 18. After logging in, the administrator can select from various processes.....	92
Figure 19. After logging in, the employee can select from various processes..	93
Figure 20. Shows how admin views dashboards.	93
Figure 21. Shows how admin create classroom and assessment.	94
Figure 22. Shows how admin can create custom mock exam.	94
Figure 23. Shows how admin can create and view the schedule & evets.....	95
Figure 24. Shows how admin view the all list of users.....	95

Figure 25. Shows how employee views dashboards.....	96
Figure 26. Shows how employee can view the modules, topic and quiz.....	96
Figure 27. Shows how employee can view the assessment.	97
Figure 28. Shows how employee can view the all performance and analytic.	97
Figure 29. Use Case 1: The researcher's web-based system's first users.	98
Figure 30. Use Case 2: The researcher's web-based research study's second user.	98
Figure 31. Use Case 3: The researcher's web-based research study's last user type...	98
Figure 32. System Context Diagram.....	99
Figure 33. This figure presents the Data Flow Diagram.....	100
Figure 34. Admin (System Admin) HIPO.	126
Figure 35. Employees HIPO.	127
Figure 36. Hardware Requirements	129
Figure 37. Software Requirements.....	129

LIST OF APPENDICES

Appendix

A. TECHNICAL BACKGROUND

Transcribed Interview
Current Process Flowchart
Proposed Flowchart
Use Case Diagram
System Context Diagram
Data Flow Diagram
Data Dictionary
Hierarchical Input Process Output Model
Project Schedule
Hardware and Software Resources
Input/Output/Reports Screenshot
Testing and Evaluation Instruction
User's Manual
Relevant Source Code

B. COMMUNICATION LETTERS AND FORMS

Request Letter
Summary of Recommendation
Research Colloquium Rating Sheet
10 Pager Similarity and AI

C. CURRICULUM VITAE

System Evaluators' Curriculum Vitae
Researchers' Curriculum Vitae

Definition of Terms

The researchers employed the following terms to better comprehend the study, as well as the mechanisms, systems, and materials used to develop the system's features and functions.

Operational Terms

Black box Testing

A type of software testing where the main objective is to evaluate a system's functioning without looking at its underlying code or implementation details.

Content Manager

The person assigned to manage learning materials within the system, including uploading lessons, organizing modules, and ensuring content accuracy.

Developmental Research

Defined as the systematic study of designing, developing, and evaluating the Learning Management System to ensure it meets usability, functionality, and effectiveness requirements.

External

Refers to authorized resource persons, reviewers, or content contributors outside the university who may provide review

	materials or expertise for the Learning Management System.
Instructors	Responsible for creating, uploading, organizing, and updating review materials, quizzes, and mock examinations in the Learning Management System. They monitor learners' progress and provide guidance when necessary.
Learning Management System (LMS)	A web-based platform developed to deliver, manage, and track Civil Service Examination review materials and learner progress.
Personnel	Employees of Laguna State Polytechnic University who are preparing for the Civil Service Examination and are enrolled in the Learning Management System to access learning materials and assessments.
System Admin	A person who has access to user management, and overall system maintenance within the Learning Management System.
User	The LSPU employee who will use the developed web-based Learning

Management System to access lessons, take mock examinations, and monitor their performance.

Technical Terms

Bootstrap

Front-end-developers can utilize this framework to improve the visual attractiveness of the system's interface.

JavaScript

A widely-used programming language that is primarily known for its ability to add interactivity to web pages

MySQL

Open-source relational database management system that uses Structured Query Language (SQL). It is widely used for managing and manipulating databases.

PHP

A server-side scripting language designed for web development.

Visual Studio Code

A source-code editor develops by Microsoft that provides a lightweight and extensible development environment for various programming languages.

CHAPTER I

INTRODUCTION

The Civil Service Examination (CSE) is a requirement for individuals seeking permanent employment and career advancement in the Philippine government. The Civil Service Commission (CSC) administers the test, which evaluates the competencies, knowledge, and abilities needed to provide successful public service. Passing the CSE grants government personnel civil service eligibility, which is necessary for job security, advancement, and professional development in the public sector. As a result, Civil Service Examination preparation has become a primary priority for government employees, particularly those in state universities and colleges. This project proposes the development of Learning Management System for LSPU Employees, a Web-based designed to enhance the Civil Service exam preparation experience by integrating game mechanics into a comprehensive reviewer. The reviewer aims to improve user engagement, motivation, and learning effectiveness through module, interactive quizzes, mock exam and progress tracking.

Laguna State Polytechnic University (LSPU) is a public higher education institution that employs both teaching and non-teaching personnel. As government employees, they are expected to obtain civil service eligibility in order to qualify for permanent appointments and career advancement. However, many LSPU employees find it difficult to prepare for the Civil Service Examination. Their work responsibilities often take up most of their time, leaving little opportunity to attend review centers or participate in structured review programs. In addition, differences in educational background and limited access to review materials make examination preparation more challenging for some employees.

When creating a Web-based system for the Civil Service Examination,

monitoring and recording are crucial components. They function as a watchful eye, monitoring both the learner's progress and the system's activities to make sure no crucial information is missed. User information, scores, review history, and performance statistics are kept safe, structured, and readily available whenever needed through appropriate monitoring and recording.

Project Context

The preparation of Laguna State Polytechnic University (LSPU) employees for the Civil Service Examination is the primary focus of this project. The objective of this project is to address the challenges faced by LSPU employees in preparing for the examination, such as limited time due to work responsibilities, lack of access to structured review materials, and difficulty in maintaining consistent review schedules. The project aims to improve employees' readiness for the Civil Service Examination by providing a more organized and accessible preparation approach. Those who offered reviewers to those who were reviewing were the ones who were recruited to study for the Civil Service exam. There was a chance that the review would be dull because the reviewer was very straightforward and included only text. He will therefore be able to prevent getting bored while reviewing by using the system that will be developed, which will keep him as a reviewer but make it more like a game. You will be able to review while equally playing and having fun.

A software program called the Learning Management System for LSPU Employees: Civil Service Examination Preparation was developed to help examinees prepare more effectively for the exam. The system aims to overcome traditional reviewer limitations and improve both accuracy and efficiency.

The researcher will use Rapid Application Development (RAD) to build the web-based application, which will be developed using HTML, CSS, JavaScript, and PHP. PHP will serve as the database for storing user information and review content. Testing and user evaluation to ensure the system works well and is user-friendly before deployment. This Learning Management System will follow development vide examinees with a structured and accurate review experience. This Learning Management System provides a structured and accurate review experience, efficient management of review materials and real-time tracking of user progress and performance. It reduces errors, repetition, and delays in monitoring scores and learning activities. Analytics and reporting features support progress evaluation, decision-making, and instructional improvement. By moving to a digital, web-based platform, this Learning Management System will reduce the reliance on traditional paper-based review methods. This makes exam preparation more efficient and effective for LSPU employees.

Research Objective

The study aims to develop an LSPU Employees: Civil Service Examination Preparation system designed to improve and simplify the review process for Civil Service Examination examinees. Specifically, this study seeks to:

The study specifically sought to perceive the following:

1. Design and develop a user-friendly web-based Learning Management System for the Civil Service Examination with the following functionalities:

For System Administrator

1. Learning Content and Modules Management
 - 1.1.1 View the master list of learning modules in a table
 - 1.1.2 Add new modules, lessons, and learning materials
 - 1.1.3 Edit modules, lessons, and uploaded materials
 2. User Management
 - 1.2.1 Approve and disapprove new users
 - 1.2.2 View user details
 - 1.2.3 View archived users
 - 1.2.4 Update user information
 - 1.2.5 Retrieve user accounts
 - 1.2.6 View all accounts in table format
 - 1.2.7 Search users in the table
 - 1.2.8 Monitor user activity
 3. Quiz and Assessment Management
 - 1.3.1 Create and manage quizzes for each module
 - 1.3.2 Add, edit, and delete questions
 - 1.3.4 View quiz results and analytics
 - 1.3.5 Randomized questions for exams
 4. Performance Monitoring
 - 1.4.1 View performance of employees per department
 - 1.4.2 Identify top performers and employees needing improvement
2. To design and develop a system that would help the personnel prepare effectively for the exam by improving their knowledge, skills, and confidence.

For Employees**2. Learning and Review System**

2.2.1 Access review materials for Civil Service Examination

2.2.2 Study modules

2.2.3 Take quizzes and mock examinations

2.2.4 Track learning progress and scores

3. Evaluate user acceptance and behavioral intention to use the developed system using the following criteria:

3.3.1 System quality and reliability.

3.3.2 Perceived ease of use.

3.3.3 Perceived usefulness in improving review efficiency and exam readiness.

3.3.4 User experience with the web-based features.

3.3.5 Attitude toward using the e-reviewer.

3.3.6 Behavioral intention to adopt and continue using the system.

Conceptual Framework

In order to demonstrate the necessary requirements, the researchers employed the input, process, and output strategy. Additionally, the research methodology made use of the study's output, software testing, and technological acceptance of the built system.

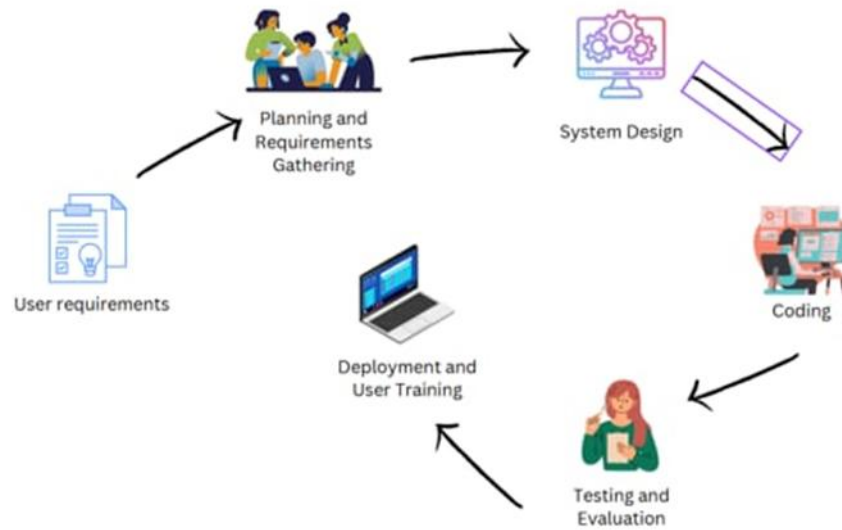


Figure 1. Conceptual Framework of the study “Learning Management System for LSPU Employees: Civil Service Examination Preparation”

the conceptual framework of the LSPU Employees: Civil Service Examination Preparation. It illustrates the systematic process followed in developing the proposed system, starting from identifying user requirements up to deployment and user training. The framework begins with user requirements, where the needs and expectations of examinees and reviewers are gathered. During the planning and requirements gathering phase, the study looked into the challenges faced by users with traditional review methods, such as lack of engagement, limited access to practice materials, and low motivation. Data were collected through interviews, brainstorming sessions, and observations. Next is the system design phase, where the structure and features of the web-based reviewer were planned. This was followed by the coding phase, where the planned features were developed into a functional web-based Learning Management

System using appropriate development tools and technologies. After development, the system went through testing and evaluation to ensure its functionality, usability, and the accuracy of the review content. The system was then deployed, and user training was conducted to help examinees navigate and use the Learning Management System effectively for Civil Service Exam preparation.

The researcher used IT solutions designed to address the problems identified from the findings. Black-box testing was conducted to assess the system's functionality and ensure its reliability. This framework provides a clear and methodical guide for developing, testing, and implementing the system.

Project Purpose

The purpose of this project is to develop and implement a Learning Management System (Learning Management System) tailored specifically for LSPU employees to facilitate efficient and organized preparation for the Civil Service Examination (CSE). The system aims to provide a centralized platform for employees to access study materials, track progress, and receive notifications about important exam-related deadlines and updates. This Learning Management System will support self-paced learning, enhance exam readiness, and promote continuous professional development for employees, enabling them to meet the standards required for successful CSE performance.

Scope and Limitation of the Study

In this capstone, there were three main points focused on namely:

1. Design and Development of the System
2. System Testing and
3. User Acceptance Testing.

The capstone has its scope and set limitations in terms of investigation, content and methodology to name a few.

The following are the scope of the study:

- The project for the Learning Management System for LSPU Employees: Civil Service Examination Preparation was initiated through a preliminary investigation involving literature review, observation, and interviews. The primary objective was to identify the current challenges faced by examinees in preparing for the Civil Service Examination.
- Develop a notification and reminder feature to provide users with updates on upcoming quizzes, scheduling, review schedules, and progress milestones.
- The preliminary investigation led to the proposal for the development of an integrated, web-based Learning Management System. Comprehensive testing was conducted to ensure the functionality, reliability, and performance of the system in supporting Civil Service Exam preparation.

Limitation of the Study:

- The system's effectiveness could be impacted by resistance to change and difficulties in adapting to the digital platform.
- There are people who don't know how to use a Computer/Devices

CHAPTER II

REVIEW OF RELATED LITERATURE

The research and related study in this chapter are cited from a range of sources. The researchers were able to look at a range of concepts, understandings, and ideas, as well as generalizations and conclusions, and many types of development according to these studies and literature. The main goal, concept, and design of the project, as well as its use and functionality, are all explained to the readers in this area of the research, along with the importance of developing the system.

Learning Management System (LMS)

According to Asrida, Y. N., Amanda, F., & Ul Fadilah, J. (2024). This study aims to analyze the effectiveness of the Learning Management System (LMS) in enhancing students' independence and learning outcomes in the digital era by employing a literature-based Classroom Action Research (CAR) approach. The applied CAR model emphasizes a reflective cycle consisting of planning, implementation, observation, and evaluation stages in the use of Learning Management System as a learning tool. This approach was chosen because it allows for a systematic examination of the learning process through the synthesis of findings from previous studies. The results indicate that the consistent implementation of Learning Management System effectively fosters students' autonomy in managing their learning activities, increases intrinsic motivation, and improves academic performance through structured and interactive learning experiences. Furthermore, the study reveals that the effectiveness of Learning Management System implementation is significantly influenced by teachers' digital competence, the relevance of instructional design, and adequate institutional support. Therefore, Learning Management System has the potential to

serve as a strategic instrument in promoting adaptive, independent.

In study of Kadir, A., Terra., Ch. Triwahyuni. (2013). *Pengenalan Teknologi Informasi*. Yogyakarta. One of the online learning platforms that adopts a conventional learning approach but is managed online is the Learning Management System (LMS). Learning Management System is not only easy to use but also encourages independent learning in a social environment for students, providing them with full control over their learning process. This study aims to explore the user experience of Learning Management System post-pandemic. The method used in this study is descriptive qualitative research, which aims to uncover the facts, conditions, variables, and phenomena occurring at present. The subjects of this study are all teachers and students from high schools in the Dewantara District. The results of the study indicate that students are very enthusiastic about the learning process using the Learning Management System application, as they find it easier to understand the material, which positively impacts their learning outcomes. Additionally, teachers show greater creativity in delivering learning materials. The use of Learning Management System also facilitates teachers in assigning tasks to students, making the learning process more varied and no longer monotonous.

According to Abaricia, C. P., & Delos Santos, M. L. C. (2023). *Enhancing e-learning system through Learning Management System technologies: learner experience study*. Shows how Learning Management System design reshapes learner experience. Competence is essential in forming attitudes and behaviors that contribute to achieving superior performance. This has prompted the Human Resource Development Agency of the East Java Provincial Government to develop the L-Bos Learning Management System platform to enhance the competencies of Civil Servants (ASN) in East Java. This study intends to examine how utilizing the Learning

Management System affects platform on the competencies of ASN within the Human Resource Development Agency of the East Java Provincial Government. The research employs a qualitative method with descriptive qualitative methods of analysis. Methods of gathering data include observations and interviews. The findings reveal that the implementation of Learning Management System satisfies the metrics for perceived utility and usability, effectively enhancing the competencies of civil servants in East Java Province.

According to Michail N. Giannakos, Patrick Mikalef & Ilias O. Pappas (2022), *Systematic Literature Review of E-Learning Capabilities to Enhance Organizational Learning*. E-learning systems are receiving ever increasing attention in academia, business and public administration. Major crises, like the pandemic, highlight the tremendous importance of the appropriate development of e-learning systems and its adoption and processes in organizations. Managers and employees who need efficient forms of training and learning flow within organizations do not have to gather in one place at the same time or to travel far away to attend courses. Contemporary affordances of e-learning systems allow users to perform different jobs or tasks for training courses according to their own scheduling, as well as to collaborate and share knowledge and experiences that result in rich learning flows within organizations. The purpose of this article is to provide a systematic review of empirical studies at the intersection of e-learning and organizational learning in order to summarize the current findings and guide future research. Forty-seven peer-reviewed articles were collected from a systematic literature search and analyzed based on a categorization of their main elements. This survey identifies five major directions of the research on the confluence of e-learning and organizational learning during the last decade. Future research should leverage big data produced from the platforms and investigate how the incorporation of

advanced learning technologies (e.g., learning analytics, personalized learning) can help increase organizational value.

According to Muhammad Iqbal, Iskandarini Iskandarini, Vivi Gusrini Rahmadhani (2025), *The Influence of Learning Management System (LMS)-Based Training and Career Development on Employee Competence at PT. Perkebunan Nusantara IV Regional II Medan*. This study examines the influence of Learning Management System (LMS)-based training and career development on employee competence at PT. Perkebunan Nusantara IV Regional II Medan. The research addresses the ongoing challenge of improving employee competence despite the implementation of Learning Management System and career development programs. A quantitative research design was used, and data were collected through structured questionnaires distributed to 123 employees. Multiple linear regression analysis was applied to examine the relationships among variables. The findings reveal that Learning Management System -based training has a significant and dominant effect on employee competence, with greater impact than career development programs. The Learning Management System provides flexibility and accessibility that enhance learning engagement, yet limitations remain, such as the lack of contextual content and alignment with job-specific needs. Career development programs also positively affect competence but show uneven access across employee groups, limiting their full impact. The study concludes that a structured, integrated approach combining Learning Management System training with equitable career development pathways is essential to maximize employee competence. The research highlights the need for organizations to align Learning Management System content with workplace requirements and develop inclusive career strategies. These findings contribute to the human resource development field by offering practical insights for enhancing digital learning systems

and career growth in the plantation sector.

In study of Blessy Doe & Vivekanandan. (2023). A study on Learning Management System Course effectiveness in workplace user experience. Reviews Learning Management System user experience and learning satisfaction. The goal of this study is to examine the effectiveness of a Learning Management System (LMS) Course among workplace users in terms of user experience. Precisely, to inspect the effectiveness of learning through the experiences and satisfaction gained from encountering Learning Management System quality factors. For analysing the relationship among learner experience and learner satisfaction on Learning Management System quality factors such as pedagogical design, interface design, content presentation format, transfer of learning, and feedback of learning, their correlation significance were evaluated. Data was gathered from 474 banking professionals working in both the public and private sectors through a questionnaire. The banks chosen were those that already have Learning Management System platforms in place for employee training. The study discovered a significant relationship in the correlation analysis of the factors pedagogical design, interface design, content presentation format, transfer of learning, and feedback of learning, as well as learner experience and satisfaction. It was also discovered that among all the Learning Management System quality factors studied, there found a strong significant relationship between the factors content presentation format and the interface design of the Learning Management System platform.

According to Krishna p. R. (2025) The Impact of E-Learning Platforms on Employee Performance and Development. The incorporation of e-learning platforms into business training and development processes has overhauled conventional methods for upskilling the workforce. This research paper examines the role of e-learning

platforms in employee performance and development, focusing on how the digital learning factor is changing organizational development. Analyzing a mix of empirical evidence, case studies, and scholarly writings, the study pinpoints the most significant e-learning benefits including enhanced accessibility, tailored learning processes, immediate feedback, and economy. These qualities not only augment retention of knowledge but also yield enhanced employee productivity and job performance. The study further explores the tech innovations—artificial intelligence, gamification, and mobile learning—that are transforming the learning process, making it more engaging and responsive to individual needs. But the paper also considers the impediments to the adoption of e-learning such as digital literacy disparities, absence of personal touch, and content relevance and quality issues. It also considers how organizational infrastructure, leadership backing, and corporate culture impact e-learning success of implementation.

According to Ray M.K.(2013) *Effective Skills Transfer Through A Learning Management System: A Case Of Eleckom Corporate Services*. Africa in general is said to still be lagging behind the developed nations in terms of workforce skills, and South Africa is not an exception. The problem of skills inadequacy, it is argued, has a negative effect on enterprises as it presents hurdles to the country's economic development objectives. Small, medium and large organizations tend to suffer from the said issue. In the context of South Africa, parastatals, as vital enterprises to the growth of the economy are also exposed to the adverse effects of relevant skills shortage. To this point, parastatals have to continual find ways to curb the inadequacy of relevant skills. This paper aimed to show how relevant skills, in a large South African parastatal, could be transferred among employees through a Learning Management System. This is done by following a case study method on the ways in which ElecKom corporate services

promotes and facilitates relevant skills transfer through its adopted learning management system. Following the interpretivist research paradigm, Activity Theory underpinned the study. The paper concludes that the issues of relevant skills shortage may be curbed through knowledge and skills transfer using learning management systems as enabling tools, so long as the systems are contextualized sensitive to the organizational culture and the background of employees.

According to Melva Z. (2024 Journal of Education and e-Learning Research. In Indonesia, educational activities have shifted to home-based learning after the COVID-19 epidemic broke out. Consequently, digital learning has become a necessary option for many teachers. This study aims to understand teachers' experiences, digital platform mastery, digitalbased learning pedagogy and the role of parents in learning at home in implementing the teaching process using learning management system-based blended learning media during the COVID-19 epidemic. A questionnaire was used in this study to gather data on the experiences of the teachers, parents' participation in home learning during COVID-19, the digital pedagogy and the mastery of digital platforms using a qualitative approach. This study involved 16 students' parents and 96 elementary school teachers. The data was analyzed using NVivo Pro software. The testing of data validity was carried out through data triangulation. Results of this recent study revealed that based on their experience, teachers still used videos as the main teaching material in digital learning and they carried out the learning assessment process using digital platforms. This was impacted by the process of changing from distant learning to the new normal. Additionally, the selection of digital media was carried out by considering media effectiveness, the learning process, available facilities and digital literacy. Moreover, parents not only provided learning facilities at home but were also more active as facilitators and supervisors during the learning process at home. This study

contributes to provide some considerations for mathematics teachers in designing learning management system-based blended learning media in elementary schools during the COVID-19 epidemic.

In Study of A. Aman (2024), A Review on E-learning and Employability Skills. This study presents a systematic literature review to ascertain the extent of exploration of e-learning and employability in extant research. Through a systematic review of literature published between 2008 and 2022, the author analyzed 88 articles published in 67 academic journals. The following were identified: key publishers, publication trend and citation, most contributing research areas, major contributing countries, leading researchers, major methodological approaches, key focus of studies and major outcomes. Future research prospects are also discussed. Study reviewed the impact of e-learning to employability, examining various areas, publications, citations, countries, and authors publishing e-learning and employability research. The present study explored the types of data collection method used, statistical analyses performed, and contributions of e-learning and employability skills literature in terms of their purpose and outcomes. Investigation observed that most of the research on this topic has indicated a positive impact of e-learning in enhancing the employability skills of higher education students. Additionally, this study provides insights to future researchers to address other potential factors that might influence employability and evaluate digital learning techniques that can meet the needs of diverse sections of society. Study reveals significant gaps in research, which can advance this significant research agenda for future investigation. This would facilitate highlighting the unexplored aspects of e-learning and employability. The topic of review is a cross-disciplinary research area, attracting scholars from different domains.

According to Revyathi, A., & Tselios, N. (2017). Extension of Technology

Acceptance Model using System Usability Scale to assess behavioral intention to use e-learning. This study examines the acceptance of technology and behavioral intention to use learning management systems (LMS). In specific, the aim of this research is to examine whether students ultimately accept and use educational learning systems such as e-class and the impact of behavioral intention on their decision to use them. An extended version of technology acceptance model has been proposed and used by employing the System Usability Scale to measure perceived ease of use. 345 university students participated in the study and the data analysis was based on partial least squares method. The results were confirmed in most of the research hypotheses. In particular, social norm, system access and self-efficacy significantly affect behavioral intention to use. As a result, it is suggested that e-learning developers and stakeholders should focus on these factors to increase acceptance and effectiveness of learning management systems.

According to Waris, I. & Hameed, I. (2022) Modeling teachers acceptance of learning management system in higher education during COVID-19 pandemic: A developing country perspective. The ubiquity of the World Wide Web has provided massive opportunities to teachers and learners around the globe to share knowledge anytime students; universities have arranged the facilities of the learning management system. This study is an empirical evaluation of university faculty members' intention to use a learning management system. We extended Davis's (1989) Technology Acceptance Model (TAM) to evaluate faculty members' intention to use learning management system amid COVID-19. Data were obtained from the faculty members of Pakistani universities through the convenience sampling technique and analyzed using PLS-SEM. The outcomes of the study revealed that perceived ease of using a learning management system, user-interface design, and faculty members'

innovativeness have a favorable impact on their intention to use learning management system. Facilitating conditions positively influenced intention to use learning management system. However, the positive relationship between facilitating conditions and perceived ease of using the learning management system was insignificant. The results indicate that the extended TAM model has effectively predicted faculty members' intention to use the learning management system. The findings of the current study can be used for the training and development of faculty members by institutions and regulatory bodies in developing countries.

Black Box Testing

When testing a web-based system, black box testing is used. According to the (Emanuele Viglianisi et al., 2020) entitled “Resttestgen: Automated black-box testing of restful APIs” the chance to utilize program information that is available through its program code is lost when using a black-box technique. Actually, the incredibly intricate inner workings of how components are used and operated do not need to be addressed by a black-box approach. Rather, to use the REST API, it merely needs a well-defined interface.

Automated testing for RESTful web APIs often uses a black-box strategy, neglecting inter-parameter dependencies. This inefficiency and inability to handle complex, pervasive inter-parameter dependencies make it unlikely for real-world services, requiring brute force to generate valid test cases. In addition, in the study of (Martin-Lopez et al., 2020) entitled “RESTest: Black-Box Constraint-Based Testing of RESTful Web APIs” In this article they present RESTest, a framework supports for automated black-box testing of RESTful APIs. RESTest is a tool that automates the specification and analysis of inter-parameter dependencies, enabling the use of

constraint solvers for valid test cases, enabling faster fault detection through deeper evaluation of input parameter combinations.

As mentioned by (Corradini et al., 2022) entitled “Automated black-box testing of nominal and error scenarios in RESTful APIs” in this article RestTestGen is a new black-box approach that generates test cases for REST APIs based on their interface definition, using input values and requests for each operation. The goal is to test nominal execution scenarios and error scenarios. Two distinct oracles are deployed to detect implementation defects. This approach is universally applicable across different programming languages or external libraries used in REST APIs. Validation on over 100 real-world APIs has shown its effectiveness in revealing faults in deployed services. The approach is mainly targeted at the research community.

Moreover, according to (Alan Budi Kusuma & Novri Hadinata, 2022) entitled “The Implementation of the Black Box Method for Testing Smart Hajj Application Ministry of Religion” they mentioned that the Haji Pintar application is functioning properly, but there are still errors in the cancellation and payment information menus, causing issues when checking the status of submitted cancellation processes and congregation portions. The Black Box approach will be used to comprehensively test the Haji Pintar application with regard to its functionality, interfaces, data structures, or external database access; performance; initiation and termination faults; usage; and advantages and outcomes derived from utilizing the program.

(Long & Zhao, 2023) explained in this study entitled “Equivalence, Identity, and Unitarity Checking in Black-Box Testing of Quantum Programs” that the Quantum programs' non-deterministic behavior presents significant error discovery challenges, and while various testing methods have been proposed, they often overlook

fundamental questions in black-box testing. In this article the authors present three new algorithms for equivalence, identity, and unitarity checking in black-box testing of quantum programs. They explore optimization techniques and parameter selection for maximum performance. Experimental evaluations show these methods can rigorously perform these checks, providing robust support for black-box testing of quantum programs.

PHP for Web Development

PHP frameworks are essential for web development due to their efficiency, dependability, maintainability, and scalability. These frameworks provide bulletproof, maintainable code and robust, secure online apps by expediting the development process, cutting down on time, and adhering to coding standards. Comprehending the many alternatives is necessary to choose the optimal framework. According to the (Majida Laaziri et al., 2019) entitled “A Comparative study of PHP frameworks performance” In this research has conducted a method of evaluation and performance testing to compare three PHP frameworks most used in web development to help the developer to better choose the most effective PHP framework for future web development.

In addition, according to the (Stenly Ibrahim Adam & Stevani Andolo, 2019) a study entitled “A New PHP Web Application Development Framework Based on MVC Architectural Pattern and Ajax Technology” states that to create a dynamic and real-time web application, they mentioned that the developers use Ajax technology due to what it is capable of. It is necessary for users to develop their own Ajax requests because the majority of PHP frameworks currently in use lack integrated Ajax libraries. Therefore, in this study they create an innovative framework for developing PHP online applications by utilizing the Model View Controller (MVC) design pattern in

conjunction with Ajax technologies. Ajax technology is implemented by the framework itself through its integrated library. They came to the conclusion after doing this research that users may construct dynamic and real-time web applications with the aid of the new PHP web application framework, which could be utilized as a web application development tool.

Also, in the study of (Thomas et al., 2022) entitled “Development of PHP and MySQL based Digital Asset Management System for Secure Organizations” this study organizations face significant challenges in monitoring asset details like quantity count, working conditions, maintenance status, and depreciation. This tool helps track asset locations, quantities, working conditions, and depreciation. Inventory is used to keep track of an organization's assets in records. There will be a comprehensive inventory list that includes every asset owned by the business, together with information on its description, inventory number, location, and state of operation. In this article discusses a system designed to manage an organization's mobile assets, devices, fixtures, inventory, and maintenance activities. The backend of the system is built using MySQL and PHP, and many frontend technologies—PHP, React, Bootstrap, and others—as well as a double security feature. Anywhere with reliable internet access and working login credentials can access it.

The vulnerabilities are located in the programmer's coding style as well as the syntax and foundation of the programming language. For every programmer and student of programming languages, it is imperative to embrace secure coding practices. In study, it was also mentioned of (Ramadevi Yenduri & Mahmood Al-khassaweneh, 2022) entitled “PHP: Vulnerabilities and Solutions” they mentioned in this study a 5-steps model is proposed that can improve the coding style of the programmers and also help build secure web applications. In this study examines server-side and client-side

vulnerabilities in two PHP-developed web applications. The first application follows poor programming, while the second follows a 5-steps model with recommendations. Both applications were scanned using an open-source proxy tool, revealing that following the 5-steps model significantly improves coding practice and results in a secure web application.

As stated by (Yu, 2022) entitled “Design and Application of Digital Network Education Platform for Economic Management Major in Colleges and Universities Based on PHP Language” in this study highlights the significance of teaching reform in colleges and universities for economic management majors, analyzes current teaching conditions, and proposes solutions based on the "four new" construction perspective. A digital network education platform for economic major was developed using B/S architecture, MVC design, PHP, and Laravel framework. Functional modules were built for course content and specialty, with MySQL data handling data storage, addition, deletion, and checking. The system improves teaching quality and students' mastery of economic management courses, optimizing the teaching mode of the economic management major.

Similarity, in the project (Yin, 2023) entitled “Design and Application of Online Teaching Platform for Ideological and Political Education in Colleges and Universities Based on PHP” they mentioned that college students are in the development stage of their thoughts, and they require correct guidance to ensure their well-being. According in this article the project aims to create an online teaching platform for ideological and political education in colleges and universities using PHP. The ThinkPHP framework is created to enhance the system features of the teaching platform, PHP is utilized as the script language for platform development, and Linux is chosen as the system development environment. In order to further optimize the ideological and political

education teaching system in colleges and universities, it employs a functional linear regression model as the statistical foundation for the practical data of ideological and political education. Finally, it uses both qualitative and quantitative analytic hierarchy process (AHP) to calculate the practical results of ideological and political education.

In the study of (Sangay Tenzin, 2022) entitled “PHP Framework for Web Application Development” they mentioned PHP is a widely used web development language, but its unstructured development model can cause confusion and problems for developers. PHP frameworks provide a basic structure for building web applications, promoting rapid application development (RAD). They save time, reduce repetitive code, and build applications with more stability. The framework uses the Model View Controller (MVC) concept to overcome the issue of plain PHP. This article presents basic concepts of PHP frameworks and an overview of the top 5 based on usage stats.

In addition, according to the (Haris & Nurdattillah Hasim, 2019) entitled “PHP Frameworks Usability in Web Application Development” in this study discussed on the current practice to understand PHP language and PHP frameworks in building web applications. Any developer who wants to use the PHP framework to create web applications should be aware of the fundamental ideas behind the framework and acquire the skills and knowledge needed to develop web-based applications effectively. The study analyzed the suitability of PHP frameworks using three approaches: systematic approach, score criteria evaluation, and technical factors. It also compared 23 different frameworks, focusing on features like ORM, Code Generator, Template Engine, and CRUD Generator. Understanding the basic core PHP for web application development is crucial, and new developers should explore various frameworks. Moreover, new developers should not limit themselves to a particular PHP framework

only but also allow themselves to explore various PHP frameworks in the development of web application projects.

Furthermore, in the study of (L. Nefedov et al., 2019) entitled “Model of choice of PHP-framework for development of web applications informatization of the project organization” the goal of this study is to enhance project organization management efficiency and reliability by developing a project organization information system using an effective PHP framework. A model for selecting a PHP framework for the project organization's digitization was created using decision-making techniques under the circumstances of ambiguous input data. The paper presents a novel model for selecting PHP frameworks for web application development, incorporating fuzzy information for more objective choices. An example is provided, demonstrating its efficiency and feasibility. Further research will focus on its practical implementation in a program-methodological complex. The proposed model for selecting a PHP framework for project organization informatization enhances reliability and reduces development time.

However, in the study of (Khan & Khanam, 2023) the project entitled “Study on MVC Framework for Web Development in PHP” they mentioned that The Model–View–Controller (MVC) design pattern is widely used in web development and ERP to separate data layer, presentation layer, and user interaction concerns. The Web developers face challenges in selecting suitable frameworks and libraries for their projects, making a clear understanding of various frameworks essential. The author mentioned in this article analyzed various research papers on various PHP frameworks, revealing that PHP offers various MVC frameworks that offer a structured approach to web application development. Explore and review some of the popular MVC frameworks for web development in PHP is the purpose in this study.

CHAPTER III

RESEARCH METHODOLOGY

This chapter covers the research design utilizing in the system's creation, the study's population, location, software use, and requirement documentation, as well as the persons and activities engage in the system's development.

Research Design

According to Muhammad (2023) says research design is just the plan a researcher makes for their study. In social science, you need to collect good information that fits your research question stuff that helps you look at a problem, check a theory, or explain something. A good research plan means your findings will be true, dependable, and fair.

In this study on "LSPU Employees: Civil Service Examination Preparation," the researcher used descriptive research as the main method. According to Deckert (2023), descriptive research provides information based on observation and describes situations or phenomena "as they are" using tools such as surveys, questionnaires, interviews, or rankings. This type of research is useful for exploring new areas or understanding specific conditions, and it can use both qualitative and quantitative data. For this study, the researcher collected data through interviews, surveys, and observation of LSPU employees to identify their challenges, needs, and preparation strategies for the Civil Service Examination. Assistance was also sought from university staff to ensure accurate and relevant information was gathered.

Locale of the Study

The study was conducted at Laguna State Polytechnic University (LSPU), Bubukal, Santa Cruz, Laguna. Specifically, it focused on the employees of the university who are preparing for the Civil Service Examination. The location was chosen by the researcher because this type of system would be highly beneficial to LSPU employees, as it provides an organized and accessible way to manage review materials, track progress, and improve overall preparation for the examination.

Population of the Study

The population of this study consisted of non-teaching employees of LSPU, Bubukal, Santa Cruz, Laguna who are preparing for the Civil Service Examination. They were selected because they are the intended users of the proposed Learning Management System and can provide relevant information about their review habits and needs.

Table 1. Respondents for the User Evaluation

Respondents	Total Population	Sample Size	Percentage
Administrator	1	1	1.56%
Employees	126	63	98.44%
Total	127	64	100%

From the study's results, this section has discussed about the total number of people in the designated areas, which was the Laguna State Polytechnic University Santa Cruz Campus (LSPU-SCC) at Santa Cruz, Laguna that was used to test how well the new system works.

The table indicates how many target respondents were selected to conduct the evaluation in response to the researchers' platforms. The table shows the total population, target population and accessible population, with the target population representing the quota that the researchers must meet, and the accessible population, which was the actual number of respondents who participated in the survey and were analyzed by the researchers as a result of unexpected events.

Sampling Design

A sort of sampling known as "purposive sampling," which is assessed, selective, or subjective, was used by the researchers to pick the study participants. The researcher selects the qualities to examine using this kind of non-probability sampling technique based only on their own discretion. Therefore, in order to satisfy the goals of the researchers, they can select a sample that accurately reflects the entire population or concentrate on individuals who fulfill particular qualifications. One advantage of this strategy is that it may quickly be implemented and yield a wide range of research answers.

Tester Profiles

The researchers wanted to check and evaluate the functionality of the developed system, which was titled " Learning Management System for LSPU Employees: Civil Service Examination Preparation." As part of the testing process, the researchers invited ten (10) credible IT experts to serve as testers.

Table 2. Testers Profile

SYSTEM EVALUATORS	NAME	CREDENTIALS
1.	Padua, Arteo Don Gerrard	Associate Manager Software Engineering
2.	Ramos, Richard D.	IT Support
3.	Besande, Gretchen R.	Software Developer
4.	Yosoña, Reynaldo A.	System Developer
5.	Arisga, Patrick Dan Alan	Web Developer
6.	Etron, Patricia Mae O.	Software Engineer
7.	Bautista, Mathew	IT Support
8.	Maglapuz, Ronaldo S.	IT Support
9.	Domingo, John Lester	IT Support
10.	Wesley Dyron C. Gumatao	Web Developer

The table above shows the testers' name as well as their credentials which are required to establish their ability to examine the developed system 's performance and functionality. Furthermore, the inclusion of tester names and credentials is a great resource for future reference and review. As projects progress of the system develop, having a recorded record of tester credentials allows project managers to evaluate the success of previous testing efforts. Assessing previous testers and their expertise can aid project developer in making informed decisions about testing techniques and resource allocation, ultimately enhancing system quality throughout its lifespan.

Data Collection Instruments

In this section, the methods and techniques used by the researcher in collecting data for the study are discussed. These methods are essential for the development of the Learning Management System for LSPU Employees: Civil Service Examination Preparation system. The researcher used the following data-gathering methods to collect important information from the non-teaching employees, who are the intended users of the system.

Interview

The researchers have used interview as a data gathering method in the target locale for gathering information about the current state and processes of the client. Using this method, they have successfully gathered essential information that helped them to formulate a system that served as a solution to the client's current issues.

Library Research

The researchers visited the university library to study previous research papers to obtain relevant data, and to look for other materials that would help them with their research, come up with fresh concepts, and comprehend the system's operation completely.

Online Research and Technology Gap Analysis

To improve the associated literature and study of the developed system, the researchers searched the Internet for information and data. Researchers can utilize the Internet to get further knowledge about the system and associated research and

literature, in addition to determining the potential capabilities and operational mechanisms of the study's planned result, which is the present system.

Brain storming

The researchers used brainstorming in deciding the selection of the locale, project plan creation, determining how to begin and go about the research, and encourage new thoughts and solutions.

Observation

An essential tool in development of the MAO-FCRMS is visual observation. The researchers gained information through observing the client's institution and its processes. Using the data gathered, the researchers have gained essential insights to help them further understand the subject.

Interview

The researchers have used interview as a data gathering method in the target locale for gathering information about the current state and processes of the client. Using this method, they have successfully gathered essential information that helped them to formulate a system that served as a solution to the client's current issues.

Internet research

The researchers have scoured various trusted academic studies and journals in order to obtain important data and information which served to aid in the current project.

Consultation

The researchers sought various experts' viewpoints regarding the related topics and technologies, providing them with a strong foundation for the creation of the proposed system. With this, the researchers can easily create process diagrams, check the project's readiness, and prepare a backup plan in case the system fails to perform its tasks.

Validation of Questionnaire

Based on ISO/IEC 25010 quality indicators as criteria and the Technology Acceptance Model (TAM) as a category, the researchers developed a survey questionnaire for user evaluation. The questionnaires were approved by the designated research consultant of the researchers before being sent to participants in the usability test. It is necessary to verify every question before to distribution and assessment. The questionnaire's validity as a measure of the verification principle was attested to by the research adviser. The validity of the questionnaire was assessed by the researchers using "pass" or "fail" answers.

Table 3. Criteria for Quality Factors

QUALITY FACTORS	SOURCE
1. Appropriateness. The develop system contains all of the functions and features that the user need.	ISO/IEC 25010:2011
2. Confidentiality. The develop system has secured and protected the user's privacy.	ISO/IEC 25010:2011
3. User Interface. I believe that interface of the developed system is user-friendly.	ISO/IEC 25010:2011

The table 3 displays the questionnaire for the intended users based on the TAM criteria, Quality Factors. It is made up of three (3) sub metrics, each having its own source. Appropriateness, Confidentiality, and User Interface were utilized to assess the system's quality criteria.

Table 4. Criteria for Perceived Ease of Use

PERCEIVED EASE OF USE	SOURCE
1. Suitability. LSPU employees can easily navigate the LMS to access Civil Service Examination (CSE) review materials, track their learning progress, and manage their study schedule effectively.	ISO/IEC 25010:2011
2. Efficiency. The LMS enables employees to quickly access learning modules, complete practice tests, making the review process faster and more organized compared to traditional methods.	ISO/IEC 25010:2011
3. Usability. The LMS is user-friendly and intuitive, allowing employees to use the system with minimal effort and without requiring advanced technical knowledge.	ISO/IEC 25010:2011

The table 4 based on the TAM criterion of Perceived Ease of Use; the questionnaire is shown for the intended users. It is comprised of three sub-metrics, each having a matching source. The system's perceived usability was assessed using three criteria: Suitability, Efficiency, and Usability.

Table 5. Criteria for Perceived Usefulness

PERCEIVED USEFULNESS	SOURCE
1. Functionality. The Learning Management System (LMS) includes organized review modules, practice exams, learning materials, and user management particularly created for Civil Service Examination (CSE) preparation of LSPU personnel.	ISO/IEC 25010:2011
2. Effectiveness. The LMS promotes employees' preparedness for the Civil Service Examination by giving organized material, self-paced learning, and continuous evaluation, decreasing the need for traditional review techniques.	ISO/IEC 25010:2011
3. Performance Efficiency. Through the LMS, employees can simply access review materials, take mock examinations, evaluate their progress, and enhance their performance effectively	ISO/IEC 25010:2011

The table 5 presents a questionnaire based on TAM's Perceived Usefulness criteria, comprising three sub-metrics: Functionality, Effectiveness, and Performance Efficiency. This structured approach assesses users' perceptions of the system's utility succinctly, aiding in targeted improvements for enhanced user experience and system performance. A thorough assessment of the system's utility is made possible by the

division of perceived usefulness into several sub-metrics, which inform iterative changes aimed at maximizing user satisfaction and adoption.

Table 6. Criteria for Experience

EXPERIENCE	SOURCE
1. The develop system was more effective and easier to use than the present method.	Abdekhoda, Ahmadi, Dehnad, & Hosseini (2014)
2. I never encountered any errors or problems while using the develop system.	Haindl (2018)
3. The functions of the develop system work properly.	ISO/IEC 25010:2011

The table 6 shows the questionnaire for the intended users based on the TAM criteria, Experience. It consists of three (3) sub metrics with corresponding sources.

Table 7. Criteria for Attitude Towards Using

ATTITUDE TOWARDS USING	SOURCE
1. The LMS provides Civil Service Examination (CSE) review materials and assessments in a simple and effective manner.	Phichitchaisopa, N., & Naenna, T. (2013)
2. The built LMS helps LSPU staff save time in reviewing by offering accessible and structured learning resources.	Becker M., Luckey, S. (2013)
3. The LMS helps employees in tracking their review progress, completing practice examinations, and enhancing their preparation for the Civil Service Examination.	Phichitchaisopa, N., & Naenna, T. (2013)

The table 7 the questionnaire for the intended users based on the TAM criterion, Attitude Towards Using, is given. It is made up of three (3) sub metrics, each with its own source.

Table 8. Criteria for Behavioral Intention to Use

BEHAVIORAL INTENTION TO USE	SOURCE
1. I am willing to use and accept the system in order to complete my work tasks.	Abdekhoda, Ahmadi, Dehnad, & Hosseini (2014)
2. I am interested to learn how the development system works.	Abdekhoda, Ahmadi, Dehnad, & Hosseini (2014)
3. I am satisfied in terms of the overall flow of the develop system.	ISO/IEC 25010:2011

The table 8 the questionnaire for the intended users is based on the TAM criteria, Behavioral Intention to Use. It is made up of three (3) sub metrics, each with its own source.

The Likert scale was employed by the researchers to gauge the participants' degree of satisfaction. According to Taherdoost, H. (2019) asks respondents to rate how much they agree or disagree with a series of statements about an attitude, item, person, or event using a Likert scale. The Likert scale may also yield a very dependable scale and is quite simple to construct. Furthermore, from the participants' perspective, it was easy to read and complete. The table below displays the Likert scale, which goes from "Highly Acceptable" to "Highly Not Acceptable."

Table 9.Likert Scale

SCALE	RANGE	VERBAL INTERPRETATION
5	4.20 – 5.00	HA-Highly Acceptable
4	3.40 - 4.19	A-Acceptable
3	2.60 - 3.39	M-Moderately Acceptable
2	1.80 - 2.59	NA- Not Acceptable
1	1.00 - 1.79	HN-Highly Not Acceptable

The Likert scale is used in the table above to measure respondents' satisfaction levels. In addition to the scale and grade, the table provides an equivalent verbal explanation that goes from "Highly Acceptable" to "Highly Not Acceptable."

Likert Scale Formula:

$$\text{Weighted Mean} = \frac{(\text{HA} \times 5) + (\text{A} \times 4) + (\text{M} \times 3) + (\text{NA} \times 2) + (\text{HN} \times 1)}{\text{Total Number of Respondents}}$$

Statistical Treatment

The frequency and percentage distribution were utilized by the researchers to handle the data acquired throughout the testing phase in order to assess the built Web-based system's functionality. The following was the formula that the researchers used:

Percentage formula:

$$\% = \frac{f}{N} \times 100$$

Where:

% = Percentage

f = Frequency of passed and failed

N = number test done

In terms of getting technical approval of the developed system depending on the level of usability, the Weighted Mean and Standard Deviation was used by the researchers.

Weighted Mean formula:

$$\mu = \frac{\sum x}{N}$$

Where:

$\bar{X}$ = Mean

$\sum x$ = Sum of all data values

N = Total Number of Respondents

The standard deviation formula was used as the statistical treatment from the data gathered is:

$$SD = \sqrt{\frac{\sum (r_i - r_{avg})^2}{n - 1}}$$

Where:

SD = Standard Deviation

n = Population size

r_i = Each value from Population

r_{avg} = Mean

The standard deviation measures the amount of variation or dispersion in a set of numbers.

Project Design

The system architecture and software technique employed by the researchers to conduct the study are discussed in this part. It also demonstrates the operation of the designed study's testing and assessment procedure. The project design made it easier

for the researchers to keep an eye on whether the web-based system was reaching the study's objectives and staying on course in accordance with the client's needs. The software's functioning was investigated in the project design, which proved to be successful.

System Architecture

The term "system architecture" describes the organizational structure of software applications used for task automation. It creates the software application's structure, written by Spacey (2019). This usually came in the form of a series of diagrams showing the different services, components, and interconnections. For this reason, system architecture was quite helpful in helping to visualize the created system's structure.

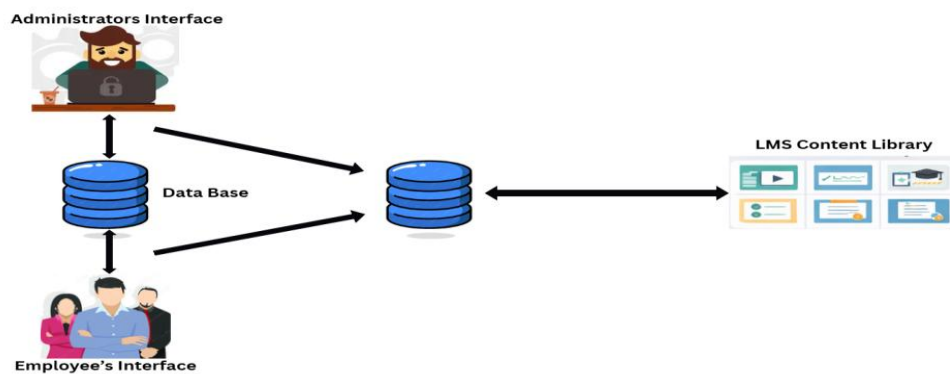


Figure 2. System Architecture of the study “Learning Management System for LSPU Employees: Civil Service Examination Preparation”

The employees access the system in a self-paced setup. They can log in, view courses, study the modules, take quizzes, and track their progress. All data such as results and completion records are stored in the central database.

The diagram shows how the administrator and employees are connected to the system, ensuring smooth management and access to learning materials without the need for an instructor.

Software Development Model

In order to create the web-based system, the researchers employed the Rapid Application Development (RAD) approach as their software process model. The rigidity of other conventional software development methods, which makes it challenging to make modifications after the original development has been finished, led to the creation of the RAD approach, a program design strategy. When a client recommends new features that meet their demands, the researchers can quickly implement improvements thanks to this design.

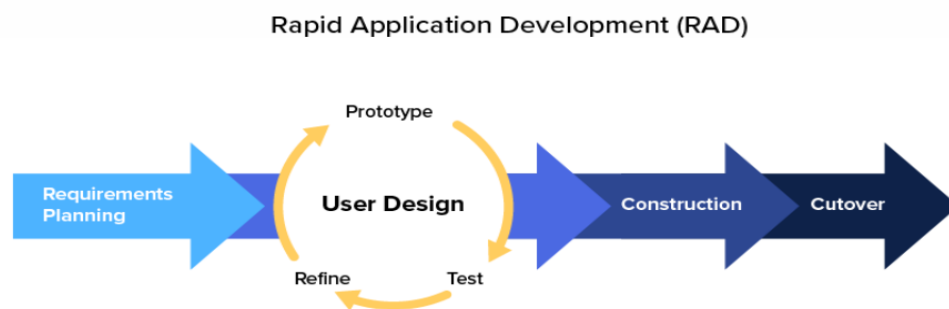


Figure 3. Rapid Application Development Methodology

According to Flora (2018) RAD is a software development method that speeds up software production by reducing the complicated and time-consuming planning process. RAD can be used as a reference for developing an information system that is superior in terms of speed, accuracy and lower cost.

Phase 1. Project requirement and planning.

The project's scope should be defined before starting development activities, ensuring no unnecessary tasks. The planning phase involves stakeholders, managers, developers, and clients discussing requirements, issues, timelines, and budgets. Collective decision-making, involving approvals from stakeholders and the entire

discussion team, helps avoid controversies and miscommunications, preventing costly changes later in the process.

Phase 2. Prototype.

The Rapid Application Development process involves rapid prototyping, where developers work with clients to design and test prototypes based on requirements. Feedback is then iterated until the application reaches a satisfactory stage, making it user-centric and meeting user expectations. This approach reduces the risk of rapidness.

Phase 3. Construction and feedback.

The team begins the development phase by converting prototypes into working models, building the application through coding and integrations. The application is continuously tested for bugs and issues, and if found, the team resolves them promptly. This process is faster due to the discussion and implementation of changes in the previous stage. Users are also involved in the process, allowing them to suggest changes and fill the gap between design and development.

Phase 4. Implementation.

After developing a working model, developers finalize it and move it to a live environment. They refine and recheck the model, conduct full scale testing to eliminate bugs, and finalize the application. The team then moves on to user onboarding and continues to provide support post-launch, ensuring a smooth transition to the live environment.

Design and Development of Testing Tool

Black box testing was employed by the researchers to determine the efficacy of the produced technology. As stated by Chaudhary, S., Khatana, A., and Verma, A. Black box testing is a software testing method that was utilized in 2017 to specify the functioning of the application. This technique's main focus is on a system's expected outputs and accessible inputs. Black box testing is done from the user's perspective, and its ability to accept both valid and invalid inputs is its most important feature.

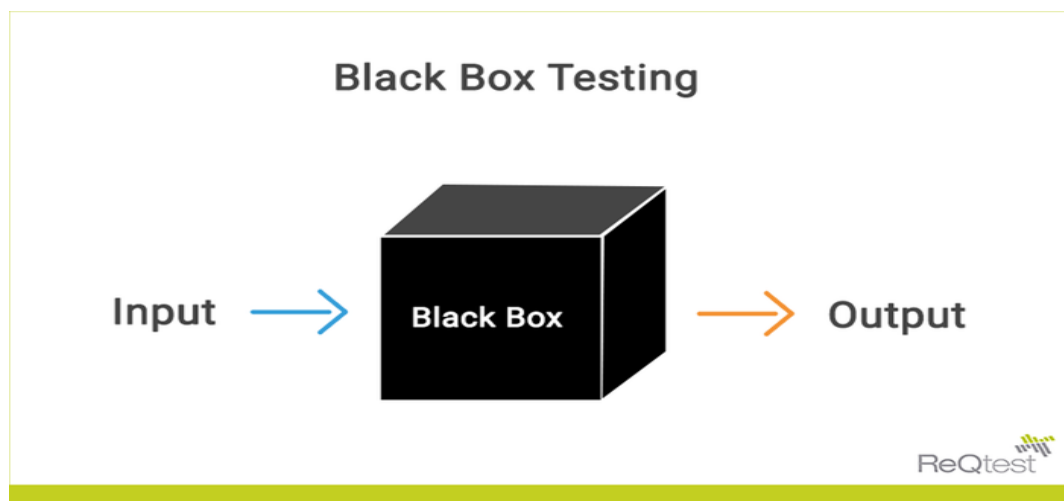


Figure 4. Black Box Testing. ReQtest (2019). Black box testing – Understanding the Basics

The functions of the developed system were tested by the researchers using the above-described approach. Testing a system without any prior knowledge of its internal operations is known as "black box" testing. An input is given by the tester, who then monitors what the system being tested produces as an output.

Testing Tool

The researchers employed black box testing to evaluate the generated web-based system's functionality. Testing without being aware of how an application functions inside is known as "black box" testing. Input-output-driven testing was another name for functional testing. A kind of software testing where the tester does not know how the product, they are evaluating functions within.

The developed systems' results during the IT experts' testing were recorded by the researchers in a test sheet for system testing. The testing sheet, which displays the step number, step specifics, and expected outcome, is below.

Table 10. *Test Case Description: Test the Functionality of Learning Modules.*

STEP NO.	STEP DETAILS	RESULTS
1	Input Username and Password	A. If the username and password are correct, the system would direct to the dashboard of the Admin. B. Can't log in if the username and password is incorrect
2	Dashboard	It shows learning analytics dashboard.
3	My Courses	It shows Test Components or Exam Section
4	Filter	It allows users to choose between Professional and Sub-Professional modules, and select the type of module they want.
5	Log Out	Logout from the system.

The table 3 above displays the steps in the Learning Management System login and log-out process. Users log in using their username and password, and if correct,

they are directed to the dashboard. If incorrect, an error message is displayed. The user can then navigate to the My Courses section to view available test components or exam sections. The filter feature allows users to select their preferred module category. The session is securely ended by logging out of the system, ensuring proper access and security of user data.

Table 11. *Test Case Description: Test the Functionality of Assessments.*

STEP NO.	STEP DETAILS	RESULTS
1	Input Username and Password	A. If the username and password are correct, the system would direct to the dashboard of the Admin. B. Can't log in if the username and password is incorrect
2	Dashboard	It shows learning analytics dashboard.
3	Assessments	It shows the available Exam
4	Take Assessments	It allows user to take Exam
5	Submit Test	Tt gives users the option to review their answers or confirm submission by selecting "Yes, Submit."
6	Log Out	Logout from the system.

The table 4 above presents the steps in accessing and completing assessments in the Learning Management System. The user logs in using a valid username and password to access the dashboard. From there, the user navigates to the Assessments section to view available exams. The system allows the user to take the exam and submit their answers. Before final submission, the system provides an option to review answers or confirm submission. The process ends by logging out of the system to ensure account security.

Table 12. Test Case Description: Test the Functionality of Analytics

STEP NO.	STEP DETAILS	RESULTS
1	Input Username and Password	A. If the username and password are correct, the system would direct to the dashboard of the Admin. B. Can't log in if the username and password is incorrect
2	Dashboard	It shows learning analytics dashboard.
3	My Progress	It shows Overall statistics and user need to focus areas.
4	Back to test	Returns you to the Assessments page so you can view and take available tests.
5	Log Out	Logout from the system.

The table 12 above shows the steps involved in accessing and monitoring progress in the Learning Management System. The user logs in using a valid username and password to access the dashboard. The dashboard displays learning analytics. In the My Progress section, the user can view overall performance statistics and focus areas. The user can return to the Assessments page to take available tests. The session is properly ended by logging out of the system.

Testing Method

A dichotomous scale with the options "pass" or "fail" was employed in this study to assess if the built web-based system produced the desired result based on the actual test inputs. as Birkett A. mentioned. Dichotomous scales (2019) have two completely opposed possibilities. "Yes" or "No" and "Agree" or "Disagree" are two examples. Although a dichotomous scale offered a more lucid binary response, it is also prone to tiredness. Respondents will naturally lean toward giving affirmative responses when this happens. The built-in real testing tool used to conduct the test is shown in the image below.

In this study, a dichotomous scale with the choices "pass" or "fail" was used to evaluate if the developed web-based system yielded the intended outcome based on the real test inputs. as A. Birkett. stated. There are two polar opposite alternatives for dichotomous scales (2019). Here are two examples: "Yes" or "No" and "Agree" or "Disagree". While providing a more clear-cut binary response, a dichotomous scale is also prone to fatigue. When this occurs, respondents will instinctively tend to answer affirmatively. The picture below depicts the actual testing instrument that was integrated and utilized to carry out the test.

TEST CASE ID: _____ Test Case Description: Test the functionalities of the admin login page
 Tester's Name: _____ Date Tested: _____ Test Case (Pass/Fail/Not Executed)
 PRECONDITIONS:
 • The tester should know how to handle and use basic functionalities of a computer
 • Computer/Laptop/PC must have access to internet
 TEST SCENARIO: Admin account login

STEP NO.	STEP DETAILS	EXPECTED RESULTS	ACTUAL RESULTS	STATUS	
				PASS	FAIL
1.	Open website	Website opens.		☐	☐
2.	Enter username and password	Display the inputted information correctly.			
3.	Click "Login"	Proceeds to the dashboard.			

Basis for Measurement → Actual Result → Dichotomous Scale

Figure 5. The Actual Structural Format of the Testing Tool

The actual testing tool that was built to conduct the test is shown in the above figure. The testing tool's system functioning was assessed based on the intended outcome.

Software Evaluation Phase

Through the process of building and distributing the assessment questionnaire, the intended users of the web-based system were involved in the evaluation process to ascertain their viewpoint, attitude, and actions regarding the system. The researchers performed the subsequent actions in order to ascertain and assess the technology acceptability of the system:

- a) The Design and Development of the Evaluation Tool
- b) Evaluation method
- c) The Validation of Questionnaire
- d) The Distribution of Questionnaire

Evaluation Tool

The Technology Acceptance Model (TAM) is used by the researchers to formulate the survey's questions.

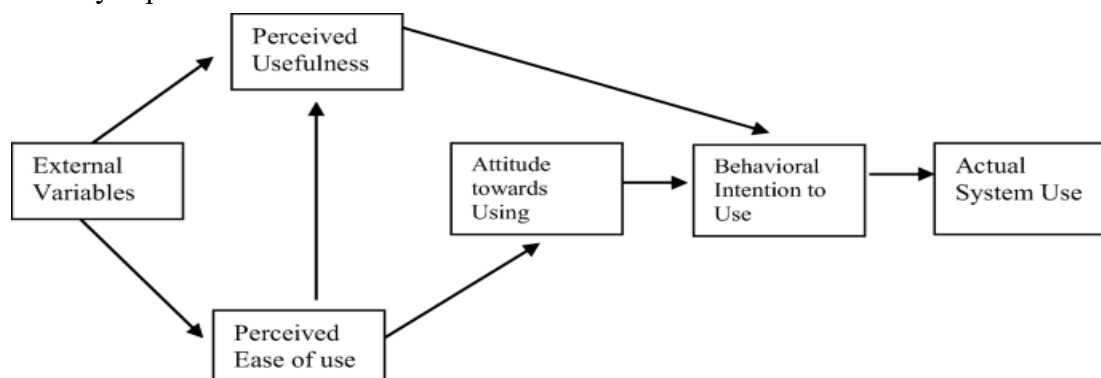


Figure 6. Technology Acceptance Model adapted from Revyathi & Tselios (2019)

The Technology Acceptance Model (TAM) is shown in the above picture.

The researchers utilized the TAM as a guide for creating the survey questions.

In the field of information systems modeling, a theory called the Technology Acceptance Model (TAM) was developed to describe how people come to accept and utilize a technology. The model indicates that when consumers are presented with a new technological tool, a multitude of factors influence their choice regarding how and when to use it. These criteria served as a means of ascertaining the users' willingness to employ the developed system.



Figure 7. ISO/IEC 25010 Software Quality Metrics

Source Image: ISO 25010. (2022). Iso25000.com. <https://iso25000.com/index.php/en/iso-25000-standards/iso-25010>

ISO 25010 is a model for assessing and estimating software quality. The standard is relevant for us since it is internationally known and used among practitioners and academic communities alike. Studies have for example used ISO 25010 to investigate what software architecture should be used for different types of software development (Haoes et al., 2016), and how to evaluate the quality of specific software products (Idri, Bachiri & FernándezAlemán, 2016; Izzatillah, 2019). ISO 25010 is also deemed a generic and broad software quality model that offers flexibility and customization to better fit practitioners' specific needs (Azouz & Lefdaoui, 2018).

Distribution of Questionnaires

The questionnaires were distributed to the study's target respondents in the form of online surveys with the use of face-to-face surveys. Before the researchers let the respondents answer the test questions, the researchers first gave a demonstration of the system that they will be using. After the researchers have finished surveying all of the target respondents, they collected all of the evaluation questionnaires in order to determine the level of satisfaction with the system that was designed. Then, the information that has been gathered was analyzed and interpreted.


 Republic of the Philippines
Laguna State Polytechnic University
 Province of Laguna
COLLEGE OF COMPUTER STUDIES


TECHNOLOGY ACCEPTANCE QUESTIONNAIRE

Dear Respondents,

Greetings in the name of the Lord!

We are students from the College of Computer Studies of the Laguna State Polytechnic Technology (BSIT) program and we are in the process of completing a research project with The working title — **“Web-Based Document Tracking System and Management for National Irrigation Administration Regional Office IV-A (CALABARZON) of Pila, Laguna”** as a partial requirement on our degree.

In this regard, we kindly request that you take the time to respond to all of the questionnaire's items, which are vital to the study's completion. The first section consists of respondent's data, while the second section evaluates user acceptability and behavioral intention to engage with the platform.

As a respondent, we seek your aid and support with this study. Your information and comments will be safeguarded and treated with the utmost discretion. Also, your positive response to this request will help the study's success.

Thank you so much for your cooperation and God bless!

Respectfully yours,

De Leon, Melvin R.
 Esmas, Dan Kenneth D.
 Sotto, Jerwin A.

RESPONDENT'S DATA

Direction: Please fill the required information in the space provided and mark (✓) the appropriate checkboxes according to your level of computer literacy.

Name (optional): _____ **Date:** _____

Respondents: ☐ Internal **Sex:** ☐ Male
 ☐ External ☐ Female

Level of Computer Literacy:

☐ **Basic:** Fundamental digital competency and basic language proficiency.
☐ **Proficient:** Comfortable with digital tools and proficient in language skills.
☐ **Advanced:** Highly skilled in digital technology and advanced language proficiency.

Figure 8. Permission Letter that was given to the respondents

The sample permission letter is shown above. This letter served as proof that the respondents gave their consent to finish the questionnaire. It contains greetings, as well as the title of the study together with the explanation of the relevance of the respondent's involvement in the study.

The next part was the self-structured test questionnaires that were distributed to the Offices of LSPU Employee, that served as the respondents of the study.

Direction: Please indicate your level of acceptance with each question by placing a checkmark (✓) corresponding to the appropriate number. Kindly use the following scale to indicate your acceptability with the following statement.

5 – Acceptable (A), 4 – Slightly Acceptable (SA), 3 – Undecided (UD),
2 – Slightly Unacceptable (SU), and 1 - Unacceptable (UA)

You may also provide your feedback and suggestions underneath the questionnaire.

QUALITY FACTORS	5	4	3	2	1
1. Appropriateness. The develop system contains all of the functions and features that the user need.					
2. Confidentiality. The develop system has secured and protected the user's privacy.					
3. User Interface. I believe that interface of the develop system is user-friendly.					
PERCEIVED EASE OF USE	5	4	3	2	1
1. Suitability. The NIA Pila, Laguna office employees would find easier to track the documents with assistance of develop system.					
2. Efficiency. The develop system produces accurate document processing as compared to the present process.					
3. Usability. The develop system is simple to use and does not require much effort in understanding the function of the system.					
PERCEIVED USEFULNESS	5	4	3	2	1
1. Functionality. The system manages documents and accounts of office employees and external.					
2. Effectiveness. The develop system reduces the responsibilities of NIA office employees in their specific task.					
3. Performance Efficiency. I believe by using develop system, I will able to create, submit, receive, send, confirmed, update, view, achieve, generate, and retrieve more efficiency.					
ATTITUDE TOWARDS USING	5	4	3	2	1
1. The designed system operates well and efficiently.					
2. The develop system will assist user save time when utilizing the system.					
3. The system will help NIA Pila, Laguna employees in tracking and process the documents in every office.					
BEHAVIORAL INTENSE OF USE	5	4	3	2	1
1. I am willing to use and accept the system in order to complete my-work tasks.					
2. I am interested to learn how the development system works.					
3. I am satisfied in terms of the overall flow of the develop system.					
EXPERIENCE	5	4	3	2	1
1. The develop system was more effective and easier to use than the present method.					
2. I never encountered any errors or problems while using the develop system.					
3. The functions of the develop system work properly.					

Figure 9. Questionnaire to assess LSPU Employee user satisfaction and intent to use

The figure above depicts the questionnaire for the LSPU Employee, it includes the direction, three (3) statements per criteria, as well as the scale that was evaluated for the TAM criteria.

CHAPTER IV

RESULT AND DISCUSSION

This chapter presents the analysis, testing, and evaluation of the developed system. It shows how the proposed web-based platform meets the needs of Laguna State Polytechnic University – Sta. Cruz Campus (LSPU-SCC) by providing an organized and efficient way for employees to review and prepare for the Civil Service Examination. The system focuses on delivering structured learning modules and conducting quizzes and examinations to assess users' knowledge and readiness.

Conceptualization

Technology has become an essential tool in education and professional development. It improves efficiency, accessibility, and organization of learning processes. One of its key applications is in examination preparation, where structured content and assessment tools play a crucial role in enhancing learning outcomes.

At Laguna State Polytechnic University – Sta. Cruz Campus, employees preparing for the Civil Service Examination often rely on traditional and unstructured review methods such as self-study and printed materials. These methods may not effectively support knowledge retention, progress monitoring, or self-assessment.

To address these challenges, the researchers developed the “Learning Management System for LSPU Employees: Civil Service Examination Preparation.” The system is designed to provide employees with organized learning modules and interactive quizzes and examinations. Through this platform, users can review lessons, take assessments, and evaluate their performance, making the review process more structured, engaging, and effective.

Research Objective

The study aims to develop an LSPU Employees: Civil Service Examination Preparation system designed to improve and simplify the review process for Civil Service Examination examinees. Specifically, this study seeks to:

The study specifically sought to perceive the following:

1. Design and develop a user-friendly web-based Learning Management System for the Civil Service Examination with the following functionalities:

For System Administrator

1. Learning Content and Modules Management
 - 1.1.1 View the master list of learning modules in a table
 - 1.1.2 Add new modules, lessons, and learning materials
 - 1.1.3 Edit modules, lessons, and uploaded materials
2. User Management
 - 1.2.1 Approve and disapprove new users
 - 1.2.2 View user details
 - 1.2.3 View archived users
 - 1.2.4 Update user information
 - 1.2.5 Retrieve user accounts
 - 1.2.6 View all accounts in table format
 - 1.2.7 Search users in the table
 - 1.2.8 Monitor user activity
3. Quiz and Assessment Management

1.3.1 Create and manage quizzes for each module

1.3.2 Add, edit, and delete questions

1.3.4 View quiz results and analytics

1.3.5 Randomized questions for exams

4. Performance Monitoring

1.4.1 View performance of employees per department

1.4.2 Identify top performers and employees needing improvement

2. To design and develop a system that would help the personnel prepare effectively for the exam by improving their knowledge, skills, and confidence.

For Employees

1. Learning and Review System

2.2.1 Access review materials for Civil Service Examination

2.2.2 Study modules

2.2.3 Take quizzes and mock examinations

2.2.4 Track learning progress and scores

2. Evaluate user acceptance and behavioral intention to use the developed system using the following criteria:

3.3.1 System quality and reliability.

3.3.2 Perceived ease of use.

3.3.3 Perceived usefulness in improving review efficiency and exam readiness.

3.3.4 User experience with the web-based features.

3.3.5 Attitude toward using the e-reviewer.

3.3.6 Behavioral intention to adopt and continue using the system.

Research Objectives No. 1: To design and develop a web-based Learning Management System for LSPU Employees: Civil Service Examination Preparation that provides a centralized platform for delivering structured learning modules, enables users to take quizzes and examinations for self-assessment, facilitates efficient monitoring of learning progress, ensures accurate recording of users' performance and results, and ultimately enhances employees' readiness, engagement, and effectiveness in preparing for the Civil Service Examination at Laguna State Polytechnic University – Sta. Cruz Campus.

The Design

The researchers utilized various data collection methods to gather the necessary information in developing the proposed web-based system in order to achieve the first objective. These methods helped the researchers understand the needs of the users and identify the problems encountered in the current review practices of employees.

The preliminary investigation was the initial step undertaken in the study. The researchers conducted an interview on October 23 with Sir Jhon Jhon Zotomayor, who served as the client and provided relevant insights regarding the current situation of employees preparing for the Civil Service Examination. Through this interview, the researchers identified the challenges encountered by employees, such as the lack of structured review materials, absence of a system for tracking progress, and limited access to organized quizzes and examinations.

These findings became the basis for designing and developing a Learning Management System that focuses on modular learning and assessment, aiming to improve the overall

preparation and performance of LSPU-SCC employees.



Figure 10: This shows the picture taken at the locale of the study and proof that the researcher conducted an interview.

Following the interview, the researchers conducted online research to gather information related to Learning Management Systems and Civil Service Examination preparation. They reviewed different websites, articles, and online sources to understand how similar systems work and to get ideas for the development of the proposed system.

After gathering the needed information, the researchers worked together to come up with possible solutions to the identified problems. They also asked for guidance from their thesis adviser and a system expert to improve the features and overall performance of the system.

The researchers used the collected online information to support the study and ensure that the developed system follows current practices in web-based learning and assessment systems.

The Development

After gathering all essential data, the researchers proceeded to the second stage, which involved developing a web-based system using the Rapid Application Development (RAD) approach.

Development of the Platform

The initial stage in designing the web-based system was requirements planning, which was the first part of the RAD Methodology process. At this point, the researchers found out the client's problem and challenges. Following the requirements planning, the researchers develop ICT solution that will address off all the issue identified. The web-based system interface was developed by the researchers using PHP, HTML5, CSS, JavaScript, and Bootstrap, with MySQL serving as the database. Following system development, the researchers moved on to the third stage, when they worked with IT professionals to evaluate the developed system using a black box. Following the users' and IT professionals approved of the web-based system, the researchers will set up the system and provide user training to demonstrate how it works. This is the last phase in the Rapid Application Development process.

All of the figures below illustrate the project's final output interface, as well as its functions and features.

User Interface

A. Learning Management System for LSPU Employees: Civil Service Examination Preparation.



Figure 11: This Figure shows the log in section

The above image displays the web-based system's user login page. Users who may access the web-based system include System Admin, Employees, and Faculty.

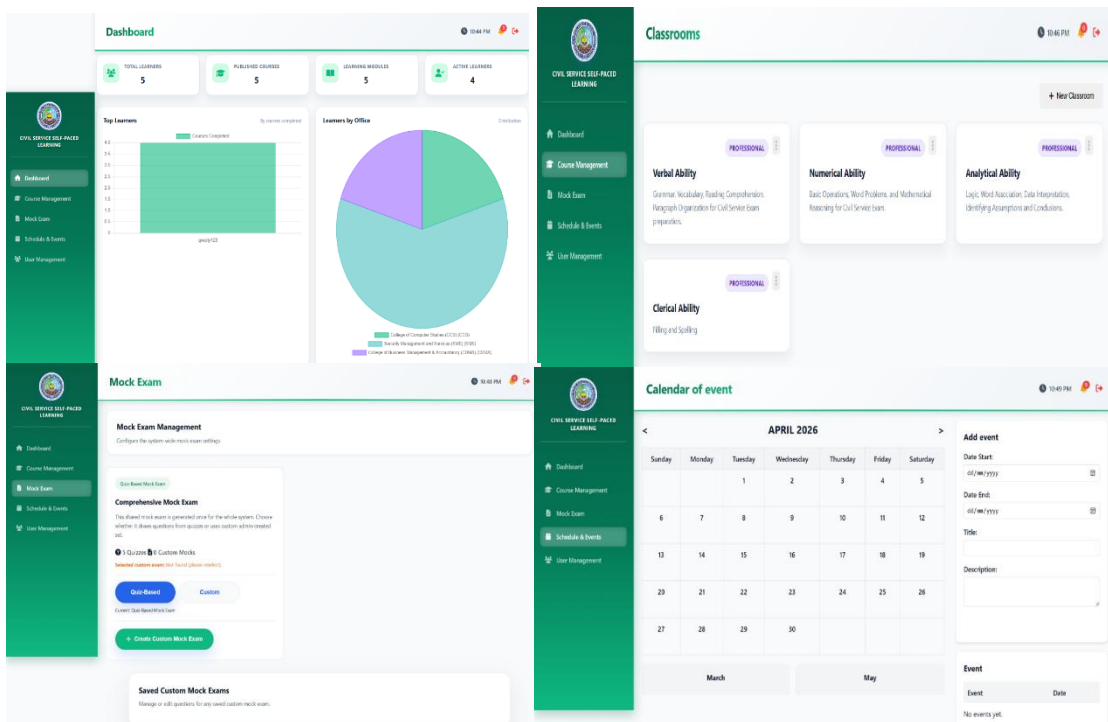


Figure 12: System Admin Account

The figure above displays the System Admin functions and processes on the system. It has several features such as a dashboard, Course Management, Mock Exam, Schedule & Event and User Management.

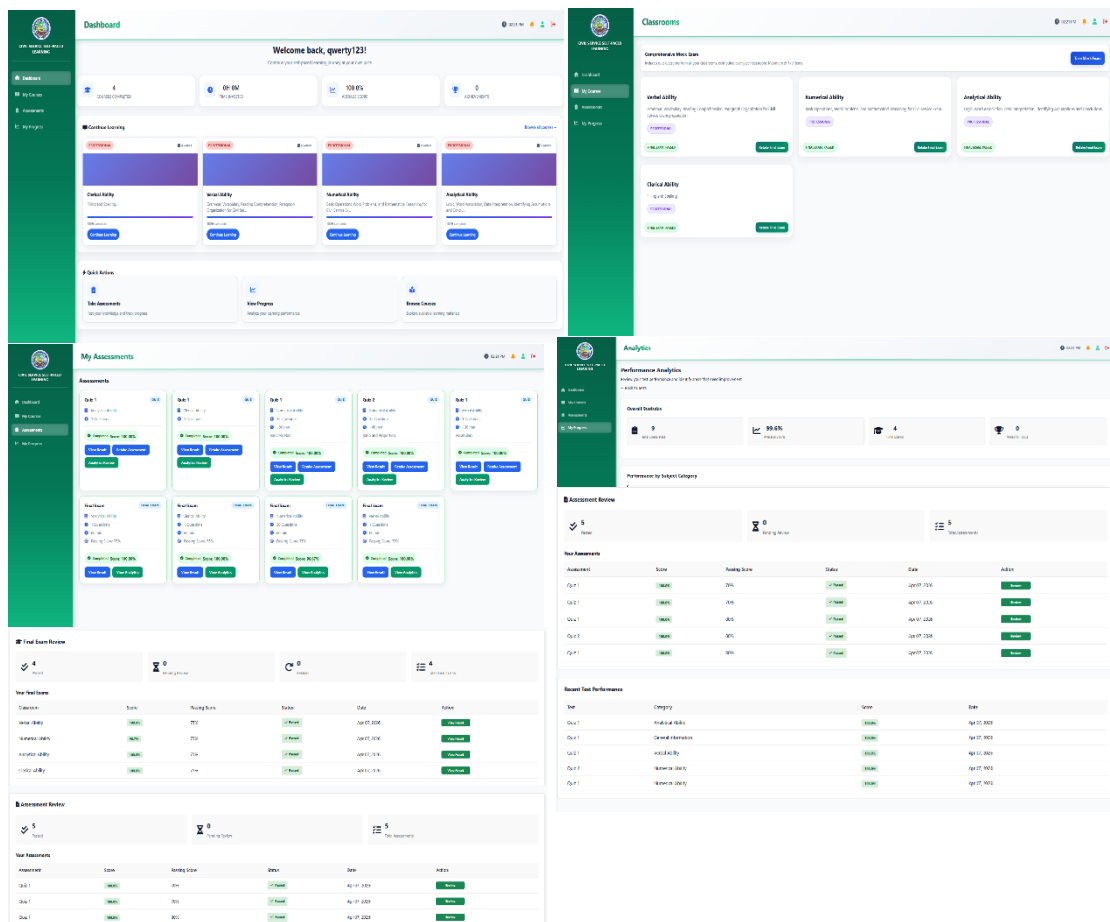


Figure 13. External Client Account

The figure above displays the Employees & Faculty functions and processes inside the system. It contains of several features such as dashboard, my courses, assessment and my progress.

Research Objectives No. 2. To design and develop a system that would help the personnel prepare effectively for the exam by improving their knowledge, skills, and confidence.

For Employees

1. Learning and Review System

2.2.1 Access review materials for Civil Service Examination

2.2.2 Study modules

2.2.3 Take quizzes and mock examinations

2.2.5 Track learning progress and scores

- 3 The researchers used the black box testing approach to examine the functions of their developed system. The researchers find ten (10) IT experts who had significant expertise and expertise in the field. The system testers were provided a testing tool with a dichotomous scale of "pass" or "fail" to provide clear distinct outcomes.

Figure 14. Face-to-face testing with Mr. Yasoña and Ms. Besande



The initial testing was also conducted face to face on April 28, 2026, with Mr. Reynaldo A. Yasoña and Ma'am Gretchen R. Besande, who began the meeting at 2:00 p.m. After the presentation and demonstration of the system, they gave their comments and suggestions for further improvement. They recommended enhancing the system's features to better support the Learning Management System for LSPU Employees: Civil Service Examination Preparation. After the testing, they expressed their appreciation for the researchers' effort and wished them success in the completion of the study.



Figure 15. Screenshots of face-to-face testing with Mr. Ramos

The second testing was also conducted face to face on April 28, 2026, with Sir Richard D. Ramos, who started the meeting at 3:00 p.m. The system was tested during the session, and after the demonstration, Sir Richard D. Ramos stated that he had no recommendation for improvement because the flow of the system was already good and easy to follow. He commended the researchers for developing a well-organized system and conveyed his best wishes for the success of the project.



Figure 16. Screenshots of face-to-face testing with Mr. Domingo

The next testing was conducted face to face with **John Lester Domingo** at **11:00 a.m.** on April 28, 2026. After the system was presented and demonstrated, he shared his observations regarding the system's overall flow and functionality. He expressed that the system was well organized and easy to navigate, and he had no additional recommendations for improvement. He then wished the researchers success in completing the study.



Figure 17. Screenshots of face-to-face testing with Ms. Etron

The testing was conducted face to face on April 26, 2026, with Ms. Patricia Mae O. Etron. After the system was presented and demonstrated, she suggested improving the appearance of the buttons to make them look more polished and visually appealing.

The tables below show the summary of the results of the Black Box Tests for each user of the developed web-based system.

Table 13. Black Box Testing Results for System Administrator

Functionality	First Trial			Second Trial		
	Pass	Fail	%	Pass	Fail	%
Dashboard	10	0	100%	10	0	100%
Sign up	10	0	100%	10	0	100%
Dashboard	10	0	100%	10	0	100%
Classroom	10	0	100%	10	0	100%
Classwork	10	0	100%	10	0	100%
User Management	9	1	90%	10	0	100%

The provided table illustrates the outcomes of black box testing conducted by 10 IT experts. It confirms that the primary functions of the system operated successfully. Specifically, all functions related to System Administrator Account achieved a flawless pass rate of 5 and 100% overall success rate. The performance of all functions collectively demonstrated effectiveness based on

the overall percentage.

Table 14. Black Box Testing Results for Users Section

Functionality	First Trial			Second Trial		
	Pass	Fail	%	Pass	Fail	%
Login	10	0	100%	10	0	100%
Sign up	10	0	100%	10	0	100%
Dashboard	10	0	100%	10	0	100%
User Classroom	10	0	100%	10	0	100%
Stream	10	0	100%	10	0	100%
Classwork	10	0	100%	10	0	100%
Read Module	10	0	100%	10	0	100%
Take Exam/Quiz	9	1	90%	10	0	100%
User Assessment	10	0	100%	10	0	100%
Take Assessment	10	0	100%	10	0	100%
Retake Assessment	10	0	100%	10	0	100%
View Result	10	0	100%	10	0	100%
My Progress	9	1	90%	10	0	100%
User Analytics	10	0	100%	10	0	100%
Display Recent Quiz/Exam	10	0	100%	10	0	100%
Display Percentage of Each Course	10	0	100%	10	0	100%

The outcomes of the black box testing conducted by the ten IT experts serve as a crucial validation of the system's functionality and reliability. The flawless pass rate for all functions related to the employees' section, coupled with the 100% overall success rate in the second trial, underscores the system's robustness in supporting learning activities and user interaction. This high level of performance not only instills confidence in the system's ability to meet

operational requirements but also reflects positively on the expertise and diligence of the development team. Furthermore, the comprehensive success across all functions indicates a thorough understanding of user needs and meticulous attention to detail during the development process.

Research Objectives No. 3. To assess and evaluate the acceptance of the user using Technology Acceptance Model in terms of:

- a. Quality Factors
- b. Perceive Ease to Use
- c. Perceived Usefulness
- d. Experience
- e. Attitude Towards to Use
- f. Behavioral Intention to Use

Demographics of the Respondents

The entire figure below presents the demographics of the respondents based on their role (admin vs. employee)



Figure 18. Demographic of the respondents based on role

The diagram depicted illustrates the breakdown of the sixty-three (63) participants surveyed. The demographic breakdown was determined by the role categories established in the survey questionnaire by the researchers. According to the graph, 1.56% of the participants identified as admin, while 98.44% identified as employees.

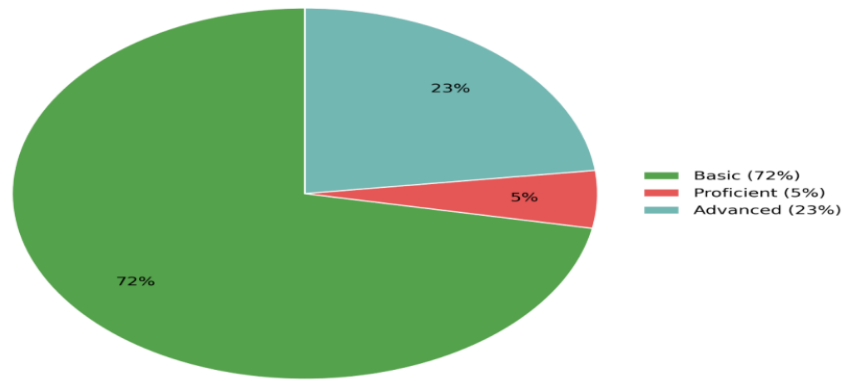


Figure 19. Demographics of the respondents based on Level of Proficiency

The illustration above displays the distribution of the sixty-three (63) respondents surveyed during the assessment. The demographic breakdown was determined by the level of computer proficiency indicated in the survey questionnaire. According to the graph, 72% of respondents are basic in using computer applications, 5% are proficient in navigating computer applications, and 23% are advanced in using computer applications.

Technology Acceptance Evaluation Survey Results and Interpretation

Below are the results of the evaluation of TAM in terms of Quality Factors, Perceived Ease of Use, Perceived Usefulness, Experience, Attitude Towards Using, and Behavioral Intention to Use.

Table 15. Evaluation of TAM in terms of Quality Factors

Identifiers	Mean	Standard Deviation	Interpretation
1. Appropriateness	4.79	0.46	Highly Acceptable
2. Confidentiality	4.71	0.50	Highly Acceptable
3. User Interface	4.76	0.43	Highly Acceptable
Total Mean	4.76		Highly Acceptable

The table provided presents the perceptions of the respondents regarding the quality factors of the developed Learning Management System for LSPU Employees:

Civil Service Examination Preparation. The first factor, **Appropriateness**, obtained a mean score of 4.79 and a standard deviation of 0.46, indicating a **Highly Acceptable** interpretation. The second factor, **Confidentiality**, received a mean score of 4.71 and a standard deviation of 0.50, also corresponding to a **Highly Acceptable** interpretation. The third factor, **User Interface**, achieved a mean score of 4.76 with a standard deviation of 0.43, again signifying a **Highly Acceptable** interpretation. The overall result yielded a mean score of 4.76, which likewise corresponds to a **Highly Acceptable** interpretation.

Table 16. Evaluation of TAM in terms of Perceived Ease to Use

Identifiers	Mean	Standard Deviation	Interpretation
1. Suitability	4.55	0.56	Highly Acceptable
2. Efficiency	4.48	0.58	Highly Acceptable
3. Usability	4.59	0.54	Highly Acceptable
Total Mean	4.53		Highly Acceptable

The table provided presents the perceptions of the respondents regarding the perceived ease of use of the developed Learning Management System for LSPU Employees: Civil Service Examination Preparation. The first factor, **Suitability**, obtained a mean score of 4.55 and a standard deviation of 0.56, indicating a **Highly Acceptable** interpretation. The second factor, **Efficiency**, received a mean score of 4.48 and a standard deviation of 0.58, also corresponding to a **Highly Acceptable** interpretation. The third factor, **Usability**, achieved a mean score of 4.59 with a standard deviation of 0.54, again signifying a **Highly Acceptable** interpretation. The overall result yielded a mean score of 4.53, which likewise corresponds to a **Highly Acceptable** interpretation. This overall high mean score affirms the positive perception of the respondents toward the system, further reinforcing its usability and convenience in supporting Civil Service examination preparation.

Table 17. Evaluation of TAM in terms of Perceived Usefulness

Identifiers	Mean	Standard Deviation	Interpretation
1. Functionality	4.60	0.55	Highly Acceptable
2. Effectiveness	4.55	0.54	Highly Acceptable
3. Performance Efficiency	4.65	0.54	Highly Acceptable
Total Mean	4.60		Highly Acceptable

The table provided presents the perceptions of the respondents regarding the perceived usefulness of the developed Learning Management System for LSPU Employees: Civil Service Examination Preparation. The first factor, **Functionality**, obtained a mean score of 4.60 and a standard deviation of 0.55, indicating a **Highly Acceptable** interpretation. The second factor, **Effectiveness**, received a mean score of 4.55 and a standard deviation of 0.54, also corresponding to a **Highly Acceptable** interpretation. The third factor, **Performance Efficiency**, achieved a mean score of 4.65 with a standard deviation of 0.54, again signifying a **Highly Acceptable** interpretation. The overall result yielded a mean score of 4.60, which likewise corresponds to a **Highly Acceptable** interpretation.

Table 18. Evaluation of TAM in terms of Attitude Towards Using

Identifiers	Mean	Standard Deviation	Interpretation
1. The designed system operates well and efficiently.	4.64	0.50	Highly Acceptable
2. The develop system will assist user save time when utilizing the system.	4.61	0.49	Highly Acceptable
3. The system will help NIA Pila, Laguna employees in tracking and process the documents in every office.	4.64	0.54	Highly Acceptable
Total Mean	4.63		Highly Acceptable

The table provided presents the perceptions of the respondents regarding their attitude towards using the developed Learning Management System for LSPU Employees: Civil Service Examination Preparation. The first indicator, **The designed system operates well and efficiently**, obtained a mean score of 4.64 and a standard deviation of 0.50, indicating a **Highly Acceptable** interpretation. The second indicator, **The developed system will assist users in saving time when utilizing the system**, received a mean score of 4.61 and a standard deviation of 0.49, also corresponding to a **Highly Acceptable** interpretation. The third indicator, **The system will help LSPU employees in tracking and processing their review materials and activities**, achieved a mean score of 4.64 with a standard deviation of 0.54, again signifying a **Highly Acceptable** interpretation. The overall result yielded a mean score of 4.63, which likewise corresponds to a **Highly Acceptable** interpretation.

Table 19. Evaluation of TAM in terms of Behavioral Intense to Use

Identifiers	Mean	Standard Deviation	Interpretation
1. I am willing to use and accept the system in order to complete my-work tasks.	4.46	0.56	Highly Acceptable
2. I am interested to learn how the development system works.	4.53	0.52	Highly Acceptable
3. I am satisfied in terms of the overall flow of the develop system.	4.59	0.55	Highly Acceptable
Total Mean	4.53		Highly Acceptable

The table provided presents the perceptions of the respondents regarding their behavioral intention to use the developed Learning Management System for LSPU Employees: Civil Service Examination Preparation. The first indicator, **I am willing to use and accept the system in order to complete my tasks and review activities**, obtained a mean score of 4.46 and a standard deviation of 0.56, indicating a **Highly Acceptable** interpretation. The second indicator, **I am interested to learn how the developed system works**, received a mean score of 4.53 and a standard deviation of 0.52, also corresponding to a **Highly Acceptable** interpretation. The third indicator, **I am satisfied with the overall flow of the developed system**, achieved a mean score of 4.59 with a standard deviation of 0.55, again signifying a **Highly Acceptable** interpretation. The overall result yielded a mean score of 4.53, which likewise corresponds to a **Highly Acceptable** interpretation.

Table 20. Evaluation of TAM in terms of Experience

Identifiers	Mean	Standard Deviation	Interpretation
1. The develop system was more effective and easier to use than the present method.	4.60	0.57	Highly Acceptable
2. I never encountered errors or problems while using the develop system.	4.55	0.56	Highly Acceptable
3. The functions of the develop system work properly.	4.61	0.53	Highly Acceptable
Total Mean	4.53		Highly Acceptable

The table provided presents the perceptions of the respondents regarding their experience in using the developed Learning Management System for LSPU Employees: Civil Service Examination Preparation. The first indicator, **The developed system was more effective and easier to use than the present method**, obtained a mean score of 4.60 and a standard deviation of 0.57, indicating a **Highly Acceptable** interpretation. The second indicator, **I never encountered errors or problems while using the developed system**, received a mean score of 4.55 and a standard deviation of 0.56, also corresponding to a **Highly Acceptable** interpretation. The third indicator, **The functions of the developed system work properly**, achieved a mean score of 4.61 with a standard deviation of 0.53, again signifying a **Highly Acceptable** interpretation. The overall result yielded a mean score of 4.53, which likewise corresponds to a **Highly Acceptable** interpretation. Such positive feedback serves as valuable validation of the system's design and functionality, providing a solid foundation for future enhancements and improvements.

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATION

This chapter discusses the primary findings on the level satisfaction with the developed web-based system. The conclusions were based on the study's goal, research questions, and results. The implications of these findings, as well as the recommendations that follow, are explained. The recommendations were based on the study's conclusions and purpose.

Summary

This study focused on the design and development of a Learning Management System for LSPU Employees: Civil Service Examination Preparation. The system was created to provide accessible, flexible, and organized review materials aligned with the competencies required for the exam.

The research examined the challenges faced by employees, such as limited time for review, lack of structured materials, and difficulty accessing reliable resources. To address these concerns, the LMS was developed with key features including user-friendly navigation, categorized review modules, practice quizzes, performance tracking.

The system was evaluated based on usability, functionality, reliability, and overall user satisfaction. Feedback from respondents indicated that the LMS significantly improved their review experience by making learning more convenient and structured.

The researchers were able to obtain the necessary data via interviews, library research, web research, technological gap analysis, and brainstorming. The first survey was conducted to establish the problems that employees experience, as well as the present procedures for document tracking, document processing, and user management. The

researchers used the Rapid Application Development (RAD) methodology when developing the web-based solution. With this methodology, the researchers can review and enhance the program without having to restart the development process from the beginning.

To evaluate the performance and usefulness of the develop system, the researchers created a questionnaire using Black Box testing. The develop system was tested virtually with the help of IT experts who served as a system tester. The tests show that all of the system's functionalities work properly.

During the system evaluation, the researchers develop and prepared a questionnaire based from the Technology Acceptance Model (TAM) and ISO/IEC 25010 Software Quality Metrics, which used a 5-point Likert Scale response to assess user satisfaction. Survey Questions were tested and sent to assess the system's usability, effectiveness, and user satisfaction. In this study, the researchers did a face-to-face evaluation to determine the user's level of satisfaction.

The data gathered from the 63 respondents were evaluated and interpreted. The interpreted data resulted in a response scale that was measured against each evaluation criteria, and the outcome was "Highly Acceptable," indicating that the users were satisfied with the developed LEARNING MANAGEMENT SYSTEM for LSPU EMPLOYEES: Civil Service Examination Preparation.

Conclusions

The feedback gathered after the testing and evaluation phase proved that the developed web-based system is working well and the respondents were satisfied and willing to use the developed system. In conclusion, the study was able to achieve its objective: (1) Develop a Learning Management System for LSPU Employees: Civil Service Examination Preparation; (2) To design and develop a system that would help

the document handler and guest to track their documents; (3) To assess and evaluate the acceptance of the user using Technology Acceptance Model in terms of Quality Factors, Perceived Ease to Use, Perceived Usefulness, Experience, Attitude towards Using, and Behavioral Intention to Use towards the develop system.

In the study, the researchers therefore conclude the following:

1. After gathering data through a comprehensive preliminary investigation, and interviews with employees of the National Irrigation Administration of Pila, Laguna, the researchers concluded that the current document tracking process needs to be improved in terms of efficiency and reliability, and that developing a web-based system can be a better solution to the difficulties that employees face when doing their specific tasks.
2. The Rapid Application Development (RAD) methodology used by the researchers is beneficial for developing the web-based system.
3. Using Black Box Testing, the researchers discovered that the developed web-based system met its specified functional requirements.

Recommendations

To further enhancement of “Learning Management System for LSPU Employees: Civil Service Examination Preparation”, the following are recommended below:

1. To future researchers, the web-based system should be more responsive on the web but not responsive on mobile. Future researchers should make their work responsive to mobile devices so that mobile users can access their work more easily.

LITERATURE CITED

- A Production Inventory System with Service Time and Production Vacations | Journal of Systems Science and Systems Engineering (springer.com)
- A serial inventory system with lead-time-dependent backordering: A reduced- state approximation (repec.org)
- A. D. AYU, ET AL. (2022) Development of Autonomous Control System using Self-Organizing Map and Autoregressive Self-Organizing Map. (2022, October 6). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9946634/>
- A. D. Reina, ET AL. (2021) Animals and Technology in Space: A perspective from Aerospace Engineering to Veterinary Medicine. (2021, July 14). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9537247>
- A. DATTA, ET AL. (2022) Sub-GHz In-Body to Out-of-Body communication channel modeling for ruminant animals for smart animal agriculture. (2023, April 1). IEEE Journals & Magazine | IEEE Xplore. <https://ieeexplore.ieee.org/document/9915440>
- A. JINDAL ET AL. (2023) Constructing feedback linearizable discretizations for Continuous-Time systems using retraction maps. (2023). IEEE Journals & Magazine | IEEE Xplore. <https://ieeexplore.ieee.org/document/10153969/>
- A. M. NOMA, ET AL. (2022) Design of intelligent and secure hospital appointment scheduling system. (2022, April 17). IEEE Conference Publication IEEE Xplore. <https://ieeexplore.ieee.org/document/9803093>
- A. MISSAOUI, ET AL. (2020) Simulation and analytic hierarchy process to

implement outpatient appointment system at Habib Bourguiba Hospital Sfax Tunisia. (2020, November 8). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9317025>

A. N. MAHANTHI, ET AL. (2023) Transforming Traditional Medical Appointment Booking with Advanced Technologies and User-Centric Design. (2023, June 1). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10265954/>

A. S., N SONI, H., (2021). A Continuous Review Production Inventory System with a Variable Preparation Time in a Fuzzy Random Environment. Mathematics, 9(7), 747. Mathematics | Free Full-Text | A Continuous Review Production-Inventory System with a Variable Preparation Time in a Fuzzy Random Environment (mdpi.com)

AHMADI, E., ET AL (2019). Inventory management of surgical supplies and sterile instruments in hospitals: a literature review. Health Systems, 8(2), 134-151. Modelling and analysis of healthcare inventory management systems (repec.org)

An inventory system with demand dependent on both time and price assuming backlogged shortages (repec.org)

ASSIS, A.G., ET AL., (2022). Classification of Materials and Medicines in Hospital Inventory Management: A Multi-Criteria Analysis. Classification of medicines and materials in hospital inventory management: a multi-criteria analysis | BMC Medical Informatics and Decision Making | Full Text (biomedcentral.com)

Azimah D. Ampuan; Reymark D. Deleña (2024) A Quantitative Evaluation of Online Appointment system at Mindanao State University–Main Campus: Employing

the System Usability Scale (SUS) and Technology Acceptance Model (TAM). (2024, January 29). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10470770/>

B Kiran; S Karthikeyan; M A Suhel Pasha; K N Manjunatha; S Manoj Kumar; Sharvin Vivian Moras (2022) Design and development of autonomous mobile robot for mapping and navigation system. (2022, December 15). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10014944/>

BALUGANI, E., LOLLI, F., ET AL., (2019, January). A periodic inventory system of

Behnaz Gharakhani, Mahsa Ghandehari & Azarnoush Ansari. (2023) GeoSOT-OctoMap: An OCTree Grid Map model for autonomous driving. (2021, October 15). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9641485/>

BERK, E., & TOY A. O. (2022). A serial inventory system with lead-time-dependent backordering: A reduced-state approximation. IISE Transaction.

Chinakorn Sujimongkol; Somsri Daochai; Kuttiya Kaewsombut; Chayanis Daochai (2022) Prevalence Survey of Renal Replacement Therapy Services: A Cross-Sectional Study. (2022, November 10). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10012084>

Chinakorn Sujimongkol; Somsri Daochai; Suntharee Wichakhrueang; Chayanis Daochai (2023) Meta-analysis on the Prevalence and impact of False, nonactionable, and nuisance alarms in Critical care environments: Informing Evidence-Based Practices for Improving Patient Safety in Alarm Monitoring Systems. (2023a, October 28). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10322051/>

Chinakorn Sujimongkol; Somsri Daochai; Suntharee Wichakhrueang; Chayanis Daochai (2023) Meta-analysis on the Prevalence and impact of False, nonactionable, and nuisance alarms in Critical care environments: Informing Evidence-Based Practices for Improving Patient Safety in Alarm Monitoring Systems. (2023b, October 28). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10322051/>

Chinese journal of medical instrumentation.

Ching-Chien Yang; Ling Ling He; Yuchen Ding; Chaohui Tang (2021) Using Regression analysis to explore Sex Health Cognition of Female College Students in China. (2021, October 1). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9681138>

Christoph Klaassen; Lukas Kasper; René Hofmann (2024) Mapping SysML v2 to NGSI-LD: Enhancing Energy Systems Modeling. (2024, September 3). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10668973/>

Conceptual model of failure risk control on raw materials inventory system - IOPscience

Cristhian Saavedra; Carlos Honores; Wilfredo Ticona (2024) Implementation of a Web-Based system to improve the appointment process in a clinic. (2024, January 18). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10463205/>

De Beauvoir, S. (2023). The second sex. In Routledge eBooks (pp. 346–354). <https://doi.org/10.4324/9781003320609-44>

Edison M. Esberto; Rose Ann C. Malaborbor; Alliana M. Ablan (2023) Development and evaluation of AppSYS: an online appointment system. (2023, October 13).

IEEE Conference Publication | IEEE

Xplore. <https://ieeexplore.ieee.org/document/10509985/>

FANG, X., & CHEN, H-C. (2021, February). Using vendor management inventory system for goods inventory management in lot manufacturing. Enterprise Information Systems.

Gino P. Monrroy; Pedro S. Castañeda (2024) System to optimize the process of booking medical appointments for people with upper extremity disabilities. (2024, March 15). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10541695>

Giva Andriana Mutiara; Gita Indah Hapsari; Muhammad Rizqy Alfarisi; Periyadi; Lisda Meisaroh; Nadya Nanda Adisty Hadian (2024) Body Weight-Based Animal Laboratory Classification utilizing load cell and IoT. (2024, August 7). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10698569>

Gloria Beraldo; Silvia Di Battista; Silvana Badaloni; Emanuele Menegatti; Monica Pivetti (2018) Sex differences in expectations and perception of a social robot. (2018, September 1). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/8625826>

GUANGYIN XU, ET AL., (2019). Simulation-based optimization of control policy on multi-echelon inventory system for fresh agricultural products. International Journal of Agricultural and Biological Engineer, Vol 12, No 2 (2019). Simulation-based optimization of control policy on multi-echelon inventory system for fresh agricultural products | Xu | International Journal of Agricultural and Biological Engineering (ijabe.org)

Hualong Cai; Jingyao Luo; Yiyan Zhuang; Caiying Huang (2023) Design of

- Electronic Health Record-Based Community Information Medical Service Management System. (2023, December 16). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10727504/>
- Improvement of Inventory System Using First In First Out (FIFO) Method IOP Science
- intermittent demand items with fixed lifetimes. International Journal of Production Research, Volume 57, 2019 – Issue 22. A periodic inventory system of intermittent demand items with fixed lifetimes (repec.org)
- inventory system with demand dependent on both time and price assuming backlogged shortages. European Journal of Operational Research, Volume 270, Issue 3, 1 November 2018, Pages 889-897.
- IoT-Based Asset Management System for Healthcare-Related Industries - Lee Carman Ka Man, Cheng Mei Na, Ng Chun Kit, 2015 (sagepub.com).
- J. ZHONG, ET AL. (2012). Development of medical supplies management system.
- Jingwei Huang; Wael Khallouli; Handley Holly A. H.; William Edmonson; Trisha Ahmed; Nadew Kibret (2021) Semantic Mapping from SysML to FRP: to Enable Executable and Verifiable Systems Design. (2021, April 15). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9447075/>
- John Ivan C. Maurat; Elsie V. Isip; Jensen A. Santillan; Aileen Gail R. Lumabas; Emmanuel A. Ramirez; Roger C. Primo (2024) Point-of-Sale Solution with Built-in CRM and Appointment Scheduling Empowered by Android Mobile Application. (2024, August 10). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10668385>
- Julian Thomson; Michael Karalewich (2020) Precision Four-Dimensional digital

mapping teaches humans and machines to see in the dark. (2020, October 11).

IEEE Conference Publication | IEEE

Xplore. <https://ieeexplore.ieee.org/document/9256683/>

Kavitha K R; VijayalakshmiS VijayalakshmiS; Murali BabuB; Anusha A; Dharshini

K (2023) Efficient RFID-Integrated online appointment scheduling system.

(2023, December 18). IEEE Conference Publication | IEEE

Xplore. <https://ieeexplore.ieee.org/document/10435111/>

Krishnakumar. S; J. Refonaa; S.L. Jany Shabu; Kg Dion Paul; S. Dhamodaran; Viji

Amutha Mary (2023) MediStock: Medical Inventory Management System.

(2023, July 6). IEEE Conference Publication | IEEE

Xplore. <https://ieeexplore.ieee.org/document/10193177>

Lara M. Wierenga; Marieke G. N. Bos; Fabienne van Rossenberg; Eveline A. Crone

(2019) Sex Effects on Development of Brain Structure and Executive

Functions: Greater Variance than Mean Effects. (2019, May 1). MIT Press

Journals & Magazine | IEEE

Xplore. <https://ieeexplore.ieee.org/document/8678735>

LE CARMAN K. M., ET AL. (2015). IoT-Based Asset Management System for

Healthcare-Related Industries. International Journal of Engineering Business

Management.

Lei Lai; Xinyi Yu; Xuecheng Qian; Linlin O (2020) 3D Semantic Map Construction

System based on Visual SLAM and CNNs. (2020, October 18). IEEE

Conference Publication | IEEE

Xplore. <https://ieeexplore.ieee.org/document/9254223/>

M. ATAALILI, ET AL. (2019, March). Joint inventory control and pricing in a

service-inventory system. International Journal of Production

Economics, Volume 209, March 2019, Pages 78-91. Joint inventory control and pricing in a service-inventory system (repec.org)

Maher A. N. Agi & Hardik N. Soni. (2020) Demystifying the Transfer Learning based Detection of Animal Diseases from Images. (2023, May 11). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10150962/>

Mamiko Shimada; Kazutaka Kurihara; Takaaki Tsujii (2022) Prototype of an Augmented Reality System to Support Animal Surgery using HoloLens 2. (2022, March 7). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9754801/>

Measuring the logistics performance of internal hospital supply chains – A literature study - ScienceDirect

Mirana Angel; Anuj Patel; Haiyi Xing; Dylan Balsz; Cody Arbuckle; David Bruyette (2023) AI and Veterinary Medicine: Performance of large Language models on the North American Licensing Examination. (2023, November 21). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10375414/>

Modelling and analysis of healthcare inventory management systems (repec.org)

Mohammad M. Qabajeh; Shima Mousa; Hamam Abo Saleh; Jamila Abo Hasan (2021) Design and implementation of mobile-based application for appointments systems for vaccinations in medical centers. (2021, December 11). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9916616/>

MONDOL, E. P. (2021). The Impact of Block Chain and Smart Inventory System on Supply Chain Performance at Retail Industry. International Journal of Computations Informations and Manufacturing (IJCIM, 1(1).

MOONS, K., WAEYENBERGH, G., PINTELON, L. (2019). Measuring the logistics performance of internal hospital supply chains-a literature study. *Omega*, 82, 205-217.

Nazlı Tokatlı; Muhammed Tayyip Koçak; Seda Kirtay; Gürkan Göztepeli; İbrahim Serhat Aktaş; Halis Altun (2023) Healthcare Service Accessibility Path Planner: Unveiling a new era of intelligent appointment management systems based on outpatient prioritizing. (2023, October 11). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10296568/>

Nicole Jillian B. Day; Karmelo Antonio Lazaro R. Carranza; Lawrence Matthew S. Lin; Albert R. Ponce; Wilbur Rex O. Reyes; Nilo T. Bugtai (2018) Design of a web-based and electronic health record management system for medical teleconsultation. (2018a, November 1). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/8666370>

Nicole Jillian B. Day; Karmelo Antonio Lazaro R. Carranza; Lawrence Matthew S. Lin; Albert R. Ponce; Wilbur Rex O. Reyes; Nilo T. Bugtai (2018) Design of a web-based and electronic health record management system for medical teleconsultation. (2018b, November 1). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/8666370>

Nouçaiba Sbai; Loubna Benabbou; Abdelaziz Berrado (2020) An AHP based approach for multi-echelon inventory system selection: Case of Distribution Systems. (2020a, October 28). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9314711>

Nouçaiba Sbai; Loubna Benabbou; Abdelaziz Berrado (2020) An AHP based approach for multi-echelon inventory system selection: Case of Distribution Systems. (2020b, October 28). IEEE Conference Publication | IEEE

Xplore. <https://ieeexplore.ieee.org/document/9314711>

of Inventory System Using First In First Out (FIFO) Method. Journal of Physics: Conference Series, Volume 1361.

Pattaramanee Arsomngern; Nichakorn Numcharoenpinij; Jitpinun Piriataravet; Wutiwong Teerapan; Woranich Hinthong; Phond Phunchongharn (2019) Computer-Aided Diagnosis for Lung Lesion in Companion Animals from X-ray Images Using Deep Learning Techniques. (2019, October 1). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/8923126/>

Pradnya Bhangale; Bhavik Bhatt; Muskaan Nandu; Piyush Chavda (2021) MeetUp: An Appointment Booking System using Flutter and Django Framework : MeetUp: Meetings Made Easy. (2021, October 7). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9591786/>

Rawan Nasser Al Siyabi Engineering department, UTAS-Al Mussanah, Al Mussanah, Sultanate of Oman ; Manal Rashid Al Hattali; Basma Abdullah Al Kharusi; Jahina Rashid Al Abri; Bindu Puthentharayil Vikraman; Vanitha Mahadevan (2023) Hospital Appointment Management System for Telemedicine System Reinforced with Diagnostic Data Transfer. (2023, January 5). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10117887/>

Rusyaizila Ramli; Kristo Radion Purba; An Nur Khushairah Mohd Nor Azman Kuzaimi (2022) The Development of Clinic Management System Mobile Application with Integrated Appointment, Prescription, and Payment Systems. (2022a, July 23). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9845170/>

- Rusyaizila Ramli; Kristo Radion Purba; An Nur Khushairah Mohd Nor Azman Kuzaimi (2022) The Development of Clinic Management System Mobile Application with Integrated Appointment, Prescription, and Payment Systems. (2022b, July 23). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9845170/>
- S. ADAK, ET AL. (2020). Effect of reliability on multi-item inventory system with shortages and partial backlog incorporating time dependent demand and deterioration. *Annals of Operations Research*, 1-21. Effect of reliability on multi-item inventory system with shortages and partial backlog incorporating time dependent demand and deterioration (repec.org)
- Sabale Pankaj; Ashwin Tomar; Yadav Poojashree; Jadhav Saurabh; Shubhangi Bhatambrekar; Shashank Awasthi (2022) The study of Online Appointment System - a case study. (2022, May 20). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9844363/>
- SAHA, E., & RAY, P.K., (2019). Modelling and analysis of inventory management systems in healthcare: A review and reflections. *Computers & Industrial Engineering*, 137, 106051.
- SAN-JOSE, L. A., SICILIA, J., & LOPEZ-DE- PABLO, D. A. (2018, November). An
- SEMBIRING, A. C., TAMPUBOLON, J., ET AL. (2018, November). Improvement
- Srinivasan, A. (2024).)The right to sex. In *Feminist Theory* (pp. 195–198). https://doi.org/10.1007/978-3-031-55397-4_15
- Steve Caleb Roosevelt; Ebenezer Veemaraj; Stewart Kirubakaran (2024) Steve Caleb Roosevelt; Ebenezer Veemaraj; Stewart Kirubakaran (2024) Steve Caleb Roosevelt; Ebenezer Veemaraj; Stewart Kirubakaran (2024) Real Time Stock

- Inventory Management System. (2024, July 29). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10677773>
- Sudeep Tanwar ^a, Karan Parekh ^a, Richrd Evans (2020)Blockchain-based electronic healthcare record system for healthcare 4.0 applications. <https://www.sciencedirect.com/>
- SUWANDI, A., ET AL. (2018). Conceptual model of failure risk control on raw materials inventory system. IOP Conference Series: Materials Science and Engineering, Volume 453.
- Syed Riaz un Nabi Jafri; Sheraz Shamim; Syed Muhammad Yasir; Saad Ahmed; Muhammad Owais Ali Siddiqui; Abdul Basit (2021) Low cost backpack scanning and mapping system for indoor environments. (2021, October 26). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9651437/>
- The Impact of Block Chain and Smart Inventory System on Supply Chain Performance at Retail Industry | International Journal of Computations, Information and Manufacturing (IJCIM) (gaftim.com)
- U. KAUR, ET AL. (2021). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9629743>
- Using vendor management inventory system for goods inventory management in IoT manufacturing (researchgate.net)
- V. Kadunc Kos; M. Brložnik; A. Domanjko Petrič; V. Avbelj (2020) Simultaneous phonocardiography and electrocardiography using smartphone in dogs, cats and horses. (2020, September 28). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/9245173>
- V. MALACCO, ET AL. (2023). IEEE Journals & Magazine | IEEE

Xplore. <https://ieeexplore.ieee.org/document/9915440/>

Viacheslav Guzhov; Kyrylo Smelyakov (2024) Free DBMSs Performance for an Inventory Management System based on Spring Boot. (2024, April 25). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10542597>

Visvesvaran C; Shri Yazhini R; Sneha T; Swetha K (2023) Resource Allocation Algorithm for web based hospital Appointment management system. (2023, November 17). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10395877/>

WERAIKAT, D., ZANJANI, M. K., & LEHOUX, N. (2019). Improving sustainability in a two-level pharmaceutical supply chain through Vendor-Managed Inventory system. *Operations Research for Health Care*, 21, 44-55.

Xuchuan Yuan. (2023) High-Definition Maps: comprehensive survey, challenges, and future perspectives. (2023). IEEE Journals & Magazine | IEEE Xplore. <https://ieeexplore.ieee.org/document/10184094/>

Yang Zhou; Xi Xia; Ruhong Yang; Xiaofeng Xu; Yihan Yu; Junjie Chen (2024) Developing Multi-scale Digital Maps with Vector Tiles: A Cartographic Analysis using ArcGIS Pro. (2024, April 12). IEEE Conference Publication | IEEE Xplore. <https://ieeexplore.ieee.org/document/10581317/>

YUE, D., & QIN, Y. (2019). A Production Inventory System with Service Time and Production Vacations. *Journal of Systems Science and Systems Engineering*, 28(2), 168-180.

Zaid Farooq Pitafi; Melissa Sleda; Silvia N.J. Moreno; S. Mark Tompkins; Benjamin M. Brainard; WenZhan Song (2024) CAgeDOT: contactless animal activity monitoring system to follow infectious disease progress. (2024, June 9). IEEE

Conference Publication | IEEE

Xplore. <https://ieeexplore.ieee.org/document/10622729/>

Zhuolin Deng; Chen Xiong; Ming Cai (2022) An Autonomous Transportation System
Architecture Mapping relation generation method based on text analysis.
(2022, December 1). IEEE Journals & Magazine | IEEE Xplore.
<https://ieeexplore.ieee.org/document/9753675/>

APPENDIX A

Technical Background

Transcribed Interview

Transcribe Interview with Sir Zotomayor

Date: May 4, 2026

Interviewers: Research Team

Topic: Learning Management System for LSPU Employees: Civil Service Examination Preparation

Interviewer: Good morning, Sir Zotomayor. ask lang po namin kung pwede po ba kayong kunin na client for thesis lang po, gagawan po namin itong office nyo po na makakatulong sainyo ng system para po mapadali ang gawain nyo po ano po kaya ang medjo nahihirapan kayo na may maitulong po kami.

Sir Zotomayor: Good morning. Wala naman akong problema sa office pero meron akong gusting ipagawa sa inyo, ayon ay yung sa Civil Service.

Interviewer: Ano po bang nangyayari sa Civil Service?

Sir Zotomayor: Itong Civil Service na ito ay parang may mga employee na binibigyan ko ng mga module parang refresher nila, dito ay tinutulungan natin ang mga empleyado ng LSPU para makapasa o makakuha sila ng eligibility na Civil Service

Interviewer: Sa tingin niyo po ba ay ganito po gagawin naming sa System, gagawa po kami ng website tapos doon mag rereview ang mga empleyado po at lalagyan po natin siya ng exam at mock exam para po makapagfocus po sila sa pagrereview o mabigyan po nila to ng oras.

Sir Zotomayor: Yes. Okay lang ba yan? Magandang Idea yan

Interviewer: May Suggestion pa po ba kayo?

Sir Zotomayor: Wala na naman, okay na naman sakin yan. Pag iisipan ko muna kung anong mga pwedeng ilagay or idagdag dito

Interviewer: Sige po, Thank you po Sir Zotomayor. Balik na lang po kami kapag

meron po kaming katanungan.

Sir Zotomayor: You're welcome. Goodluck Sa inyo

Transcribed Interview with Sir Zotomayor

Date: May 7, 2026

Interviewers: Research Team

Topic: Learning Management System for LSPU Employees: Civil Service Examination Preparation

Interviewer: Good morning, Sir Zotomayor. Nagbabalik po kami

Sir Zotomayor: Good morning. Anong gawa niyo rito?

Interviewer: Bali po meron po sana kaming hingin, pwede po ba?

Sir Zotomayor: Yes naman, ano ba 'yan?

Interviewer: Gusto po sana naming hingin ay data ng mga Module po ng Ilalagay sa Civil Service Exam Preparation. Ang Title po naming ngayon ay LEARNING MANAGEMENT SYSTEM FOR LSPU EMPLOYEES: Civil Service Examination Preparation.

Sir Zotomayor: So hihingin mo ba every topic?

Interviewer: Yes po, diba po meron pong 4 na course or scope po ang Civil Service?

Sir Zotomayor: Yes, ang Civil Service ay may Numerical Ability, Verbal Ability, Analytical Ability at General Information

Interviewer: Opo, pwede po ba naming mahingi yung mga data/module nila.

Ilalagay na po kasi naming sa system para may nakikita na po

Sir Zotomayor: Yes, ayusin ko muna yung iba at ibigay ko muna sa inyo Yung Numerical Ability. Isend niyo sakín gmail niyo para doon ko na lang isesend yung Data, and the rest of the data ay bigay ko na lang mga next week.

Ayusin ko muna siya.

Sir Zotomayor: May tanong ako paano kaya yung quiz diyan sa system

Interviewer: Gusto niyo po bang makita Sir?

Sir Zotomayor: Yes, patingin ako

Interviewer: Ang nangyayari po kasi rito sa system ay diba po dito sa module 1

kapag hindi niyo pa po natatapos ang module 1 ay hindi niyo po maoopen
yung quiz 1 or next na module. Kailangan niyo pong tapusin ang mga
naunang module bago makapag proceed sa next

Sir Zotomayor: Ow nice, Okay.

Interviewer: may isa pa po sana kaming itatanong Sir

Sir Zotomayor: Sige

Interviewer: Pwede po bang malaman kung ilan po ang employee na sinesendan
niyo ng module o yung gagamit nito

Sir Zotomayor: Maraming Employee nag LSPU pero yung iba ay hindi na sila
nagttake or hindi na nila tinitignan yung mga sinesend ko na reviewer dahil
hindi sila kukuha ng Civil Service, pwedeng ilagay mo ay 126 na employee.

Interviewer: Okay po Sir, ayon lang naman po ang gusto naming mahingi. Balik na
lang po ulit kami kapag may katanungan kami, Maraming Salamat po

Sir Zotomayor: Sige, Goodluck. Send ko na lang sa gmail na nakalagay dito sa
papel

Interviewer: Okay po

Story Board

Figure Admin Interface

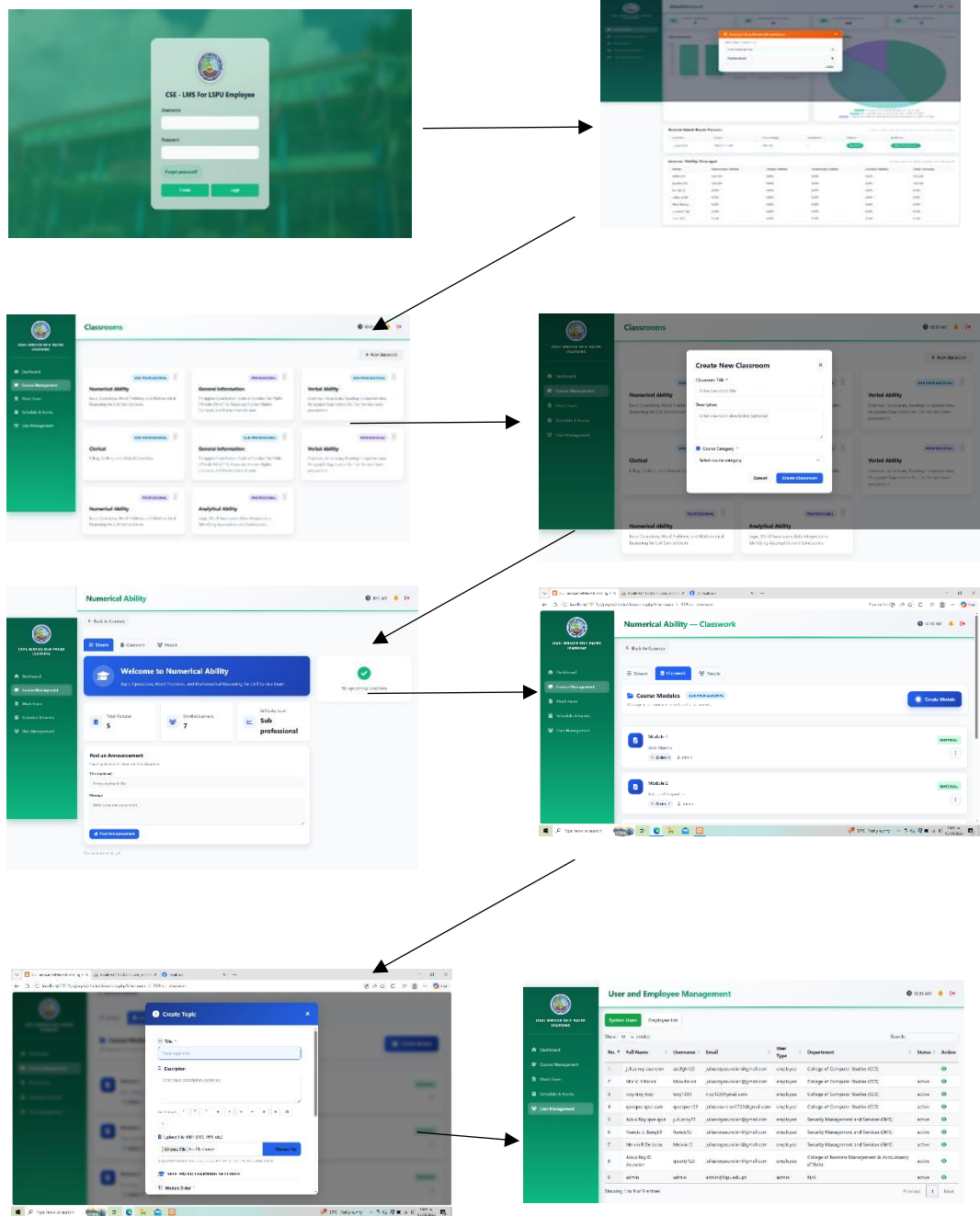
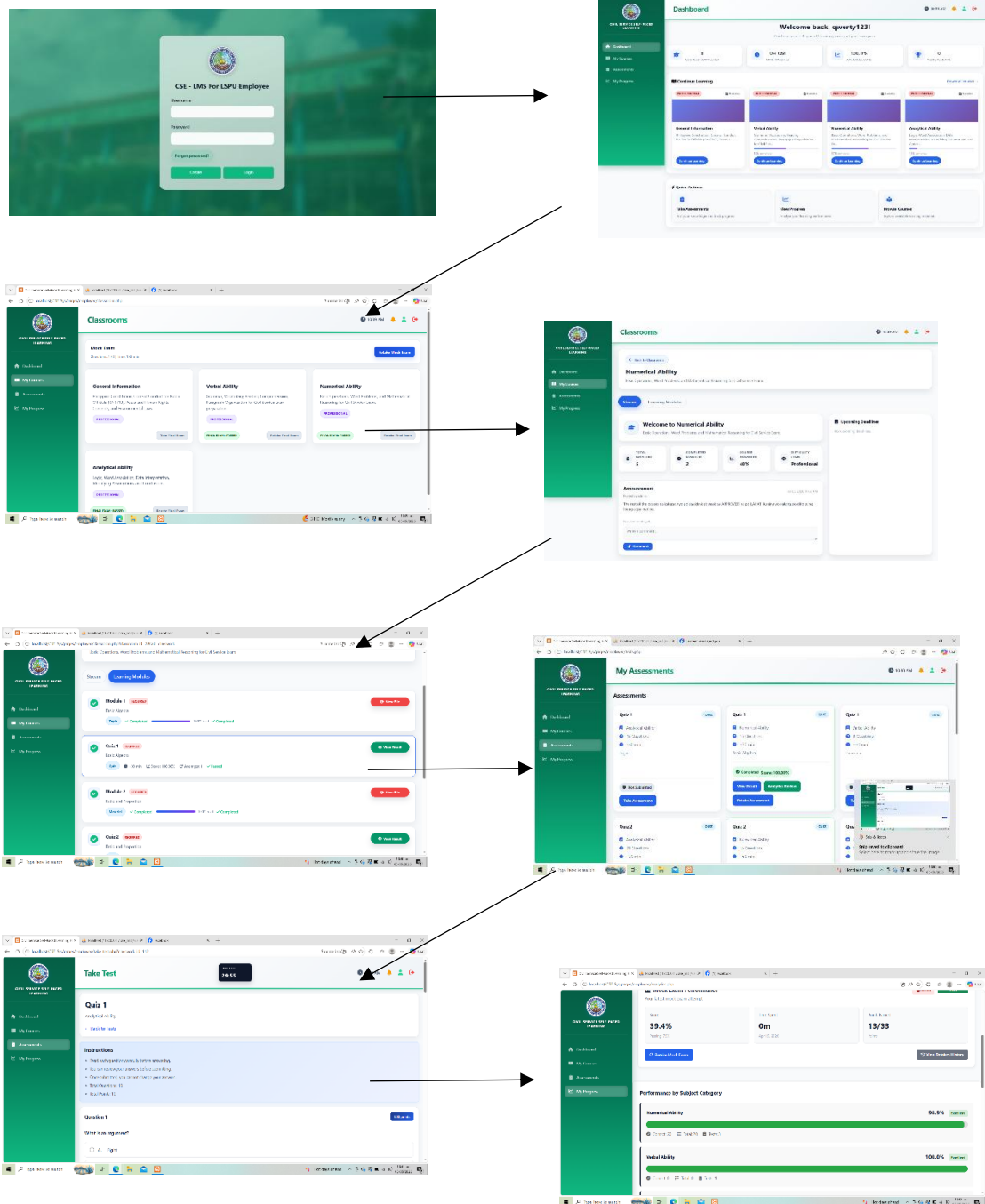


Figure User Interface



Current Process Flowchart

In this section, it shows the current process of Learning Management System for LSPU Employees Civil Service Examination Preparation.

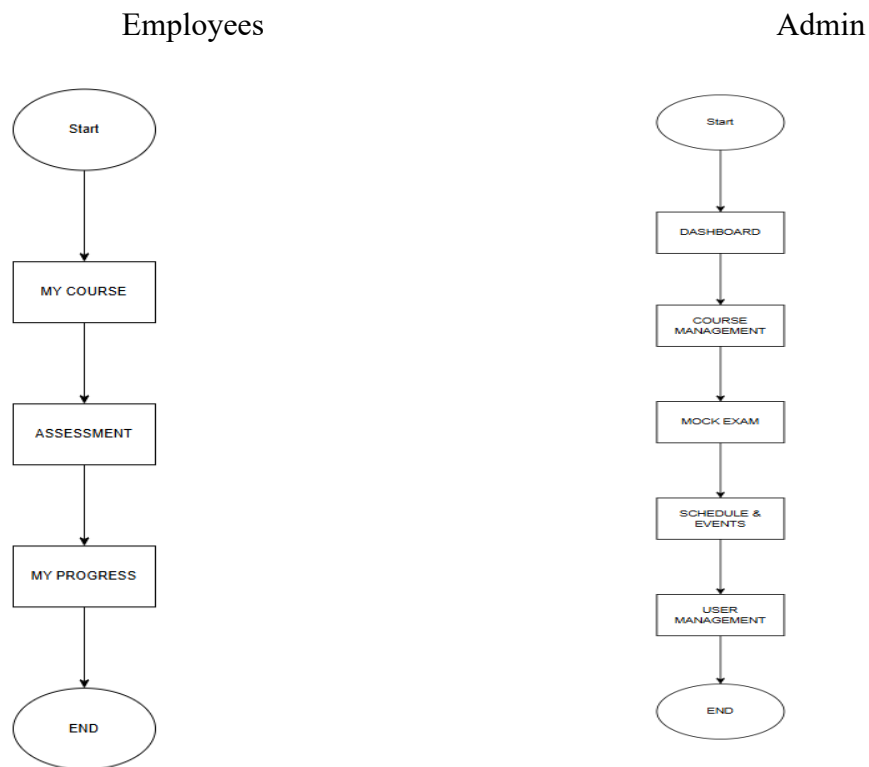


Figure 20. The image above shows the current process of Learning Management System for LSPU Employees Civil Service Examination Preparation.

Proposed Process Flowchart

This section shows the proposed process of the developed system

The Admin Homepage

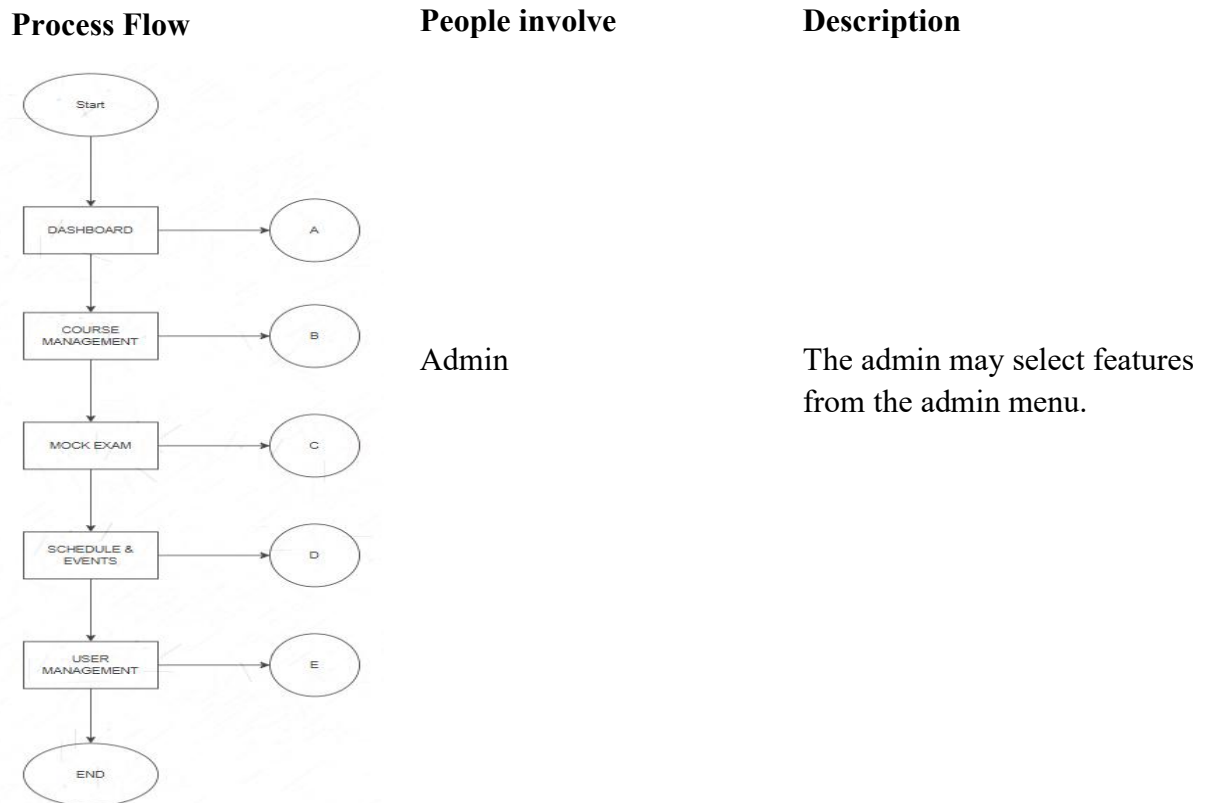


Figure 21. After logging in, the administrator can select from various processes. Each menu item has a corresponding option.

Proposed Process Flowchart

This section shows the proposed process of the developed system

The User Homepage

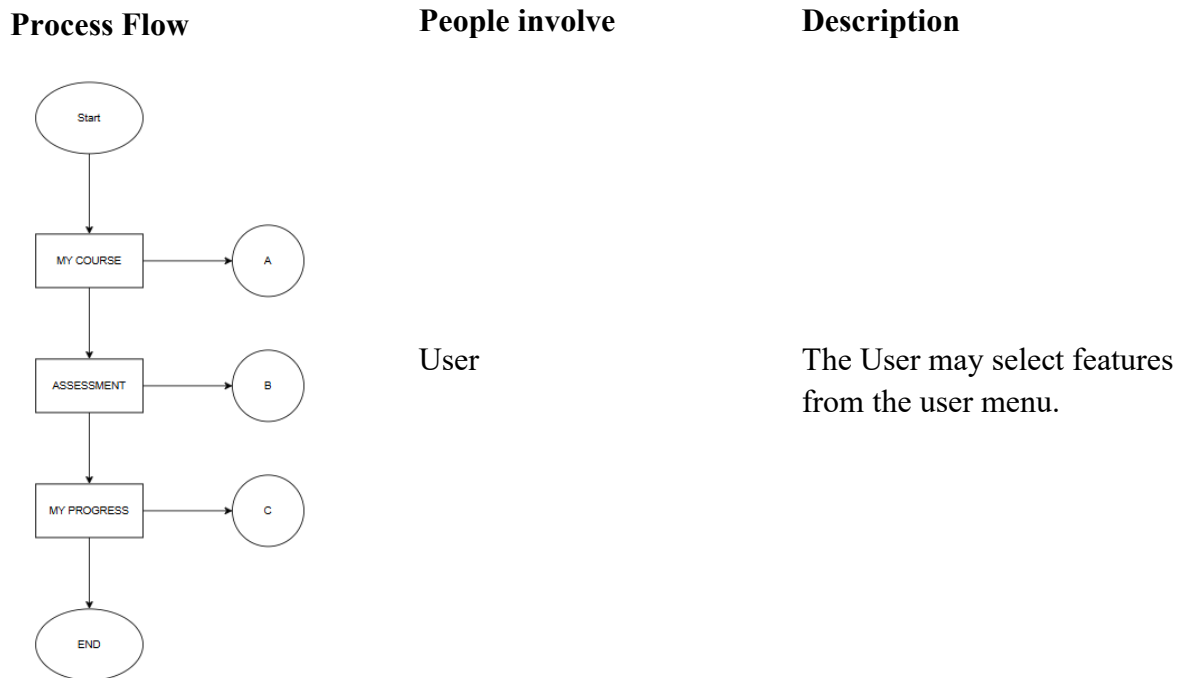


Figure 22. After logging in, the employee can select from various processes. Each menu item has a corresponding option.

The Admin Dashboard Process

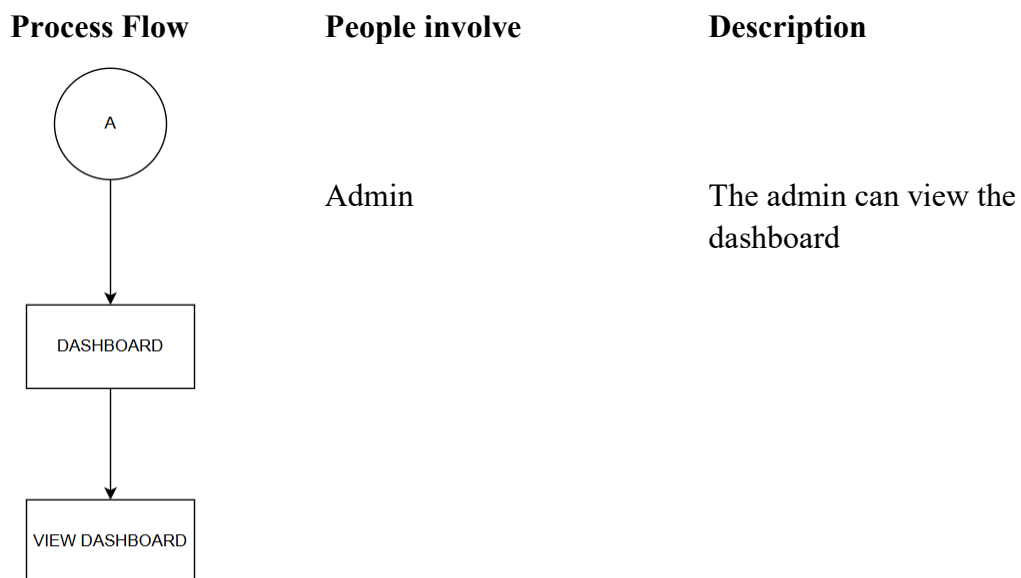


Figure 23. Shows how admin views dashboards.

The Admin Course Management Process

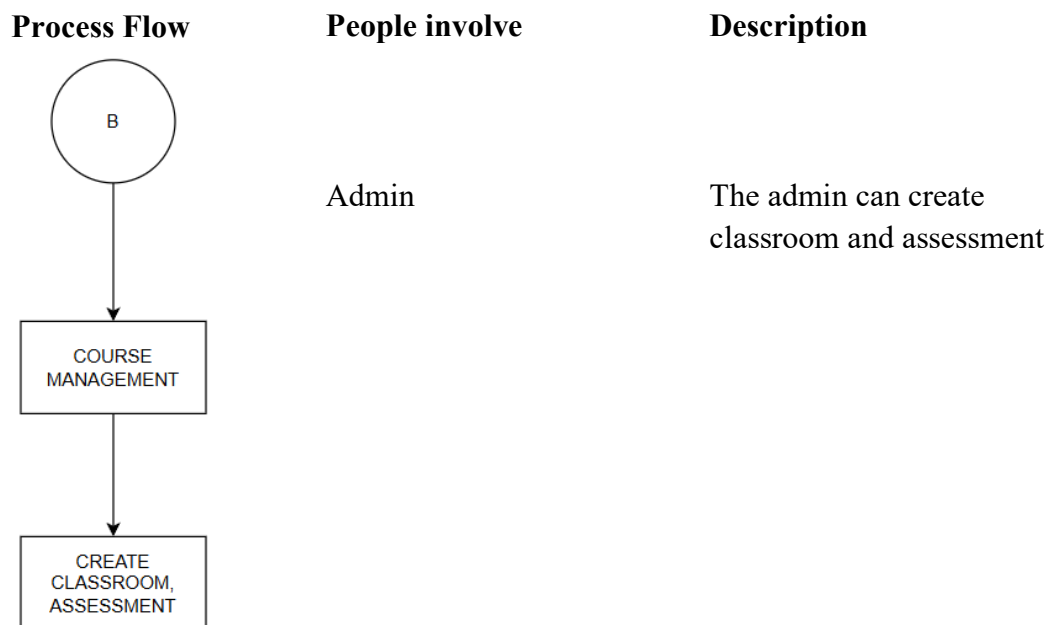


Figure 24. Shows how admin create classroom and assessment.

The Admin Mock Exam Process

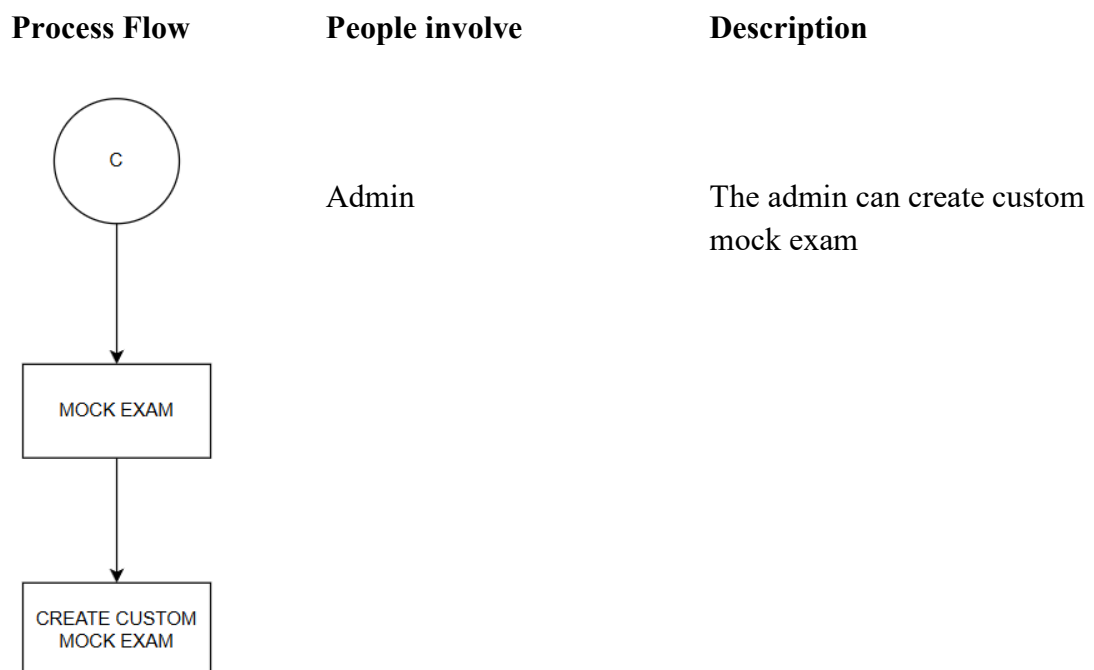


Figure 25. Shows how admin can create custom mock exam.

The Admin Schedule & Events Process

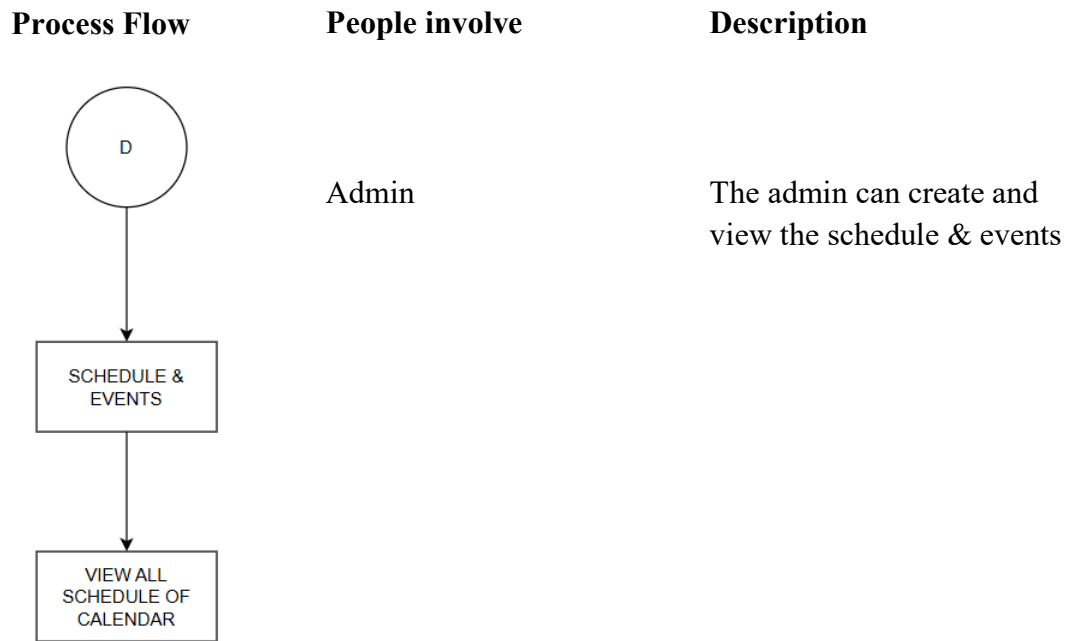


Figure 26. Shows how admin can create and view the schedule & evets.

The Admin User Management Process

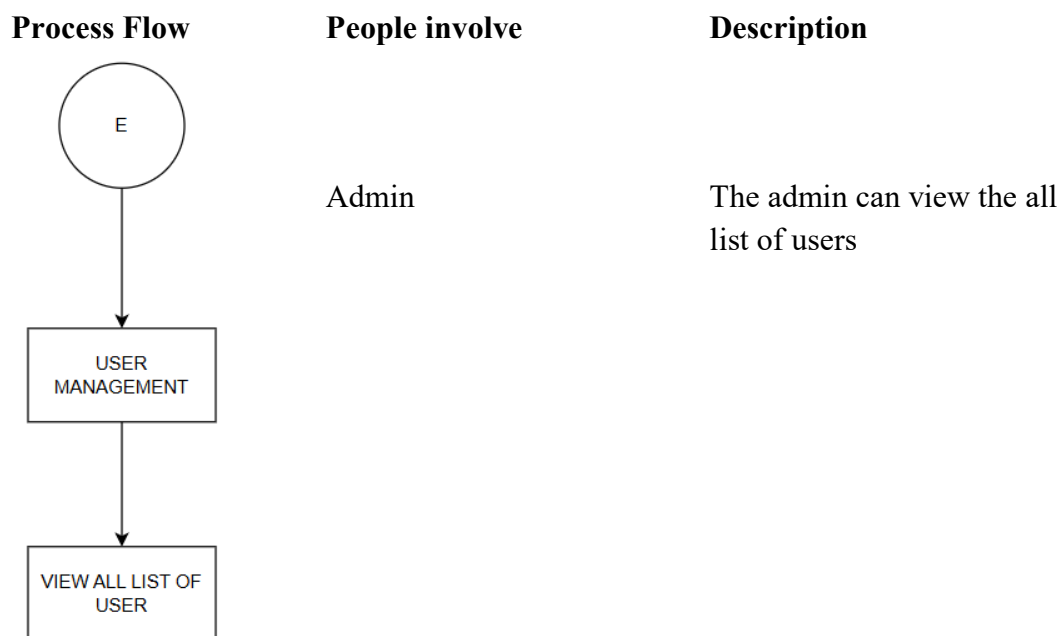


Figure 27. Shows how admin view the all list of users.

The Employee Dashboard Process

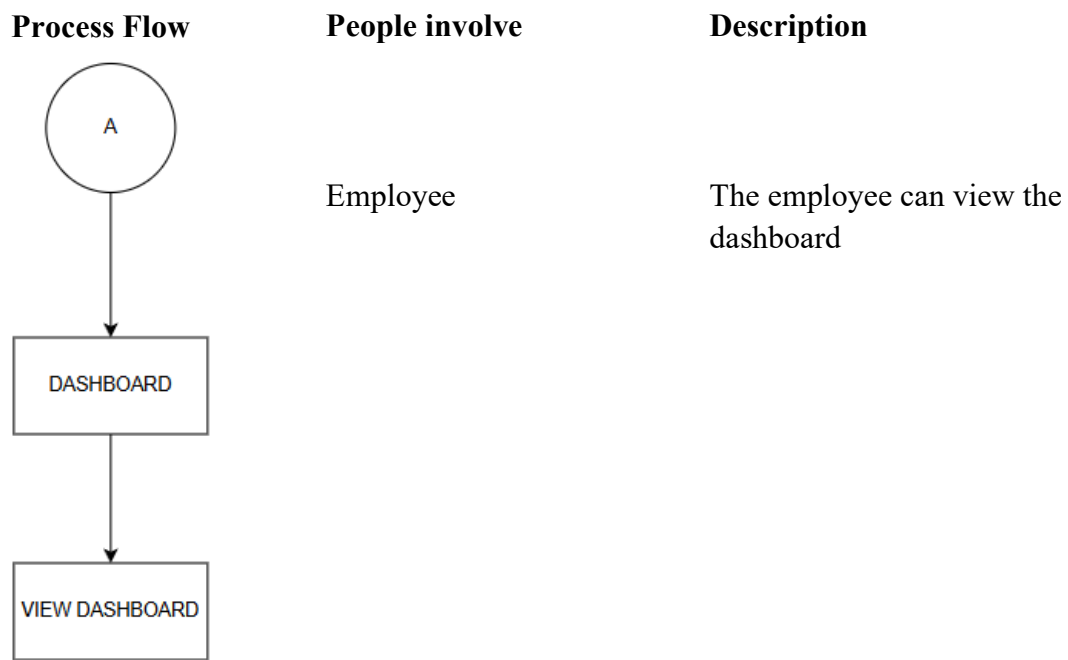


Figure 28. Shows how employee views dashboards.

The Employee My Courses Process

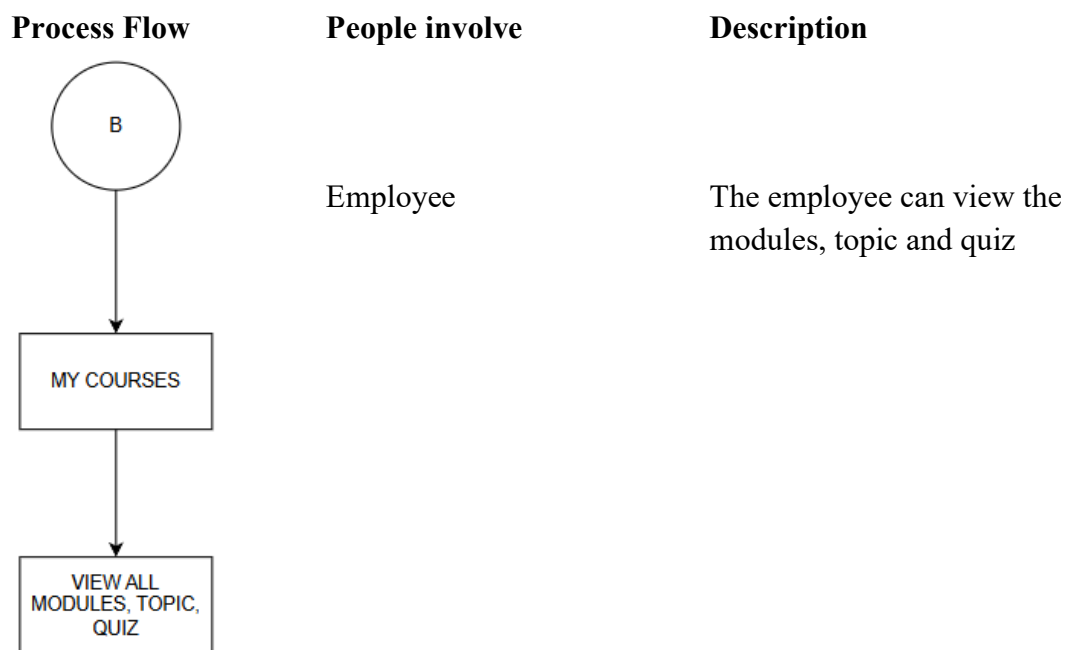


Figure 29. Shows how employee can view the modules, topic and quiz.

The Employee Assessment Process

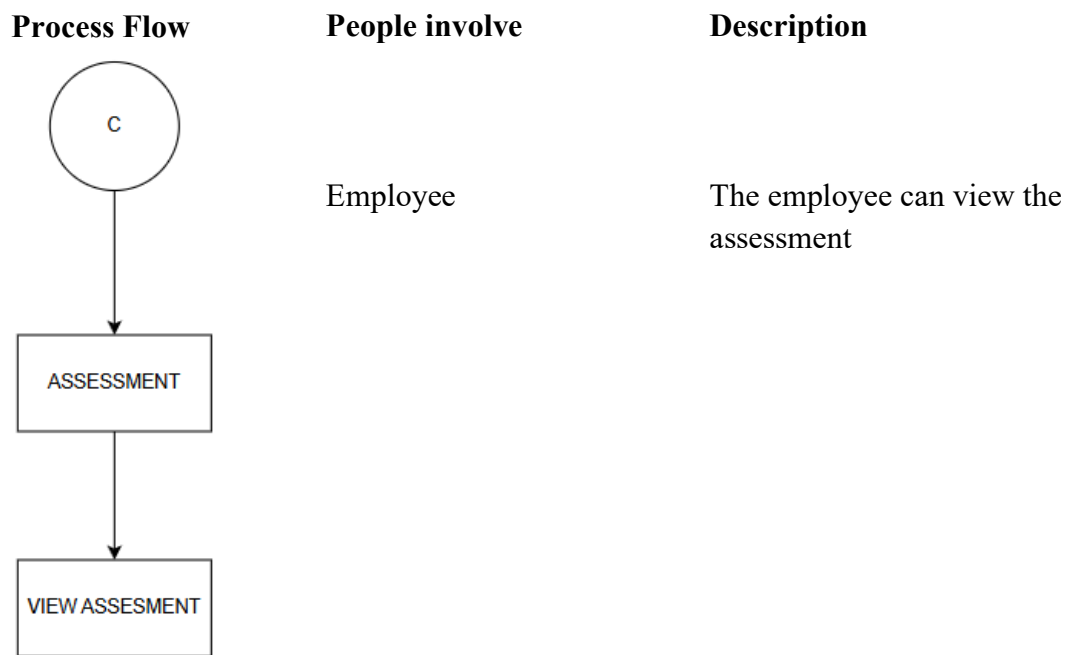


Figure 30. Shows how employee can view the assessment.

The Employee Schedule & Events Process

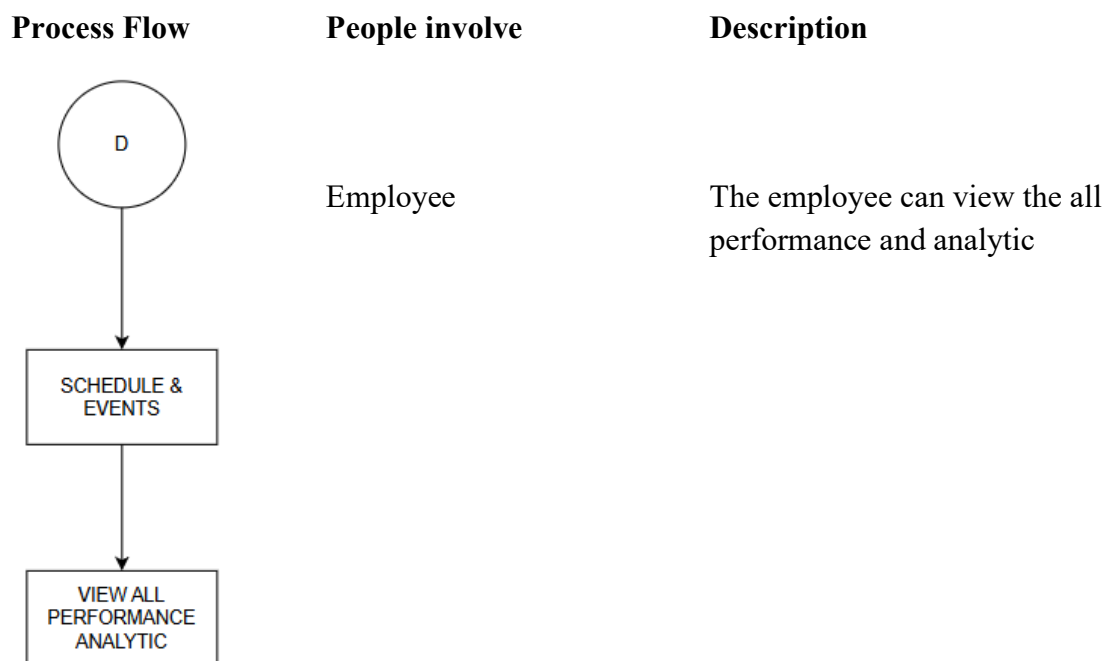


Figure 31. Shows how employee can view the all performance and analytic.

Use Case Diagram

The use case diagram that follows shows the roles and responsibilities of each user on the developed web-based system.

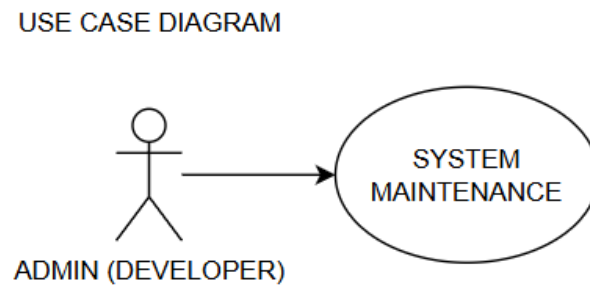


Figure 32. Use Case 1: The researcher's web-based system's first users are the administrator or system developers. It mainly concentrates on maintaining the system.

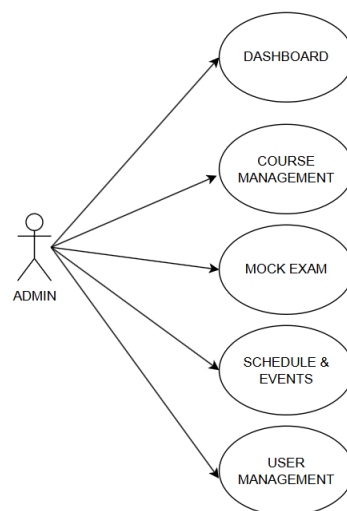


Figure 33. Use Case 2: The researcher's web-based research study's second user type is the system administration. Its responsibility is to view all the entered data.

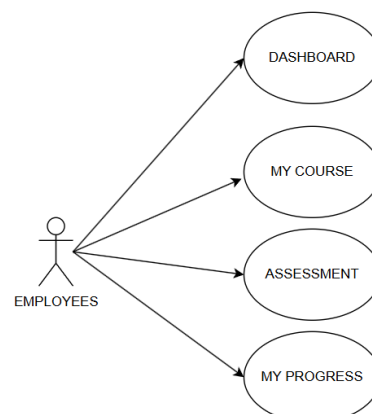


Figure 34. Use Case 3: The researcher's web-based research study's last user type is Employees. Its responsibility to learn lesson and take the assessment.

System Context Diagram

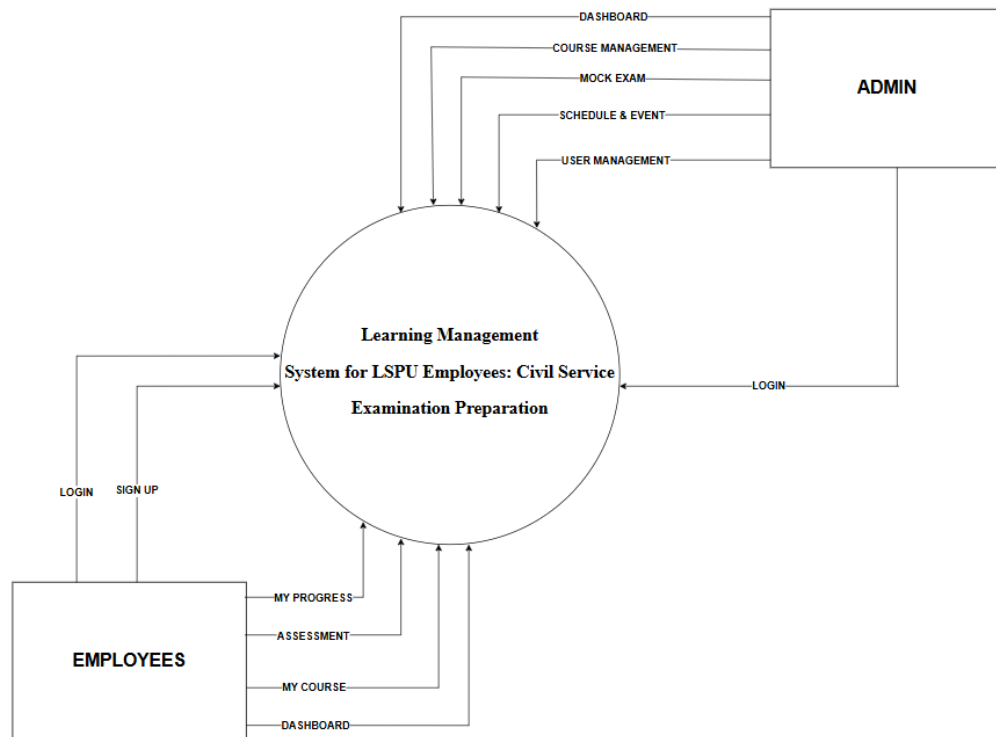


Figure 35. System Context Diagram. The figure above illustrates and demonstrates the process and function of each user who will have an account to the web-based system.

Data Flow Diagram

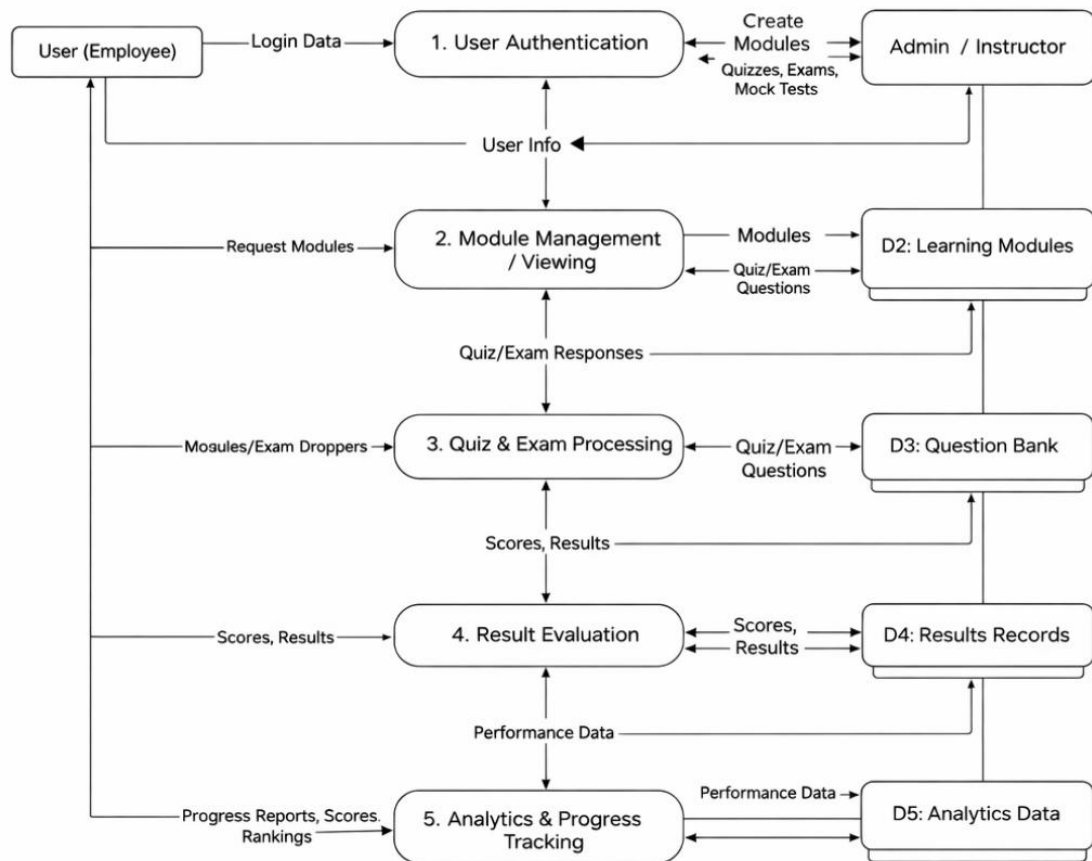


Figure 36. This figure presents the Data Flow Diagram of the Learning Management System for LSPU Employees: Civil Service Examination Preparation.

The Data Flow Diagram (DFD) illustrates how data moves within the system, showing the interaction between users, processes, and data storage components. It provides a clear representation of how LSPU employees access learning materials, take practice examinations, and track their progress through the system. The diagram also highlights how administrators manage content, user accounts, and assessment results. Through this structured flow of information, the system ensures efficient learning delivery, organized data management, and effective preparation for the Civil Service Examination.

Data Dictionary

This shows the table of database that is used in the system.

Table 21. Show the data dictionary of calendar_events.

Field	Data Type	Description
id	int(11)	id of the calendar_events
title	varchar(255)	title of the calendar_events
description	text	description of the calendar_events
date_start	date	date start of the calendar_events
date_end	date	date end of the calendar_events
created_by	int(11)	created by of the calendar_events
created_at	timestamp	created at of the calendar_events
updated_at	timestamp	updated at of the calendar_events

The table shows the data type and the calendar_events information.

Table 22. Show the data dictionary of classrooms.

Field	Data Type	Description
id	int(11)	id of the classrooms
title	varchar(255)	title of the classrooms
description	text	description of the classrooms
subject_category	varchar(100)	subject category of the classrooms
course_code	varchar(50)	course code of the classrooms
duration_weeks	int(11)	duration weeks of the classrooms
difficulty_level	enum('sub-professional'	difficulty level of the classrooms
is_published	tinyint(1)	is published of the classrooms
enrollment_type	enum('open'	enrollment type of the classrooms
created_by	int(11)	created by of the classrooms
created_at	timestamp	created at of the classrooms
updated_at	timestamp	updated at of the classrooms
mock_exam_mode	enum('quiz'	mock exam mode of the classrooms

The table shows the data type and the classrooms information.

Table 23. Show the data dictionary of classwork.

Field	Data Type	Description
id	int(11)	id of the classwork
classroom_id	int(11)	classroom id of the classwork
title	varchar(255)	title of the classwork
type	enum('pre-test'	type of the classwork
description	text	description of the classwork
content	text	content of the classwork
file_path	varchar(500)	file path of the classwork
created_by	int(11)	created by of the classwork
due_date	datetime	due date of the classwork
module_order	int(11)	module order of the classwork
estimated_duration_minutes	int(11)	estimated duration minutes of the classwork
is_mandatory	tinyint(1)	is mandatory of the classwork
unlock_after_classwork_id	int(11)	unlock after classwork id of the classwork
passing_score	decimal(5	passing score of the classwork
allow_retake	tinyint(1)	allow retake of the classwork
max_attempts	int(11)	max attempts of the classwork
created_at	timestamp	created at of the classwork
updated_at	timestamp	updated at of the classwork
question_display_count	int(11)	question display count of the classwork

The table shows the data type and the classwork information.

Table 24. Show the data dictionary of course_progress.

Field	Data Type	Description
id	int(11)	id of the course_progress
user_id	int(11)	user id of the course_progress
classroom_id	int(11)	classroom id of the course_progress
enrolled_at	timestamp	enrolled at of the course_progress
started_at	timestamp	started at of the course_progress
completed_at	timestamp	completed at of the course_progress
completion_percentage	decimal(5	completion percentage of the course_progress
total_time_spent_minutes	int(11)	total time spent minutes of the course_progress
current_module_id	int(11)	current module id of the course_progress
status	enum('enrolled'	status of the course_progress
last_accessed_at	timestamp	last accessed at of the course_progress

The table shows the data type and the course_progress information.

Table 25. Show the data dictionary of course_resources.

Field	Data Type	Description
id	int(11)	id of the course_resources
classwork_id	int(11)	classwork id of the course_resources
classroom_id	int(11)	classroom id of the course_resources
resource_type	enum('pdf'	resource type of the course_resources
title	varchar(255)	title of the course_resources
description	text	description of the course_resources
file_path	varchar(500)	file path of the course_resources
external_url	varchar(500)	external url of the course_resources
file_size_kb	int(11)	file size kb of the course_resources
duration_minutes	int(11)	duration minutes of the course_resources
uploaded_by	int(11)	uploaded by of the course_resources
created_at	timestamp	created at of the course_resources

The table shows the data type and the course_resources information.

Table 26. Show the data dictionary of department.

Field	Data Type	Description
id	int(11)	id of the department
department	text	department of the department
code	varchar(20)	code of the department
1	int(11)	1 of the department
2	int(11)	2 of the department
3	int(11)	3 of the department
4	int(11)	4 of the department

The table shows the data type and the department information.

Table 27. Show the data dictionary of discussion_replies.

Field	Data Type	Description
id	int(11)	id of the discussion_replies
thread_id	int(11)	thread id of the discussion_replies
user_id	int(11)	user id of the discussion_replies
content	text	content of the discussion_replies
is_instructor_reply	tinyint(1)	is instructor reply of the discussion_replies
created_at	timestamp	created at of the discussion_replies
updated_at	timestamp	updated at of the discussion_replies

The table shows the data type and the discussion_replies information.

Table 28. Show the data dictionary of discussion_threads.

Field	Data Type	Description
id	int(11)	id of the discussion_threads
classroom_id	int(11)	classroom id of the discussion_threads
classwork_id	int(11)	classwork id of the discussion_threads
user_id	int(11)	user id of the discussion_threads
title	varchar(255)	title of the discussion_threads
content	text	content of the discussion_threads
is_pinned	tinyint(1)	is pinned of the discussion_threads
created_at	timestamp	created at of the discussion_threads
updated_at	timestamp	updated at of the discussion_threads

The table shows the data type and the discussion_threads information.

Table 29. Show the data dictionary of employee.

Field	Data Type	Description
id	int(11)	id of the employee
user_id	int(11)	user id of the employee
employee_no	varchar(50)	employee no of the employee
first_name	varchar(100)	first name of the employee
middle_name	varchar(100)	middle name of the employee
last_name	varchar(100)	last name of the employee
name_extension	varchar(20)	name extension of the employee
email	varchar(150)	email of the employee
office_id	int(11)	office id of the employee
employment_type	enum('Personnel'	employment type of the employee
enrollment_category	enum('Sub	enrollment category of the employee
status	enum('active'	status of the employee
created_at	timestamp	created at of the employee
updated_at	timestamp	updated at of the employee

The table shows the data type and the employee information.

Table 30. Show the data dictionary of final_exam_submissions.

Field	Data Type	Description
id	int(11)	id of the final_exam_submissions
classroom_id	int(11)	classroom id of the final_exam_submissions
user_id	int(11)	user id of the final_exam_submissions
score	decimal(10	score of the final_exam_submissions
total_points	decimal(10	total points of the final_exam_submissions
percentage_score	decimal(10	percentage score of the final_exam_submissions
review_status	enum('auto_graded'	review status of the final_exam_submissions
question_count	int(11)	question count of the final_exam_submissions
time_started	timestamp	time started of the final_exam_submissions
time_completed	timestamp	time completed of the final_exam_submissions
created_at	timestamp	created at of the final_exam_submissions

The table shows the data type and the final_exam_submissions information.

Table 31. Show the data dictionary of final_exam_user_answers.

Field	Data Type	Description
id	int(11)	id of the final_exam_user_answers
submission_id	int(11)	submission id of the final_exam_user_answers
question_id	int(11)	question id of the final_exam_user_answers
answer_option_id	int(11)	answer option id of the final_exam_user_answers
answer_text	text	answer text of the final_exam_user_answers
is_correct	tinyint(1)	is correct of the final_exam_user_answers
points_earned	decimal(10	points earned of the final_exam_user_answers
display_sequence	int(11)	display sequence of the final_exam_user_answers
created_at	timestamp	created at of the final_exam_user_answers

The table shows the data type and the final_exam_user_answers information.

Table 32. Show the data dictionary of learner_achievements.

Field	Data Type	Description
id	int(11)	id of the learner_achievements
user_id	int(11)	user id of the learner_achievements
achievement_type	enum('course_completion')	achievement type of the learner_achievements
title	varchar(255)	title of the learner_achievements
description	text	description of the learner_achievements
classroom_id	int(11)	classroom id of the learner_achievements
earned_at	timestamp	earned at of the learner_achievements

The table shows the data type and the learner_achievements information.

Table 33. Show the data dictionary of learner_analytics.

Field	Data Type	Description
user_id	int(11)	user id of the learner_analytics
username	varchar(50)	username of the learner_analytics
office	varchar(255)	office of the learner_analytics
enrolled_courses	bigint(21)	enrolled courses of the learner_analytics
completed_courses	bigint(21)	completed courses of the learner_analytics
avg_test_score	decimal(9	avg test score of the learner_analytics
total_study_time_minutes	decimal(32	total study time minutes of the learner_analytics
total_tests_taken	bigint(21)	total tests taken of the learner_analytics
total_achievements	bigint(21)	total achievements of the learner_analytics
learner_bookmarks	--	learner bookmarks of the learner_analytics
learner_bookmarks	(	learner bookmarks of the learner_analytics
id	int(11)	id of the learner_analytics
user_id	int(11)	user id of the learner_analytics
classwork_id	int(11)	classwork id of the learner_analytics
created_at	timestamp	created at of the learner_analytics

The table shows the data type and the learner_analytics information.

Table 34. Show the data dictionary of learner_notes.

Field	Data Type	Description
id	int(11)	id of the learner_notes
user_id	int(11)	user id of the learner_notes
classwork_id	int(11)	classwork id of the learner_notes
classroom_id	int(11)	classroom id of the learner_notes
note_content	text	note content of the learner_notes
created_at	timestamp	created at of the learner_notes
updated_at	timestamp	updated at of the learner_notes

The table shows the data type and the learner_notes information.

Table 35. Show the data dictionary of learner_progress.

Field	Data Type	Description
id	int(11)	id of the learner_progress
user_id	int(11)	user id of the learner_progress
classwork_id	int(11)	classwork id of the learner_progress
status	enum('not_started')	status of the learner_progress
completion_percentage	decimal(5)	completion percentage of the learner_progress
time_spent_minutes	int(11)	time spent minutes of the learner_progress
started_at	timestamp	started at of the learner_progress
completed_at	timestamp	completed at of the learner_progress
last_accessed_at	timestamp	last accessed at of the learner_progress

The table shows the data type and the learner_progress information.

Table 36. Show the data dictionary of learning_paths.

Field	Data Type	Description
id	int(11)	id of the learning_paths
title	varchar(255)	title of the learning_paths
description	text	description of the learning_paths
category	enum('professional'	category of the learning_paths
created_by	int(11)	created by of the learning_paths
created_at	timestamp	created at of the learning_paths
updated_at	timestamp	updated at of the learning_paths

The table shows the data type and the learning_paths information.

Table 37. Show the data dictionary of mastery_checks.

Field	Data Type	Description
id	int(11)	id of the mastery_checks
classwork_id	int(11)	classwork id of the mastery_checks
title	varchar(255)	title of the mastery_checks
description	text	description of the mastery_checks
passing_criteria	text	passing criteria of the mastery_checks
created_at	timestamp	created at of the mastery_checks

The table shows the data type and the mastery_checks information.

Table 38. Show the data dictionary of mock_exam_settings.

Field	Data Type	Description
id	int(11)	id of the mock_exam_settings
setting_name	varchar(100)	setting name of the mock_exam_settings
setting_value	varchar(255)	setting value of the mock_exam_settings
updated_at	timestamp	updated at of the mock_exam_settings

The table shows the data type and the mock_exam_settings information.

Table 39. Show the data dictionary of mock_exam_submissions.

Field	Data Type	Description
id	int(11)	id of the mock_exam_submissions
user_id	int(11)	user id of the mock_exam_submissions
score	decimal(10	score of the mock_exam_submissions
total_points	decimal(10	total points of the mock_exam_submissions
percentage_score	decimal(10	percentage score of the mock_exam_submissions
review_status	enum('auto_graded'	review status of the mock_exam_submissions
question_count	int(11)	question count of the mock_exam_submissions
total_classrooms	int(11)	total classrooms of the mock_exam_submissions
time_started	timestamp	time started of the mock_exam_submissions
time_completed	timestamp	time completed of the mock_exam_submissions
created_at	timestamp	created at of the mock_exam_submissions

The table shows the data type and the mock_exam_submissions information.

Table 40. Show the data dictionary of mock_exam_user_answers.

Field	Data Type	Description
id	int(11)	id of the mock_exam_user_answers
submission_id	int(11)	submission id of the mock_exam_user_answers
question_id	int(11)	question id of the mock_exam_user_answers
answer_option_id	int(11)	answer option id of the mock_exam_user_answers
answer_text	text	answer text of the mock_exam_user_answers
is_correct	tinyint(1)	is correct of the mock_exam_user_answers
points_earned	decimal(10	points earned of the mock_exam_user_answers
display_sequence	int(11)	display sequence of the mock_exam_user_answers
created_at	timestamp	created at of the mock_exam_user_answers

The table shows the data type and the mock_exam_user_answers information.

Table 41. Show the data dictionary of notifications.

Field	Data Type	Description
id	int(11)	id of the notifications
user_id	int(11)	user id of the notifications
title	varchar(255)	title of the notifications
message	text	message of the notifications
type	enum('course')	type of the notifications
related_id	int(11)	related id of the notifications
is_read	tinyint(1)	is read of the notifications
created_at	timestamp	created at of the notifications

The table shows the data type and the notifications information.

Table 42. Show the data dictionary of password_reset_tokens.

Field	Data Type	Description
id	int(11)	id of the password_reset_tokens
user_id	int(11)	user id of the password_reset_tokens
email	varchar(255)	email of the password_reset_tokens
token	varchar(10)	token of the password_reset_tokens
expires_at	datetime	expires at of the password_reset_tokens
created_at	timestamp	created at of the password_reset_tokens

The table shows the data type and the password_reset_tokens information.

Table 43. Show the data dictionary of path_courses.

Field	Data Type	Description
id	int(11)	id of the path_courses
path_id	int(11)	path id of the path_courses
classroom_id	int(11)	classroom id of the path_courses
sequence_order	int(11)	sequence order of the path_courses
is_required	tinyint(1)	is required of the path_courses

The table shows the data type and the path_courses information.

Table 44. Show the data dictionary of test_answer_options.

Field	Data Type	Description
id	int(11)	id of the test_answer_options
question_id	int(11)	question id of the test_answer_options
option_text	text	option text of the test_answer_options
option_letter	varchar(2)	option letter of the test_answer_options
is_correct	tinyint(1)	is correct of the test_answer_options
option_order	int(11)	option order of the test_answer_options

The table shows the data type and the test_answer_options information.

Table 45. Show the data dictionary of test_questions.

Field	Data Type	Description
id	int(11)	id of the test_questions
classwork_id	int(11)	classwork id of the test_questions
question	text	question of the test_questions
question_type	enum('multiple_choice')	question type of the test_questions
points	decimal(5	points of the test_questions
question_order	int(11)	question order of the test_questions
solution_text	text	solution text of the test_questions
created_at	timestamp	created at of the test_questions

The table shows the data type and the test_questions information.

Table 46. Show the data dictionary of test_submissions.

Field	Data Type	Description
id	int(11)	id of the test_submissions
classwork_id	int(11)	classwork id of the test_submissions
user_id	int(11)	user id of the test_submissions
score	decimal(5	score of the test_submissions
total_points	decimal(5	total points of the test_submissions
percentage_score	decimal(5	percentage score of the test_submissions
attempt_number	int(11)	attempt number of the test_submissions
review_status	enum('pending'	review status of the test_submissions
time_spent_minutes	int(11)	time spent minutes of the test_submissions
submitted_at	timestamp	submitted at of the test_submissions
admin_reviewed_by	int(11)	admin reviewed by of the test_submissions
admin_reviewed_at	timestamp	admin reviewed at of the test_submissions

The table shows the data type and the test_submissions information.

Table 47. Show the data dictionary of test_user_answers.

Field	Data Type	Description
id	int(11)	id of the test_user_answers
submission_id	int(11)	submission id of the test_user_answers
question_id	int(11)	question id of the test_user_answers
answer_option_id	int(11)	answer option id of the test_user_answers
answer_text	text	answer text of the test_user_answers
is_correct	tinyint(1)	is correct of the test_user_answers
points_earned	decimal(5	points earned of the test_user_answers
admin_adjusted	tinyint(1)	admin adjusted of the test_user_answers
admin_notes	text	admin notes of the test_user_answers

The table shows the data type and the test_user_answers information.

Table 48. Show the data dictionary of users.

Field	Data Type	Description
id	int(11)	id of the users
prefix	varchar(10)	prefix of the users
username	varchar(50)	username of the users
password	varchar(255)	password of the users
email	varchar(100)	email of the users
office	varchar(255)	office of the users
first_name	varchar(50)	first name of the users
middle_initial	varchar(5)	middle initial of the users
last_name	varchar(50)	last name of the users
name_extension	varchar(10)	name extension of the users
employment_type	enum('Personnel')	employment type of the users
office_id	int(11)	office id of the users
enrollment_category	enum('Sub	enrollment category of the users
user_type	enum('admin'	user type of the users
created_at	timestamp	created at of the users
updated_at	timestamp	updated at of the users
status	enum('pending'	status of the users

The table shows the data type and the users information.

Table 49. Show the data dictionary of user_preferences.

Field	Data Type	Description
id	int(11)	id of the user_preferences
user_id	int(11)	user id of the user_preferences
daily_study_goal_minutes	int(11)	daily study goal minutes of the user_preferences
preferred_study_time	enum('morning')	preferred study time of the user_preferences
email_notifications	tinyint(1)	email notifications of the user_preferences
reminder_notifications	tinyint(1)	reminder notifications of the user_preferences
created_at	timestamp	created at of the user_preferences
updated_at	timestamp	updated at of the user_preferences

The table shows the data type and the user_preferences information.

Hierarchical Input Process Output Model

This diagram shows the hierarchical input process output of the Learning Management System for LSPU Employees: Civil Service Examination Preparation

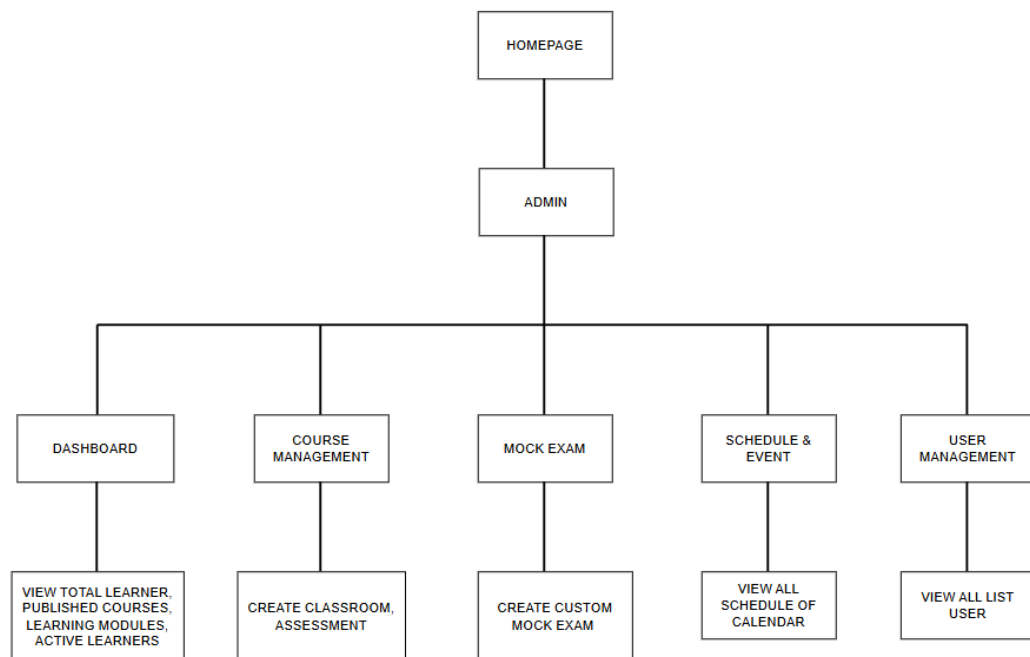


Figure 37. Admin (System Admin) HIPO. It illustrates the module of the system hierarchy to show the system design in Admin (System Admin) account panel

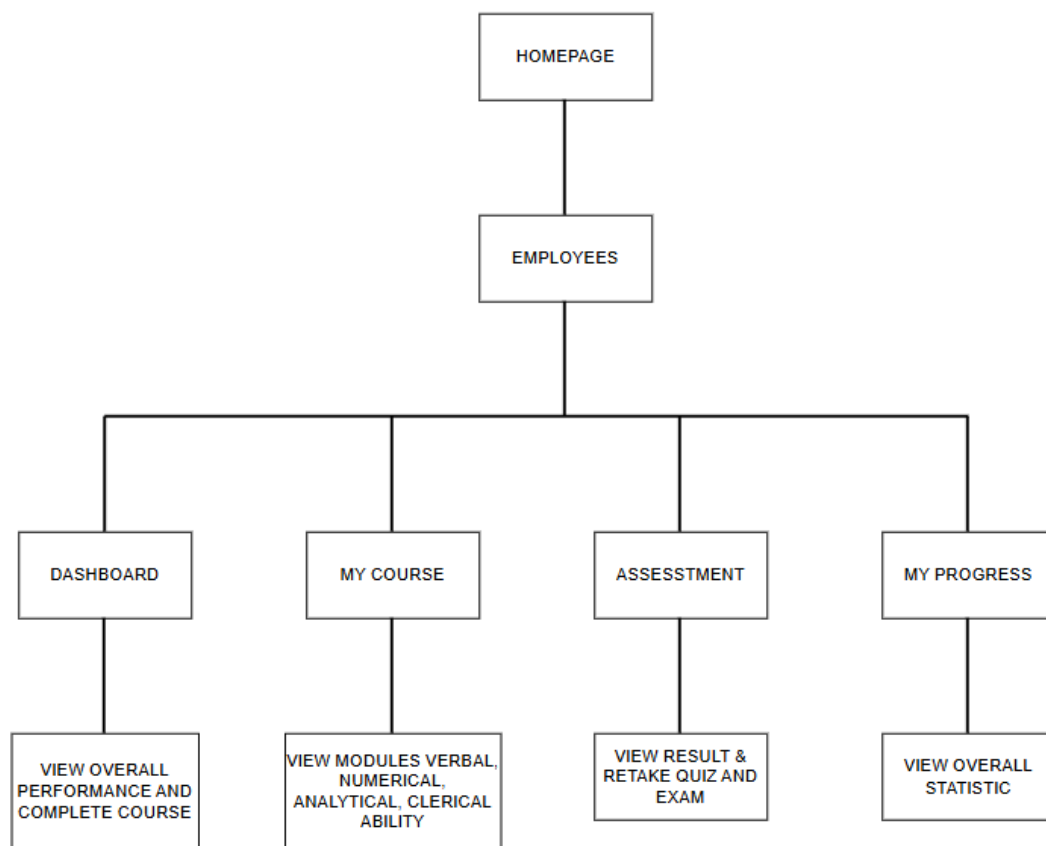


Figure 38. Employees HIPO. It illustrates the module of the system hierarchy to show the system design in Employees account panel.

Hardware and Software Requirements

This is section where the software and hardware resources that is needed to accomplish the study.

Figure 39. Hardware Requirements

Hardware Requirements	Existing
Laptop or Computer Unit	Yes
Processor Core i3 or Higher	Yes
RAM 4 Gigabyte or Higher	Yes

The table above shows the hardware requirements of the study. “Yes” if the requirements are existing, and “No” if the requirement doesn’t exist. Based on the client, the hardware is present in the NIA of Pila, Laguna.

Figure 40. Software Requirements

Hardware Requirements	Existing
Operating System Windows 10 32/64-bit	Yes
Wi-Fi	Yes
Any type of browser	Yes

The table above shows the software requirements of the study. “Yes” if the requirements are existing, and “No” if the requirement doesn’t exist. The individuals involved in the study do not have some of the resources needed, so the researchers lent their device to them for the system testing and evaluation phase.

Testing and Evaluation Instruments

This section presents the testing and evaluation tools that the researchers used in conducting the user evaluation and testing phase of the study.

TECHNOLOGY ACCEPTANCE QUESTIONNAIRE

Dear Respondents,

Greetings in the name of the Lord!

We are students from the College of Computer Studies of the Laguna State Polytechnic Technology (BSIT) program and we are in the process of completing a research project with the working title — **“Learning Management System for LSPU Employees: Civil Service Examination Preparation”** as a partial requirement on our degree.

In this regard, we kindly request that you take the time to respond to all of the questionnaire's items, which are vital to the study's completion. The first section consists of respondent 's data, while the second section evaluates user acceptability and behavioral intention to engage with the platform.

As a respondent, we seek your aid and support with this study. Your information and comments will be safeguarded and treated with the utmost discretion. Also, your positive response to this request will help the study's success.

Thank you so much for your cooperation and God bless!

Respectfully yours,

Aloba, Chrischan A.

Asuncion, Julius Roy D.

Bangkil, Francis Anselme V.

RESPONDENT'S DATA

Direction: Please fill the required information in the space provided and mark (✓) the appropriate checkboxes according to your level of computer literacy.

Name (optional): _____ **Date:** _____

Respondents: ☐ Job-Order ☐ Faculty **Gender** ☐ Male
☐ Casual Others: _____ ☐ Female
☐ On-Call Others: _____

Level of Proficiency

- ☐ **Basic:** – Fundamental digital competency and basic language proficiency.
☐ **Proficient:** – Comfortable with digital tools and proficient in language skills.
☐ **Advanced:** – Highly skilled in digital technology and advanced language proficiency.

Direction: Please indicate your level of acceptance with each question by placing a checkmark (✓) corresponding to the appropriate number. Kindly use the following scale to indicate your acceptability with the following statement.

**5 – Acceptable (A), 4 – Slightly Acceptable (SA), 3 – Undecided (UD),
2 – Slightly Unacceptable (SU), and 1 - Unacceptable (UA)**

You may also provide your feedback and suggestions underneath the questionnaire.

QUALITY FACTORS	5	4	3	2	1
3. Appropriateness. The develop system contains all of the functions and features that the user need.					
4. Confidentiality. The develop system has secured and protected the user's privacy.					
4. User Interface. I believe that interface of the developed system is user-friendly.					
PERCEIVED EASE OF USE	5	4	3	2	1
A. Suitability. LSPU employees can easily navigate the Learning Management System (LMS) to access Civil Service Examination (CSE) review materials, track their learning progress, and manage their study schedule effectively.					
2. Efficiency. The Learning Management System (LMS) enables employees to quickly access learning modules, complete practice tests, making the review process faster and more organized compared to traditional methods.					
3. Usability. The Learning Management System (LMS) is user-friendly and intuitive, allowing employees to use the system with minimal effort and without requiring advanced technical knowledge.					
PERCEIVED USEFULNESS	5	4	3	2	1
4. Functionality. The Learning Management System (LMS) includes organized review modules, practice exams, learning materials, and user management particularly created for Civil Service Examination (CSE) preparation of LSPU personnel.					
5. Effectiveness. The Learning Management System (LMS) promotes employees' preparedness for the Civil Service Examination by giving organized material, self-paced learning, and continuous evaluation, decreasing the need for traditional review techniques.					
6. Performance Efficiency. Through the Learning Management System (LMS), employees can simply access review materials, take mock examinations, evaluate their progress, and enhance their performance effectively.					

EXPERIENCE	5	4	3	2	1
4. The develop system was more effective and easier to use than the present method.					
5. I never encountered any errors or problems while using the develop system.					
6. The functions of the develop system work properly.					
ATTITUDE TOWARDS USING	5	4	3	2	1
4. The LMS provides Civil Service Examination (CSE) review materials and assessments in a simple and effective manner					
5. The built Learning Management System (LMS), helps LSPU staff save time in reviewing by offering accessible and structured learning resources.					
6. The Learning Management System (LMS) helps employees in tracking their review progress, completing practice examinations, and enhancing their preparation for the Civil Service Examination.					
BEHAVIORAL INTENTION TO USE	5	4	3	2	1
4. I am willing to use and accept the system in order to complete my work tasks.					
5. I am interested to learn how the development system works.					
6. I am satisfied in terms of the overall flow of the develop system.					

Black Box Testing Questionnaire

TEST CASE ID: Test Case Description: Test the functionalities of the
User Login Interface
Tester's Name: Date Tested: Test Case (Pass/Fail/Not Executed)

PRECONDITIONS:

- The tester should know how to handle and use basic functionalities of a computer
- Computer/Laptop/PC must have access to internet

TEST SCENARIO: System User Login

STEP NO.	STEP DETAILS	EXPECTED RESULTS	ACTUAL RESULTS	STATUS	
				PASS	FAIL
1.	Open website	The website will open and show the login page.			
2.	Enter email and password	Display the inputted information correctly.			
3.	Click Show Password	The password will be visible.			
4	Click Forgot Password	This allows users to reset their forgotten passwords.			
5	Click "Login"	• The system admin interface will show. • After logging in, the system administrator will see the Dashboard			

Black Box Testing Questionnaire

TEST CASE ID: _____ Test Case Description: Test the functionalities of the User Login Interface

Tester's Name: _____ Date Tested: _____ Test Case (Pass/Fail/Not Executed)

PRECONDITIONS:

- The tester should know how to handle and use basic functionalities of a computer
- Computer/Laptop/PC must have access to internet

TEST SCENARIO: System User Signup

STEP NO.	STEP DETAILS	EXPECTED RESULTS	ACTUAL RESULTS	STATUS	
				PASS	FAIL
1.	Click Register	The page goes to the registration form.			
2.	Click Register Button	After filling out and submitting the registration form, the user receives a notification (Email) that the account is pending and awaiting admin approval.			
3.	Click Back to Log in	After admin approval, the user receives an email notification confirming that the account has been approved.			
4	Log In	The user fills out the login form.			
5	Click Log in	The user is successfully logged in and redirected to the dashboard.			

Black Box Testing Questionnaire

TEST CASE ID: _____ Test Case Description: Test the functionalities of the User Login Interface

Tester's Name: _____ Date Tested: _____ Test Case (Pass/Fail/Not Executed)

PRECONDITIONS:

- The tester should know how to handle and use basic functionalities of a computer
- Computer/Laptop/PC must have access to internet

TEST SCENARIO: System Admin Course Management

STEP NO.	STEP DETAILS	EXPECTED RESULTS	ACTUAL RESULTS	STATUS	
				PASS	FAIL
1.	Click Course Management	Course Card will Show			
2.	Click New Classroom	It will show how to create course/classroom			
3.	Click create classroom	Classroom will create			
4	Click classroom	Stream and classwork will shown			
5	Click Stream	Can post announcement			
6	Click Classwork	Classwork will show			
6	Click New Create Module/Quiz/Exam	Admin can choose what it want to create			
7	Click Create Module	Module/Quiz/ Exam already create			

Black Box Testing Questionnaire

TEST CASE ID: _____ Test Case Description: Test the functionalities of the User Login Interface

Tester's Name: _____ Date Tested: _____ Test Case (Pass/Fail/Not Executed)

PRECONDITIONS:

- The tester should know how to handle and use basic functionalities of a computer
- Computer/Laptop/PC must have access to internet

Test Case for System Admin Schedule and Event

STEP NO.	STEP DETAILS	EXPECTED RESULTS	ACTUAL RESULTS	STATUS	
				PASS	FAIL
1.	Click Schedule and Event	Show Calendar			
2.	Click add Event/Schedule	Admin Can fill out the schedule or event			
3.	Click Add	The Event/Schedule will show in the calendar			

Black Box Testing Questionnaire

TEST CASE ID: Test Case Description: Test the functionalities of the
User Login Interface

Tester's Name: Date Tested: Test Case (Pass/Fail/Not Executed)

PRECONDITIONS:

- The tester should know how to handle and use basic functionalities of a computer
- Computer/Laptop/PC must have access to internet

TEST SCENARIO: System Admin User Management

STEP NO.	STEP DETAILS	EXPECTED RESULTS	ACTUAL RESULTS	STATUS	
				PASS	FAIL
1.	Click User Management	The all accounts of incoming registration, and User			
2	Click Action	• Shows user information. • Displays new user details • Provides options to approve or decline the user.			
3	Click Approve	The user receives an email notification confirming that the account has been approved.			
4	Click Decline	The user account is declined and removed from the pending list (or marked as declined).			

Black Box Testing Questionnaire

TEST CASE ID: _____ Test Case Description: Test the functionalities of the User Login Interface

Tester's Name: _____ Date Tested: _____ Test Case (Pass/Fail/Not Executed)

PRECONDITIONS:

- The tester should know how to handle and use basic functionalities of a computer
- Computer/Laptop/PC must have access to internet

TEST SCENARIO: User Dashboard

STEP NO.	STEP DETAILS	EXPECTED RESULTS	ACTUAL RESULTS	STATUS	
				PASS	FAIL
1.	Log In	Show dashboard			
2.	Dashboard	Show quick action			
3.	Click Quick Action	• Go to Classroom • Go to Assessment • Go to My Progress			

Black Box Testing Questionnaire

TEST CASE ID: _____ Test Case Description: Test the functionalities of the User Login Interface

Tester's Name: _____ Date Tested: _____ Test Case (Pass/Fail/Not Executed)

PRECONDITIONS:

- The tester should know how to handle and use basic functionalities of a computer
- Computer/Laptop/PC must have access to internet

TEST SCENARIO: User Classroom

STEP NO.	STEP DETAILS	EXPECTED RESULTS	ACTUAL RESULTS	STATUS	
				PASS	FAIL
1.	Click Classroom	Show Course Card			
2.	Click Course Card	Show Stream and classwork			
3.	Stream	User see the announcement			
4	Click Classwork	User will see the Modules, quiz or exam			
5	Click View Module	The User will see module			
6	Click quiz	User will take exam			
5	Click submit	User will pass the Exam and can see if it pass or fail			

Black Box Testing Questionnaire

TEST CASE ID: _____ Test Case Description: Test the functionalities of the User Login Interface

Tester's Name: _____ Date Tested: _____ Test Case (Pass/Fail/Not Executed)

PRECONDITIONS:

- The tester should know how to handle and use basic functionalities of a computer
- Computer/Laptop/PC must have access to internet

TEST SCENARIO: User Assessment

STEP NO.	STEP DETAILS	EXPECTED RESULTS	ACTUAL RESULTS	STATUS	
				PASS	FAIL
1.	Click "Assessment"	Displays available quizzes or exams.			
2.	Click "Take Assessment"	Exam page opens			
3.	Start exam	• Questions are displayed			
	Submit test (complete)	Results are shown			
	Submit test (incomplete)	System blocks submission			
	Timer ends test	Exam auto-submits			
	View results	Score is displayed			

Black Box Testing Questionnaire

TEST CASE ID: _____ Test Case Description: Test the functionalities of the User Login Interface

Tester's Name: _____ Date Tested: _____ Test Case (Pass/Fail/Not Executed)

PRECONDITIONS:

- The tester should know how to handle and use basic functionalities of a computer
- Computer/Laptop/PC must have access to internet

TEST SCENARIO: User my Progress

STEP NO.	STEP DETAILS	EXPECTED RESULTS	ACTUAL RESULTS	STATUS	
				PASS	FAIL
1.	Click My Progress	• Display User Progress • Show User Analytics • Displays recent exams/quizzes taken by the user.			
2.	Click View Result	The user can view the results of completed exams/quizzes.			
3.	Click Retake Exam	The user can view failed exams and click the "Retake Exam" button to retake them.			
4	Click "Mock Exam – View Result"	Displays the percentage score of the user's completed mock exams by course.			
5	Click Retake Mock Exam	Redirects the user to the mock exam page to retake the exam.			

APPENDIX B

Communication Letters and Forms

Communication Letter



Republic of the Philippines
Laguna State Polytechnic University
 Province of Laguna



COLLEGE OF COMPUTER STUDIES

Jhon Jhon P. Zotomayor
President of NTEA
Board Of Regents
Brgy. Bubukal, Sta. Cruz, Laguna

Dear Sir **Zotomayor**,

Good day!

We hope this letter finds you well. Our Team is currently completing our Information Technology at Laguna State Polytechnic University in College of Computer Studies. As part of our academic requirements, we are undertaking ITEP 312 Capstone Project 1 that aims to shorten or enhance the way they do their work using the system we will develop.

We are reaching out to respectfully request your participation in an initial interview and investigation phase of our project. Your expertise and insights are invaluable to our research and will greatly contribute to the depth and relevance of our findings. The interview would take approximately 5 – 10 minutes and can be conducted at your convenience via face-to-face interview.

Please be assured that all information shared will be treated with the utmost confidentiality and will only be used for academic purposes. Additionally, we are more than willing to share the findings and outcomes of our project with you, hoping that it might provide value or insights beneficial to your work.

Would it be possible to schedule a time that is convenient for you within the week? We are flexible and willing to accommodate your schedule as much as possible. Your participation would be greatly appreciated and would significantly impact the quality and success of my project.

Thank you very much for considering our request. We look forward to the possibility of speaking with you and learning from your experience.

Please feel free to contact us at 09151734336 or francisbangkil28@gmail.com if you have any questions or to arrange a meeting time.

Sincerely,

Francis Anselme V. Bangkil
 Julius Roy D. Asuncion
 Chrischan Adomado Aloba
 BSIT-SMP Students
 College of Computer Studies
 Laguna State Polytechnic University

Noted by:

Maria Laureen B. Miranda, LPT, MIT
 Subject Instructor
 Research Implementing Unit Head, CCS

Request Letter



Republic of the Philippines
Laguna State Polytechnic University
 Province of Laguna



COLLEGE OF COMPUTER STUDIES

Jhon Jhon P. Zotomayor
 President of NTEA
 Board of Regents
 Brgy. Bubukal, Sta. Cruz, Laguna

Dear Sir **Zotomayor**,

Good day!

We hope this letter finds you well. Our Team is currently completing our Information Technology at Laguna State Polytechnic University in College of Computer Studies. As part of our academic requirements, we are undertaking ITEP 312 Capstone Project 1 that aims to shorten or enhance the way they do their work using the system we will develop.

We are reaching out to respectfully request your permission to collect and use your data as part of the initial interview and investigation phase of our project. Your expertise and insights are invaluable to our research and will greatly contribute to the depth and relevance of our findings. The data collection process would take approximately 5–10 minutes and can be conducted at your convenience through a face-to-face interview.

Please be assured that all information shared will be treated with the utmost confidentiality and will only be used for academic purposes. Additionally, we are more than willing to share the findings and outcomes of our project with you, hoping that it might provide value or insights beneficial to your work.

Thank you very much for considering our request. We look forward to the possibility of speaking with you and learning from your experience.

Please feel free to contact us at 09151734336 or francisbangkil28@gmail.com if you have any questions or to arrange a meeting time.

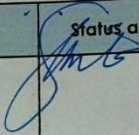
Sincerely,

Francis Anselme V. Bangkil
 Julius Roy D. Asuncion
 Chrischan Adomado Aloja
 BSIT-SMP Students
 College of Computer Studies
 Laguna State Polytechnic University

Noted by:

Maria Laureen B. Miranda, LPT, MIT
 Subject Instructor
 Research Implementing Unit Head, CCS

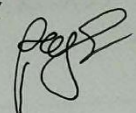
Summary of Recommendation

SUMMARY OF RECOMMENDATIONS						
___ Title Proposal	___ Research Colloquium	___ Final Oral Defense				
Group No. <u>BS10</u>	Date: _____	Time: _____				
Name of Proponents: <u>JON CHRISTIAN A. ALBA</u>						
<u>Julius Roy D. Agoncillo</u>						
<u>Francis Angelina V. Pangkil</u>						
Working Title: <u>LEARNING MANAGEMENT SYSTEM FOR LSPU EMPLOYEES:</u> <u>CIVIL SERVICE EXAMINATION PREPARATION</u>						
Panel Member	Comments Recommendations / Suggestions	Status and Signature				
<u>Nelson CA</u>	<u>- create account</u>					
Checked by: <table style="width: 100%; border: none;"> <tr> <td style="border: 1px solid black; width: 50%; padding: 2px;">Panel Member:</td> <td style="border: 1px solid black; width: 50%; padding: 2px;">Panel Member:</td> </tr> <tr> <td style="border: 1px solid black; padding: 2px;">Panel Member:</td> <td style="border: 1px solid black; padding: 2px;">Panel Member:</td> </tr> </table>			Panel Member:	Panel Member:	Panel Member:	Panel Member:
Panel Member:	Panel Member:					
Panel Member:	Panel Member:					
Noted by: <table style="width: 100%; border: none;"> <tr> <td style="border: 1px solid black; width: 50%; padding: 2px;">Instructor/Adviser/RIUH:</td> <td style="border: 1px solid black; width: 50%; padding: 2px;">Associate Dean:</td> </tr> </table>			Instructor/Adviser/RIUH:	Associate Dean:		
Instructor/Adviser/RIUH:	Associate Dean:					
Conformed by: <table style="width: 100%; border: none;"> <tr> <td style="border: 1px solid black; width: 50%; padding: 2px;">Student:</td> <td style="border: 1px solid black; width: 50%; padding: 2px;">Student:</td> </tr> <tr> <td style="border: 1px solid black; padding: 2px;">Student:</td> <td style="border: 1px solid black; padding: 2px;"></td> </tr> </table>			Student:	Student:	Student:	
Student:	Student:					
Student:						

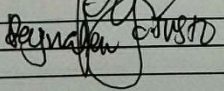
SUMMARY OF RECOMMENDATIONS

☐ Title Proposal
 ☐ Research Colloquium
 ☐ Final Oral Defense

Group No. <u>BS10</u>	Date: _____	Time: _____
Name of Proponents: <u>JON CHRISTIAN A. ALBA</u>		
<u>Jalim Rey D. Arancillo</u>		
<u>Francis Aracelis V. Binyhil</u>		
Working Title: <u>LEARNING MANAGEMENT SYSTEM FOR LEAD EMPLOYEES:</u> <u>CIVIL SERVICE EXAMINATION PREPARATION</u>		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
Reynaldo C. Justo	1. Improve the Practice Examination: 1a. in all levels - -improvement on search= 2. include all	
FOD	Analytical clerical verbal numerical	need to have a complete set on FOD
	3. test for 30 students employees	

Checked by:

Panel Member: 	Panel Member: _____
Panel Member: _____	Panel Member: _____

Noted by:

Instructor/Adviser/RIUH: _____	Associate Dean: _____
--------------------------------	-----------------------

Conformed by:

Student: _____	Student: _____
Student: _____	Student: _____

SUMMARY OF RECOMMENDATIONS

☐ Title Proposal
 ☐ Research Colloquium
 ☐ Final Oral Defense

Group No. BS10	Date:	Time:
Name of Proponents: Aloba, Crischan A.		
Asuncion, Julius Roy D.		
Bangkil, Francis Anselme V. <i>Absent</i>		
Working Title: Learning Management System for LSPU Employees: Civil Service Examination Preparation		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
M A LAURGEN MIRANDA	- CHECK THE NEW VGMIO OF LSPU - FORMAT - TIMES NEW ROMAN - ALPHABETICAL OF REL - JUSTIFY	<i>[Signature]</i>
MARK P. BERWARDINO	- SELF PACE LEARNING - DESIGN - Practice examination - Timed practice test - Performance Analytic	<i>[Signature]</i>
	- Example problem solving - Employee feedback - Surveys - Module 1, Verbal ability Module 2, Numerical ability Module 3, Analytical Ability Module 4, General Info	

Checked by:

Panel Member: <i>[Signature]</i>	Panel Member:
Panel Member: <i>[Signature]</i>	Panel Member:

Noted by:

Instructor/Adviser/RIUH:	Associate Dean:
--------------------------	-----------------

Conformed by:

Student:	Student:
Student:	

Research Colloquium Rating Sheet



Republic of the Philippines
Laguna State Polytechnic
University
Province of Laguna



COLLEGE OF COMPUTER STUDIES RESEARCH COLLOQUIUM RATING SHEET

Group Code: <u>ES10</u>	Date: <u>April 2, 2024</u>	Time: _____
Name of Proponents: <u>Francis Angelina V. Bangkil Julius Roy D. Aquino JON CATRICHAN ALQBA</u>		
Research Title: <u>Learning Management System for LSPU Employees: Civil Service Examination</u>		

MANUSCRIPT (20%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, chapter division, style including neatness and organization of details)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB-TOTAL	30	

ORAL DEFENSE (20%)	WEIGHT (%)	EVALUATED SCORE
1. Presentation (content and creativity of visual aid and/or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB-TOTAL	20	

CAPSTONE/THESIS PROJECT (60%)	WEIGHT (%)	EVALUATED SCORE
Functionality-This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability-Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability-Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	
SUB-TOTAL	50	
OVER ALL TOTAL	100	85

VERDICT

[] **PASSED:** Minor revisions are necessary, but they do not have to be presented in front and checked by all panelists. Grade range: **95-100**.

[] **PASSED WITH MINOR REVISIONS:** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94**.

[] **PASSED WITH MAJOR REVISIONS:** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84**.

[] **REDEFENSE:** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74**.

[] **DISAPPROVED:** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70**.

Role on the Defense

Signature over Printed Name of Evaluator

Ms. Miranda



Republic of the Philippines
Laguna State Polytechnic
University
Province of Laguna



COLLEGE OF COMPUTER STUDIES
RESEARCH COLLOQUIUM RATING SHEET

Group Code.: <u>BS10</u>	Date: <u>April 2, 2016</u>	Time: _____
Name of Proponents:		
<u>Francis Aracelis V. Bengkil Julius Roy D. Asuncion RISCATAN ALDREY</u>		
Research Title: <u>Learning Management System for LSPU Employees: Civil Service Examination</u>		
MANUSCRIPT (20%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, chapter division, style including neatness and organization of details.)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB - TOTAL	30	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/ or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB - TOTAL	20	
CAPSTONE/THESIS PROJECT (60%)		
Functionality- This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability- Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability- Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	
SUB - TOTAL	50	
OVER ALL TOTAL	100	89

VERDICT

- [] **PASSED.** Minor revisions are necessary, but they do not have to be presented in front and checked by all panelists. Grade range: **95-100.**
- [] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94.**
- [] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84.**
- [] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexamination. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74.**
- [] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70.**

Panel
Role on the Defense

Reginald R. Rusto
Signature over Printed Name of Evaluator



Republic of the Philippines
Laguna State Polytechnic
University
Province of Laguna



COLLEGE OF COMPUTER STUDIES
RESEARCH COLLOQUIUM RATING SHEET

Group Code: BS 10	Date: April 7, 2026	Time: _____
Name of Proponents:		
JOU CHRISCHIAN AIOBA	Julius Roy D. Abandon	Francis Anselmo V. Banzon
Research Title: LEARNING MANAGEMENT SYSTEM FOR LSPU EMPLOYEES: CIVIL SERVICE EXAMINATION PREPARATION		
MANUSCRIPT (20%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, chapter division, style including neatness and organization of details)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB - TOTAL	30	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/ or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB - TOTAL	20	
CAPSTONE/THESIS PROJECT (60%)		
Functionality-This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability- Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability- Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	
SUB - TOTAL	50	
OVER ALL TOTAL	100	

VERDICT

- [] **PASSED.** Minor revisions are necessary, but they do not have to be presented in front and checked by all panelists. Grade range: **85-100**.
- [] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: **85-94**.
- [] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: **75-84**.
- [] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: **70-74**.
- [] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: **BELOW 70**.

Role on the Defense

[Signature]
Signature over Printed Name of Evaluator



10pager.docx

 Central Luzon State University

Document Details

Submission ID**trn.oid::29324139841716****Submission Date****May 24, 2026, 10:56 PM GMT+8****Download Date****May 24, 2026, 10:59 PM GMT+8****File Name****10pager.docx****File Size****1.7 MB****11 Pages****3,638 Words****21,607 Characters**



19% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.



7 AI-generated only 19%

Likely AI-generated text from a large-language model.



0 AI-generated text that was AI-paraphrased 0%

Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.





LEARNING MANAGEMENT SYSTEM FOR LSPU EMPLOYEES: CIVIL SERVICE EXAMINATION PREPARATION

ASUNCION, JULIUS ROY D.¹ BANGKIL, FRANCIS ANSELME V.² ALOBA, JON CHRISCHAN A.³

College of Computer Studies
Laguna State Polytechnic

University Laguna, Philippines
juliusroyasuncion@gmail.com

College of Computer Studies
Laguna State Polytechnic

University Laguna, Philippines
francisbangkil28@gmail.com

College of Computer Studies
Laguna State Polytechnic

University Laguna, Philippines
chrischanaloba944@gmail.com

VILLARICA, MIA V.⁴

College of Computer Studies
Laguna State Polytechnic

University Laguna, Philippines
mia.villarica@lspu.edu.ph

ABSTRACT - The adoption of digital technologies in organizational training has become essential in enhancing employee competence and professional development. This study, entitled “Learning Management System for LSPU Employees: Civil Service Examination Preparation,” aimed to design and develop a web-based platform to support Laguna State Polytechnic University (LSPU) employees in preparing for the Civil Service Examination. The system provides centralized access to review materials, enables self-paced learning, tracks user progress, and generates performance reports for both learners and administrators. The study utilized the Rapid Application Development (RAD) methodology to ensure efficient system design and implementation. System functionality was evaluated through black box testing conducted by IT experts, yielding a 100% success rate across all tested features. Furthermore, the Technology Acceptance Model (TAM) was employed to assess user acceptance, with sixty (60) respondents participating in the evaluation. The results indicated a “Highly Acceptable” rating in terms of perceived usefulness, ease of use, and overall system performance. The system was also evaluated based on ISO/IEC 25010 standards, confirming its quality and reliability.

Keywords: *Learning Management System; Civil Service Examination Preparation LSPU Employee; Rapid Application Development; Black Box Testing; Technology Acceptance Model; ISO/IEC 25010 Software Metrics*

I. INTRODUCTION

The Civil Service Examination (CSE) is a requirement for individuals seeking permanent employment and career advancement in the Philippine government. The Civil Service Commission (CSC) administers the test, which evaluates the competencies, knowledge, and abilities needed to provide successful public service. Passing the CSE grants government personnel civil service eligibility, which is necessary for job security, advancement, and professional development in the public sector.

As a result, Civil Service Examination Service Examination preparation has become a primary priority for government employees, particularly those in state universities and colleges. This project proposes the development of Learning Management System for LSPU Employees, a Web-based designed to enhance the Civil Service exam preparation experience by integrating game mechanics into a comprehensive reviewer. The reviewer aims to improve user engagement, motivation, and learning effectiveness through module, interactive quizzes, mock exam and progress tracking. Laguna State Polytechnic University (LSPU) is a public higher education institution that employs both teaching and non-teaching personnel. As government employees, they are expected to obtain civil service eligibility in order to qualify for permanent appointments and career advancement. However, many LSPU employees find it difficult to prepare for the Civil Service Examination. Their work responsibilities often take up most of their time, leaving little opportunity to attend review centers or participate in structured review programs. In addition, differences in educational background and limited access to review materials make examination preparation more challenging for some employees.

When creating a Web-based system for the Civil Service Examination, monitoring and recording are crucial components. They function as a watchful eye, monitoring both the learner's progress and the system's activities to make sure no crucial information is missed. User information, scores, review history, and performance statistics are kept safe, structured, and readily available whenever needed through appropriate monitoring and recording.

II. RESEARCH OBJECTIVES

The study aims to develop an LSPU Employees: Civil Service Examination Preparation system designed to improve and simplify the review process for Civil Service Examination examinees. Specifically, this study seeks to:

1. To design & develop a learning management system for LSPU Sta. Cruz, Laguna.
2. To design and develop a system that would help the personnel prepare effectively for the exam by improving their knowledge, skills, and confidence:
 - a. Learning and Review System
 - b. Access review materials for Civil Service Examination
 - c. Study modules



- d. Take quizzes and mock examinations
3. To assess & evaluate the acceptance of the user using Technology Acceptance Model of:
 - a. Quality factors
 - b. Perceived ease of use
 - c. Perceived Usefulness
 - d. Experience
 - e. Attitude Towards Using
 - f. Behavioral Intention to Use

III. LITERATURE OF THE STUDY

The following information was gathered by the researchers for their assessment of pertinent studies and research that inspired their most recent inquiry. The discussions on Learning Management Systems (LMS), their features, implementation, strategies and the role of the technology in enhancing educational management are summed up here

Y Asrida, Y. N., Amanda, F., & Ul Fadilah, J. (2024) analyze the effectiveness of the Learning Management System (LMS) in promoting students' independence and learning outcomes in the digital era using a literature-based Classroom Action Research (CAR) approach. The study applies a reflective cycle of planning, implementation, observation, and evaluation and finds that consistent LMS use strengthens learners' autonomy, boosts intrinsic motivation, and improves academic performance through structured and interactive activities.

Kadir, A., Terra, & Triwahyuni (2013) describe the Learning Management System (LMS) as an online platform that adapts conventional learning into a digital environment, enabling flexible and student-centered instruction. Using descriptive qualitative research with high school teachers and students in the Dewantara District, the study shows that learners are highly engaged when using the LMS, finding the material easier to understand and their learning outcomes visibly improved.

Abaricia, C. P., & Delos Santos, M. L. C. (2023) examine how LMS-based training enhances the competencies of civil servants (ASN) through the L-Bos Learning Management System developed by the East Java Provincial Human Resource Development Agency.

Giannakos, M. N., Mikalef, P., & Pappas, I. O. (2022) conduct a systematic literature review on e-learning capabilities and their role in organizational learning, gathering 47 peer reviewed studies to map current trends and future directions. They highlight that contemporary e learning systems allow managers and employees to train flexibly, collaborate remotely, and share knowledge without fixed time or location constraints.



Iqbal, M., Iskandarini, I., & Rahmadhani, V. G. (2025) investigate how LMS-based training and career development affect employee competence at PT. Perkebunan Nusantara IV Regional II Medan using a quantitative design and structured questionnaires from 123 employees. Multiple linear regression shows that LMS-based training has a stronger and more significant impact on competence than traditional career development programs, thanks to its flexibility and accessibility.

Doe, B., & Vivekanandan (2023) evaluate LMS course effectiveness in the workplace by examining user experience and learning satisfaction among 474 banking professionals. The study analyzes relationships among LMS quality factors—pedagogical design, interface design, content presentation format, transfer of learning, and feedback—and finds that learner experience and satisfaction are strongly shaped by these dimensions

Krishna, K. P. (2025) explores how e-learning platforms transform employee performance and development by integrating digital training into organizational processes. Drawing on empirical studies, case examples, and literature, the paper emphasizes key benefits such as accessibility, personalized learning paths, instant feedback, and cost efficiency, all of which enhance knowledge retention, productivity, and job performance.

IV. METHODOLOGY

Research Design

According to Muhammad (2023) says research design is just the plan a researcher makes for their study. In social science, you need to collect good information that fits your research question stuff that helps you look at a problem, check a theory, or explain something. A good research plan means your findings will be true, dependable, and fair. In this study on "LSPU Employees: Civil Service Examination Preparation," the researcher used descriptive research as the main method. According to Deckert (2023), descriptive research provides information based on observation and describes situations or phenomena "as they are" using tools such as surveys, questionnaires, interviews, or rankings. This type of research is useful for exploring new areas or understanding specific conditions, and it can use both qualitative and quantitative data. For this study, the researcher collected data through interviews, surveys, and observation of LSPU employees to identify their challenges, needs, and preparation strategies for the Civil Service Examination. Assistance was also sought from university staff to ensure accurate and relevant information was gathered.

Locale of the Study

The study was conducted at Laguna State Polytechnic University (LSPU), Bubukal, Santa Cruz, Laguna. Specifically, it focused on the employees of the university who are preparing for the Civil Service Examination. The location was chosen by the researcher because this type of system would be highly beneficial to LSPU employees, as it provides an organized and accessible way to manage review materials, track progress, and improve overall preparation for the examination

The population of this study consisted of non-teaching employees of LSPU, Bubukal, Santa Cruz, Laguna who are preparing for the Civil Service Examination. They were selected because they are the intended users of the proposed Learning Management System and can provide relevant information about their review habits and needs.

Table 1. Respondents for the User Evaluation

Table 1. Respondents for the User Evaluation

Respondents	Total Population	Sample Size	Percentage
Administrator	1	1	1.56%
Employees	126	63	98.44%
Total	127	64	100%

The table indicates how many target respondents were selected to conduct the evaluation in response to the researchers' platforms. The table shows the total population, target population and accessible population, with the target population representing the quota that the researchers must meet, and the accessible population, which was the actual number of respondents who participated in the survey and were analyzed by the researchers as a result of unexpected events.

Sampling Design

A sort of sampling known as "purposive sampling," which is assessed, selective, or subjective, was used by the researchers to pick the study participants. The researcher selects the qualities to examine using this kind of non-probability sampling technique based only on their own discretion. Therefore, in order to satisfy the goals of the researchers, they can select a sample that accurately reflects the entire population or concentrate on individuals who fulfill particular qualifications. One advantage of this strategy is that it may quickly be implemented and yield a wide range of research answers.

System Architecture

The term "system architecture" describes the organizational structure of software applications used for task automation. It creates the software application's structure, written by Spacey (2019). This usually came in the form of a series of diagrams showing the different services, components, and interconnections. For this reason, system architecture was quite helpful in helping to visualize the created system's structure.

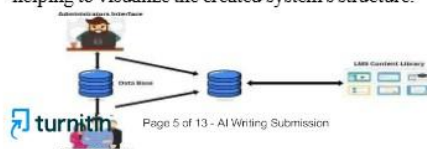


Figure 1. System Architecture of the study "Learning Management System for LSPU Employees: Civil Service Examination Preparation"

Software Development Model

In order to create the web-based system, the researchers employed the Rapid Application Development (RAD) approach as their software process model. The rigidity of other conventional software development methods, which makes it challenging to make modifications after the original development has been finished, led to the creation of the RAD approach, a program design strategy. When a client recommends new features that meet their demands, the researchers can quickly implement improvements thanks to this design.

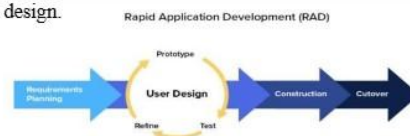


Figure 2. Rapid Application Development Methodology

According to Flora (2018) RAD is a software development method that speeds up software production by reducing the complicated and time-consuming planning process. RAD can be used as a reference for developing an information system that is superior in terms of speed, accuracy and lower cost.

Phase 1. Project requirement and planning.

The project's scope should be defined before starting development activities, ensuring no unnecessary tasks. The planning phase involves stakeholders, managers, developers, and clients discussing requirements, issues, timelines, and budgets. Collective decision-making, involving approvals from stakeholders and the entire discussion team, helps avoid controversies and miscommunications, preventing costly changes later in the process.

Phase 2. Prototype.

The Rapid Application Development process involves rapid prototyping, where developers work with clients to design and test prototypes based on requirements. Feedback is then iterated until the application reaches a satisfactory stage, making it user-centric and meeting user expectations. This approach reduces the risk of rapidness.

Phase 3. Construction and feedback.

The team begins the development phase by converting prototypes into working models, building the application through coding and integrations. The application is continuously tested for bugs and issues, and if found, the team resolves them promptly. This process is faster due to the discussion and implementation of changes in the previous stage. Users are also involved in the process, allowing them to suggest changes and fill the gap between design and development.

Phase 4. Implementation.

After developing a working model, developers finalize it and move it to a live environment. They evaluate and record the model, conduct full scale testing to eliminate bugs, and

turnitin Page 6 of 13 - AI Writing Submission
 finalize the application. The team then moves on to user onboarding and continues to provide support post-launch, ensuring a smooth transition to the live environment.

Design

The researchers employed various methods for data collection to obtain the information needed to develop the proposed web-based system, which would help accomplish the first objective. These methods enabled the researchers to gain insight into user needs and pinpoint the issues faced in the current review practices of employees. The initial phase of the study involved conducting a preliminary investigation. On October 23, the researchers interviewed Sir Zotomayor, who acted as the client and offered valuable perspectives on the existing conditions of employees studying for the Civil Service Examination. This interview allowed the researchers to uncover the difficulties faced by employees, such as the lack of organized review materials, the absence of a system to monitor progress, and restricted access to structured quizzes and examinations.

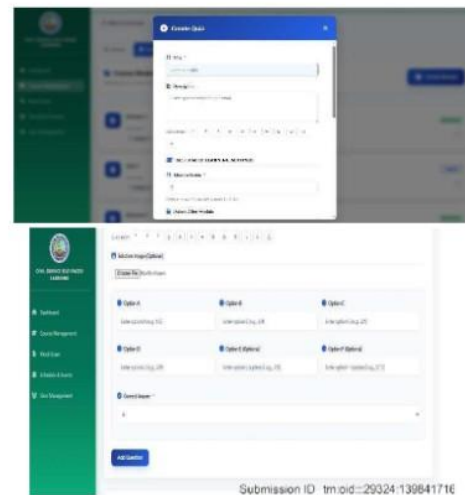
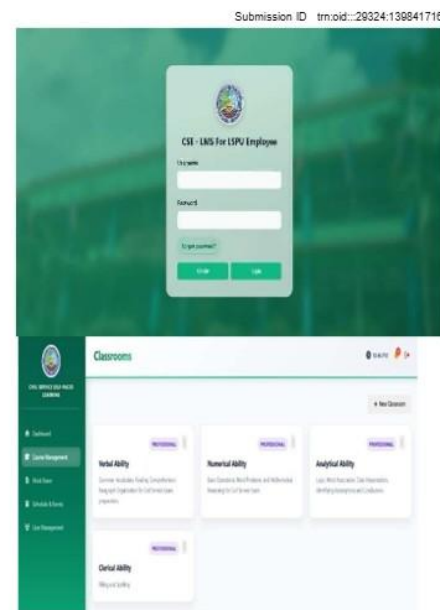
These findings became the basis for designing and developing a Learning Management System that focuses on modular learning and assessment, aiming to improve the overall preparation and performance of LSPU-SCC employees named **LEARNING MANAGEMENT SYSTEM FOR LSPU EMPLOYEES: CIVIL SERVICE EXAMINATION PREPARATION**.

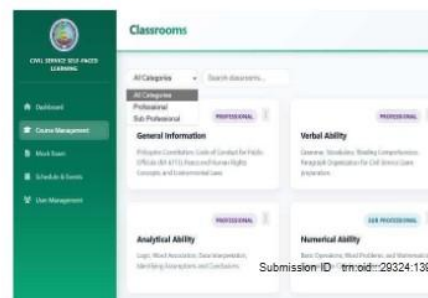
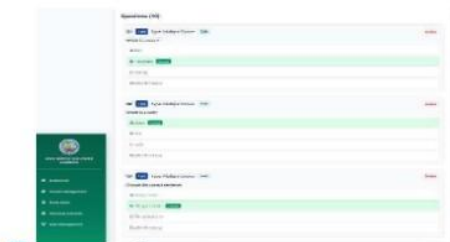
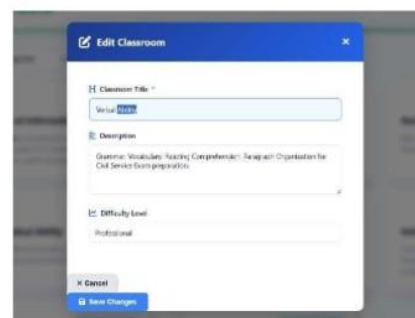
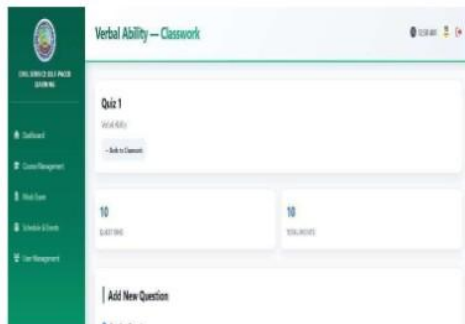
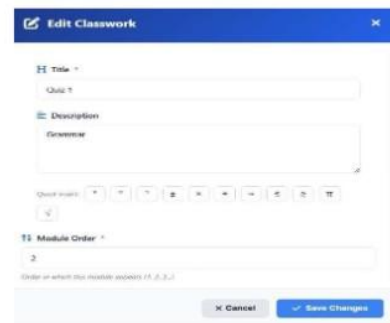
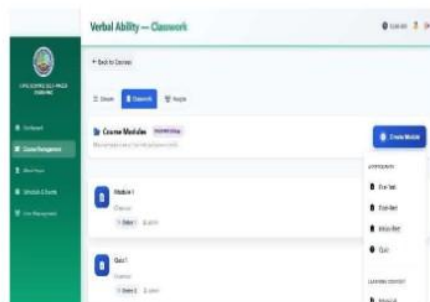
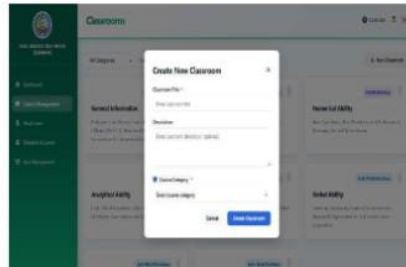
The Development

After gathering all essential data, the researchers proceeded to the second stage, which involved developing a web-based system using the Rapid Application Development (RAD) approach.

V. Result and Discussion

Research Objective No. 1: To design and developed a web-based **LEARNING MANAGEMENT SYSTEM FOR LSPU EMPLOYEES: CIVIL SERVICE EXAMINATION PREPARATION** that provides a centralized platform for delivering structured learning modules, enables users to take quizzes and examinations for self-assessment, facilitates efficient monitoring of learning progress, ensures accurate recording of users, performance and results, and ultimately enhances employees readiness, engagement and effectiveness in preparing for the Civil Service Examination at Laguna State Polytechnic University – Sta. Cruz Campus.





Mock Exam Management

Configure the parameters for mock exam activities.

Comprehensive Mock Exam

For student mock exams to generate questions for the mock exam. These activities can be used for the general assessment of student knowledge.

☒ 10 Questions ☐ Custom Mock

[View Mock](#)

[Create Comprehensive Mock Exam](#)

Sub Professional Mock Exam

For student mock exams to generate questions for the mock exam. These activities can be used for the general assessment of student knowledge.

☐ 10 Questions ☒ Custom Mock

[View Mock](#)

[Create Sub Professional Mock Exam](#)

User and Employee Management

Manage the user and employee information.

Id	Name	Username	Role	Super User	Status	Action
1	John Doe	john.doe	Admin	Yes	Active	Edit Delete

Save Mock Exam

Save the mock exam details.

Sub Professional Mock Exam

For student mock exams to generate questions for the mock exam. These activities can be used for the general assessment of student knowledge.

[View Mock](#)

[Create Sub Professional Mock Exam](#)

Custom Mock Exam

For student mock exams to generate questions for the mock exam. These activities can be used for the general assessment of student knowledge.

[View Mock](#)

[Create Custom Mock Exam](#)

Calendar of event

View the calendar of events.

Event	Start	End	Location	Category	Status
Mock Exam	2023-05-01	2023-05-02	Online	Exam	Active

Create Mock Exam

Configure mock exam display settings, time, and pricing.

Questions to Display:

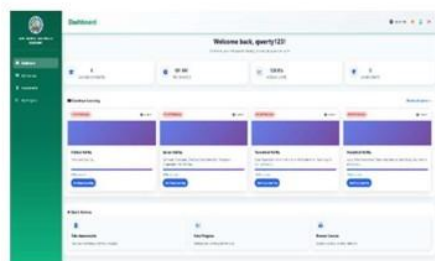
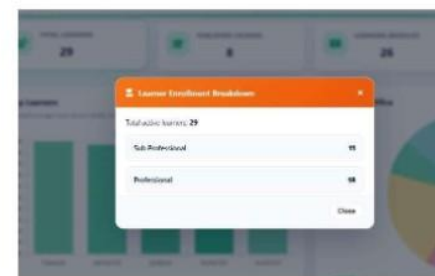
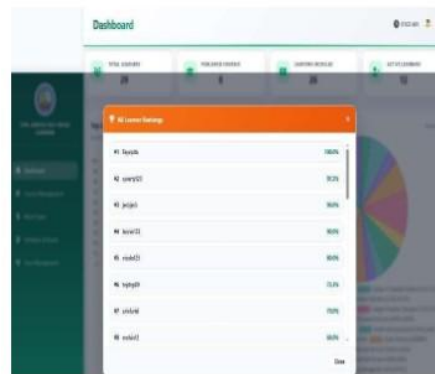
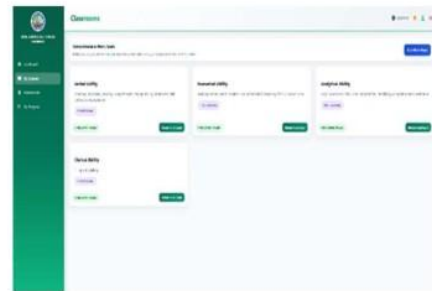
Passing Score (%):

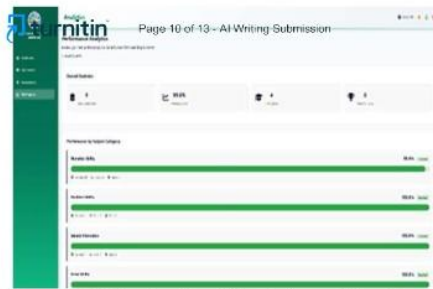
Timer (minutes):

[Cancel](#) [Create](#)

Recent Mock Exam Results

Exam	Score	Percentage	Status	Action
Mock Exam	80%	80%	Pass	View Result





Research Objective No. 2: To design and develop a system that would help the employee prepare effectively for the quiz, exam by improving their knowledge, skills, and confidence

- Create Classroom
- Add Classwork
- Create Module
- Add Quiz/Examination
- Read Module
- Take Quiz/Examination
- View Analytics
- Take Mock Exam

Table 13: Black Box Testing Results for System Administrator

Functionality	First Trial			Second Trial		
	Pass	Fail	%	Pass	Fail	%
Dashboard	10	0	100%	10	0	100%
Sign up	10	0	100%	10	0	100%
Dashboard	10	0	100%	10	0	100%
Classroom	10	0	100%	10	0	100%
Classwork	10	0	100%	10	0	100%
User Management	9	1	90%	10	0	100%

The provided table illustrates the outcomes of black box testing conducted by 10 IT experts. It confirms that the primary functions of the system operated successfully. Specifically, all functions related to System Administrator Account achieved a flawless pass rate of 5 and 100% overall success rate. The performance of all functions collectively demonstrated effectiveness based on the overall percentage.

Table 14: Black Box Testing Results for Users Section

Functionality	First Trial			Second Trial		
	Pass	Fail	%	Pass	Fail	%
Login	10	0	100%	10	0	100%
Sign up	10	0	100%	10	0	100%
Dashboard	10	0	100%	10	0	100%
User Classroom	10	0	100%	10	0	100%
Stream	10	0	100%	10	0	100%
Classwork	10	0	100%	10	0	100%
Read Module	10	0	100%	10	0	100%
Take Exam/Quiz	9	1	90%	10	0	100%
User Assessment	10	0	100%	10	0	100%
Take Assessment	10	0	100%	10	0	100%
Retake Assessment	10	0	100%	10	0	100%
View Result	10	0	100%	10	0	100%
My Progress	9	1	90%	10	0	100%
User Analytics	10	0	100%	10	0	100%
Display Recent Quiz/Exam	10	0	100%	10	0	100%
Display Percentage of Each Course	10	0	100%	10	0	100%

The outcomes of the black box testing conducted by the ten IT experts serve as a crucial validation of the system's functionality and reliability. The flawless pass rate for all functions related to the employees' section, coupled with the 100% overall success rate in the second trial, underscores the system's robustness in supporting learning activities and user interaction. This high level of performance not only instills confidence in the system's ability to meet operational requirements but also reflects positively on the expertise and diligence of the development team. Furthermore, the comprehensive success across all functions indicates a thorough understanding of user needs and meticulous attention to detail during the development process.

Research Objectives No. 3: To Assess and Evaluate the acceptance of the user using Technology Acceptance Model in terms of:

- Quality Factors
- Perceive Ease to Use
- Perceived Usefulness
- Experience
- Attitude Towards to Use
- Behavioral Intention to Use

The researcher leaned on the Technology Acceptance Model (TAM) survey to gauge how well users took to the new tech and planned to use it, right in line with the study's aims. Once the questionnaire was drafted, they ran it by the research adviser and a stats expert for approval. With their green light, it was good to go. Feedback came via a Likert scale tracking satisfaction: Very Satisfied (Highly Acceptable), Satisfied (Acceptable), Neither (Undecided), Unsatisfied (Slightly Unacceptable), or Very Dissatisfied (Unacceptable). We handed out printed surveys in person to the system's key users for straightforward, real-time responses.

Table of Level of Acceptability

Identifiers	Mean	Interpretation
Quality Factors	4.48	Acceptable
Perceived Ease of Use	4.51	Acceptable
Perceived Usefulness	4.52	Acceptable
Attitude Towards Using	4.47	Acceptable
Behavioral Intention to Use	4.53	Acceptable
Experience	4.54	Acceptable
Total Mean	4.51	Acceptable

The evaluation also focused on user acceptance and satisfaction using several key factors: system quality, ease of use, perceived usefulness, attitude towards the system, behavioral intention to use, and overall user experience.

SUMMARY OF CONCLUSIONS AND RECOMMENDATIONS

This chapter discusses the primary findings on the level satisfaction with the developed system based on the study's goal, research conclusions were based on the study's goal, research

turnitin Page 11 of 13 - AI Writing Submission
 questions, and results. The implications of these findings, as well as the recommendations that follow, are explained. The recommendations were based on the study's conclusions and purpose.

Summary

This study focused on the design and development of a Learning Management System for LSPU Employees: Civil Service Examination Preparation. The research examined the challenges faced by employees, such as limited time for review, lack of structured materials, and difficulty accessing reliable resources. To address these concerns, the LMS was developed with key features including user-friendly navigation, categorized review modules, practice quizzes, performance tracking.

The system was evaluated based on usability, functionality, reliability, and overall user satisfaction.

Feedback from respondents indicated that the LMS significantly improved their review experience by making learning more convenient and structured.

The researchers were able to obtain the necessary data via interviews, library research, web research, technological gap analysis, and brainstorming. The first survey was conducted to establish the problems that employees experience, as well as the present procedures for document tracking, document processing, and user management. The researchers used the Rapid Application Development (RAD) methodology when developing the web-based solution. With this methodology, the researchers can review and enhance the program without having to restart the development process from the beginning.

To evaluate the performance and usefulness of the developed system, the researchers created a questionnaire using Black Box testing. The developed system was tested virtually with the help of IT experts who served as a system tester. The tests show that all of the system's functionalities work properly.

During the system evaluation, the researchers developed and prepared a questionnaire based from the Technology Acceptance Model (TAM) and ISO/IEC 25010 Software Quality Metrics, which used a 5-point Likert Scale response to assess user satisfaction. Survey Questions were tested and sent to assess the system's usability, effectiveness, and user satisfaction. In this study, the researchers did a face-to-face evaluation to determine the user's level of satisfaction.

The data gathered from the 63 respondents were evaluated and interpreted. The interpreted data resulted in a response scale that was measured against each evaluation criteria, and the outcome was "Highly Acceptable." Indicating that the user were satisfied with the developed **LEARNING**

turnitin Page 11 of 13 - AI Writing Submission

MANAGEMENT SYSTEM for LSPU EMPLOYEES: Civil Service Examination Preparation.

Conclusions

The feedback gathered after the testing and evaluation phase proved that the developed web-based system is working well and the respondents were satisfied and willing to use the developed system. In conclusion, the study was able to achieve its objective: (1) Develop a Learning Management System for LSPU Employees: Civil Service Examination Preparation; (2) To design and develop a system that would help the document handler and guest to track their documents; (3) To assess and evaluate the acceptance of the user using Technology Acceptance Model in terms of Quality Factors, Perceived Ease to Use, Perceived Usefulness, Experience, Attitude towards Using, and Behavioral Intention to Use towards the developed system. In the study, the researchers therefore conclude the following:

1. After gathering data through a comprehensive preliminary investigation, and interviews with employees of the National Irrigation Administration of Pila, Laguna, the researchers concluded that the current document tracking process needs to be improved in terms of efficiency and reliability, and that developing a web-based system can be a better solution to the difficulties that employees face when doing their specific tasks.
2. The Rapid Application Development (RAD) methodology used by the researchers is beneficial for developing the web-based system.
3. Using Black Box Testing, the researchers discovered that the developed web-based system met its specified functional requirements.

Recommendations

To further enhancement of "Learning Management System for LSPU Employees: Civil Service Examination Preparation", the following are recommended below:

1. To future researchers, the web-based system should be more responsive on the web but not responsive on mobile. Future researchers should make their work responsive to mobile devices so that mobile users can access their work more easily.

Literature Cited

Datta, A., et al. (2022). Modeling sub-GHz communication channels from internal to external body environments in ruminants, targeting advancements in smart livestock management. Published in IEEE Journals & Magazine (2023, April 1). Full text available at <https://ieeexplore.ieee.org/document/9915440>.

Jindal, A., et al. (2023). This work presents a method for creating feedback-linearizable discrete models of continuous-time systems via retraction mapping. Refer to IEEE Journals & Magazine, 2023. See <https://ieeexplore.ieee.org/document/10153969/> for details.

Submission ID: 29324139841716



A. M. NOMA, ET AL. (2022) Design of intelligent and secure hospital appointment scheduling system. (2022, April 17). IEEE Conference Publication IEEE Xplore. <https://ieeexplore.ieee.org/document/9803093>





10pager.docx

 Central Luzon State University

Document Details

Submission ID**trn:oid::29324:139841716****Submission Date****May 24, 2026, 10:56 PM GMT+8****Download Date****May 24, 2026, 10:59 PM GMT+8****File Name****10pager.docx****File Size****1.7 MB****11 Pages****3,535 Words****21,607 Characters**

16% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

-  **Not Cited or Quoted** 0%
Matches with neither in-text citation nor quotation marks
-  **Missing Quotations** 0%
Matches that are still very similar to source material
-  **Missing Citation** 0%
Matches that have quotation marks, but no in-text citation
-  **Cited and Quoted** 0%
Matches with in-text citation present, but no quotation marks

Top Sources

- 14%  Internet sources
- 2%  Publications

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

-  **Not Cited or Quoted** 0%
Matches with neither in-text citation nor quotation marks
-  **Missing Quotations** 0%
Matches that are still very similar to source material
-  **Missing Citation** 0%
Matches that have quotation marks, but no in-text citation
-  **Cited and Quoted** 0%
Matches with in-text citation present, but no quotation marks

Top Sources

- 14%  Internet sources
- 2%  Publications

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	journal.unimma.ac.id	<1%
2	Internet	www.globalscientificjournal.com	<1%
3	Internet	www.myquest.co	<1%
4	Internet	serambi.org	<1%

LEARNING MANAGEMENT SYSTEM FOR LSPU EMPLOYEES: CIVIL SERVICE EXAMINATION PREPARATION

ASUNCION, JULIUS ROY D.¹ BANGKIL, FRANCIS ANSELME V.² ALOBA, JON CHRISCHAN A.³ VILLARICA, MIA V.⁴
 College of Computer Studies College of Computer Studies College of Computer Studies College of Computer Studies
 Laguna State Polytechnic University Laguna State Polytechnic University Laguna State Polytechnic University Laguna State Polytechnic
 University Laguna, Philippines University Laguna, Philippines University Laguna, Philippines University Laguna, Philippines
 juliusroyasuncion@gmail.com francisbangkil28@gmail.com chrishchanaloba944@gmail.com

ABSTRACT - The adoption of digital technologies in organizational training has become essential in enhancing employee competence and professional development. This study, entitled “Learning Management System for LSPU Employees: Civil Service Examination

Preparation,” aimed to design and develop a web-based platform to support Laguna State Polytechnic University (LSPU) employees in preparing for the Civil Service Examination. The system provides centralized access to review materials, enables self-paced learning, tracks user progress, and generates performance reports for both learners and administrators. The study utilized the Rapid Application Development (RAD) methodology to ensure efficient system design and implementation. System functionality was evaluated through black box testing conducted by IT experts, yielding a 100% success rate across all tested features. Furthermore, the Technology Acceptance Model (TAM) was employed to assess user acceptance, with sixty (60) respondents participating in the evaluation. The results indicated a “Highly Acceptable” rating in terms of perceived usefulness, ease of use, and overall system performance. The system was also evaluated based on ISO/IEC 25010 standards, confirming its quality and reliability.

Keywords: *Learning Management System; Civil Service Examination Preparation LSPU Employee; Rapid Application Development; Black Box Testing; Technology Acceptance Model; ISO/IEC 25010 Software Metrics*

I. INTRODUCTION

The Civil Service Examination (CSE) is a requirement for individuals seeking permanent employment and career advancement in the Philippine government. The Civil Service Commission (CSC) administers the test, which evaluates the competencies, knowledge, and abilities needed to provide successful public service. Passing the CSE grants government personnel civil service eligibility, which is necessary for job security, advancement, and professional development in the public sector. As a result, Civil

Service Examination preparation has become a primary priority for government employees, particularly those in state universities and colleges. This project proposes the development of Learning Management System for LSPU Employees, a Web-based designed to enhance the Civil Service exam preparation experience by integrating game

mechanics into a comprehensive reviewer. The reviewer aims to improve user engagement, motivation, and learning effectiveness through module, interactive quizzes, mock exam and progress tracking. Laguna State Polytechnic University (LSPU) is a public higher education institution that employs both teaching and non-teaching personnel. As government employees, they are expected to obtain civil service eligibility in order to qualify for permanent appointments and career advancement. However, many LSPU employees find it difficult to prepare for the Civil Service Examination. Their work responsibilities often take up most of their time, leaving little opportunity to attend review centers or participate in structured review programs. In addition, differences in educational background and limited access to review materials make examination preparation more challenging for some employees.

When creating a Web-based system for the Civil Service Examination, monitoring and recording are crucial components. They function as a watchful eye, monitoring both the learner's progress and the system's activities to make sure no crucial information is missed. User information, scores, review history, and performance statistics are kept safe, structured, and readily available whenever needed through appropriate monitoring and recording.

II. RESEARCH OBJECTIVES

The study aims to develop an LSPU Employees: Civil Service Examination Preparation system designed to improve and simplify the review process for Civil Service Examination examinees. Specifically, this study seeks to:

1. To design & develop a learning management system for LSPU
 2. To design and develop a system that would help the personnel prepare effectively for the exam by improving their knowledge, skills, and confidence:
 - a. Learning and Review System
 - b. Access review materials for Civil Service Examination
 - c. Study modules

3. To assess & evaluate the acceptance of the user using Technology Acceptance Model of:

- Quality factors
- ease use
- Perceived Usefulness
- Experience
- Attitude Towards
- Behavioral Intention to Use

III. LITERATURE OF THE STUDY

2 The following information was gathered by the researchers for their assessment of pertinent studies and research that inspired their most recent inquiry. The discussions on Learning Management Systems (LMS), their features, implementation strategies, and the role of technology in enhancing educational management are summed up here.

4 1 Y Asrida, Y. N., Amanda, F., & Ul Fadilah, J. (2024) analyze the effectiveness of the Learning Management System (LMS) in promoting students' independence and learning outcomes in the digital era using a literature-based Classroom Action Research (CAR) approach. The study applies a reflective cycle of planning, implementation, observation, and evaluation and finds that consistent LMS use strengthens learners' autonomy, boosts intrinsic motivation, and improves academic performance through structured and interactive activities.

Kadir, A., Terra, & Triwahyuni (2013) describe the Learning Management System (LMS) as an online platform that adapts conventional learning into a digital environment, enabling flexible and student-centered instruction. Using descriptive qualitative research with high school teachers and students in the Dewantara District, the study shows that learners are highly engaged when using the LMS, finding the material easier to understand and their learning outcomes visibly improved.

3 Abaricia, C. P., & Delos Santos, M. L. C. (2023) examine how LMS-based training enhances the competencies of civil servants ASN through the L-Bos Learning Management System developed by the East Java Provincial Human Resource Development Agency.

Giannakos, M. N., Mikalef, P., & Pappas, I. O. (2022) conduct a systematic literature review on e-learning capabilities and their role in organizational learning, gathering 47 peer reviewed studies to map current trends and future directions. They highlight that contemporary e learning systems allow managers and employees to train flexibly, collaborate remotely, and share knowledge without fixed time or location

investigate how LMS-based training and career development affect employee competence at PT. Perkebunan Nusantara IV Regional II Medan using a quantitative design and structured questionnaires from 123 employees. Multiple linear regression shows that LMS-based training has a stronger and more significant impact on competence than traditional career development programs, thanks to its flexibility and accessibility.

Doe, B., & Vivekanandan (2023) evaluate LMS course effectiveness in the workplace by examining user experience and learning satisfaction among 474 banking professionals. The study analyzes relationships among LMS quality factors—pedagogical design, interface design, content presentation format, transfer of learning, and feedback—and finds that learner experience and satisfaction are strongly shaped by these dimensions

Krishna, K. P. (2025) explores how e-learning platforms transform employee performance and development by integrating digital training into organizational processes. Drawing on empirical studies, case examples, and literature, the paper emphasizes key benefits such as accessibility, personalized learning paths, instant feedback, and cost efficiency, all of which enhance knowledge retention, productivity, and job performance.

IV. METHODOLOGY

Research Design

According to Muhammad (2023) says research design is just the plan a researcher makes for their study. In social science, you need to collect good information that fits your research question stuff that helps you look at a problem, check a theory, or explain something. A good research plan means your findings will be true, dependable, and fair. In this study on "LSPU Employees: Civil Service Examination Preparation," the researcher used descriptive research as the describes situations or phenomena "as they are" using tools main method. According to Deckert (2023), descriptive research provides information based on observation and such as surveys, questionnaires, interviews, or rankings. This type of research is useful for exploring new areas or understanding specific conditions, and it can use both qualitative and quantitative data. For this study, the

researcher collected data through interviews, surveys, and observation of LSPU employees to identify their challenges, needs, and preparation strategies for the Civil Service Examination. Assistance was also sought from university staff to ensure accurate and relevant information was gathered.

Locale of the Study

The study was conducted at Laguna State Polytechnic University (LSPU), Bubukal, Santa Cruz, Laguna. Specifically, it focused on the employees of the university who are preparing for the Civil Service Examination. The location was chosen by the researcher because this type of system would be highly beneficial to LSPU employees, as it provides an organized and accessible way to manage review materials, track progress, and improve overall preparation for the examination.

The population of this study consisted of non-teaching employees of LSPU, Bubukal, Santa Cruz, Laguna who are preparing for the Civil Service Examination. They were selected because they are the intended users of the proposed Learning Management System and can provide relevant information about their review habits and needs.

Table 1. Respondents for the User Evaluation

Table 1. Respondents for the User Evaluation

Respondents	Total Population	Sample Size	Percentage
Administrator	1	1	1.56%
Employees	126	63	98.44%
Total	127	64	100%

The table indicates how many target respondents were selected to conduct the evaluation in response to the researchers' platforms. The table shows the total population, target population and accessible population, with the target population representing the quota that the researchers must meet, and the accessible population, which was the actual number of respondents who participated in the survey and were analyzed by the researchers as a result of unexpected events.

Sampling Design

A sort of sampling known as "purposive sampling," which is assessed, selective, or subjective, was used by the researchers to pick the study participants. The researcher selects the qualities to examine using this kind of non-probability sampling technique based only on their own discretion. Therefore, in order to satisfy the goals of the researchers, they can select a sample that accurately reflects the entire population or concentrate on individuals who fulfill particular qualifications. One advantage of this strategy is that it may quickly be implemented and yield a wide range of research answers.

System Architecture

The term "system architecture" describes the organizational structure of software applications used for task automation. It creates the software application's structure, written by Spacey (2019). This usually came in the form of a series of diagrams showing the different services, components, and interconnections. For this reason, system architecture was quite helpful in helping to visualize the created system's structure.

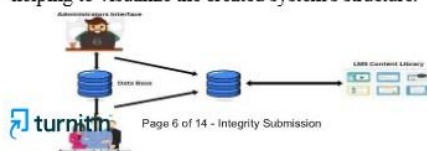


Figure 1. System Architecture of the study "Learning Management System for LSPU Employees: Civil Service Examination Preparation"

Software Development Model

In order to create the web-based system, the researchers employed the Rapid Application Development (RAD) approach as their software process model. The rigidity of other conventional software development methods, which makes it challenging to make modifications after the original development has been finished, led to the creation of the RAD approach, a program design strategy. When a client recommends new features that meet their demands, the researchers can quickly implement improvements thanks to this design.

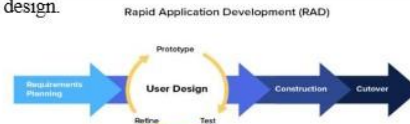


Figure 2. Rapid Application Development Methodology

According to Flora (2018) RAD is a software development method that speeds up software production by reducing the complicated and time-consuming planning process. RAD can be used as a reference for developing an information system that is superior in terms of speed, accuracy and lower cost.

Phase 1. Project requirement and planning.

The project's scope should be defined before starting development activities, ensuring no unnecessary tasks. The planning phase involves stakeholders, managers, developers, and clients discussing requirements, issues, timelines, and budgets. Collective decision-making, involving approvals from stakeholders and the entire discussion team, helps avoid controversies and miscommunications, preventing costly changes later in the process.

Phase 2. Prototype.

The Rapid Application Development process involves rapid prototyping, where developers work with clients to design and test prototypes based on requirements. Feedback is then iterated until the application reaches a satisfactory stage, making it user-centric and meeting user expectations. This approach reduces the risk of rapidness.

Phase 3. Construction and feedback.

The team begins the development phase by converting prototypes into working models, building the application through coding and integrations. The application is continuously tested for bugs and issues, and if found, the team resolves them promptly. This process is faster due to the discussion and implementation of changes in the previous stage. Users are also involved in the process, allowing them to suggest changes and fill the gap between design and development.

Phase 4. Implementation.

After developing a working model, developers finalize it and move it to a live environment. They refine and recheck the model, conduct full scale testing to eliminate bugs, and

finalize the application. The team then moves on to user onboarding and continues to provide support post-launch, ensuring a smooth transition to the live environment.

Design

The researchers employed various methods for data collection to obtain the information needed to develop the proposed web-based system, which would help accomplish the first objective. These methods enabled the researchers to gain insight into user needs and pinpoint the issues faced in the current review practices of employees. The initial phase of the study involved conducting a preliminary investigation. On October 23, the researchers interviewed Sir Zotomayor, who acted as the client and offered valuable perspectives on the existing conditions of employees studying for the Civil Service Examination. This interview allowed the researchers to uncover the difficulties faced by employees, such as the lack of organized review materials, the absence of a system to monitor progress, and restricted access to structured quizzes and examinations.

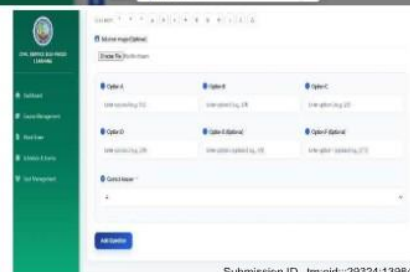
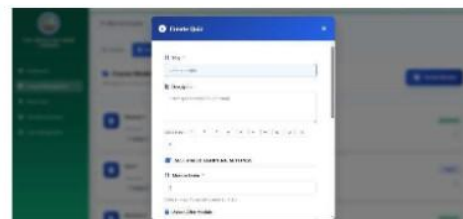
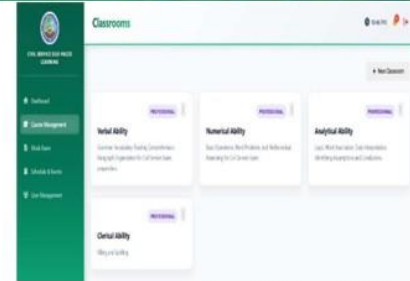
These findings became the basis for designing and developing a Learning Management System that focuses on modular learning and assessment, aiming to improve the overall preparation and performance of LSPU-SCC employees named **LEARNING MANAGEMENT SYSTEM FOR LSPU EMPLOYEES: CIVIL SERVICE EXAMINATION PREPARATION**.

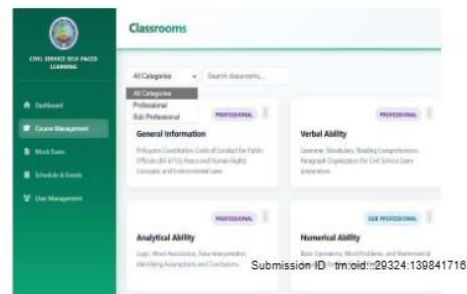
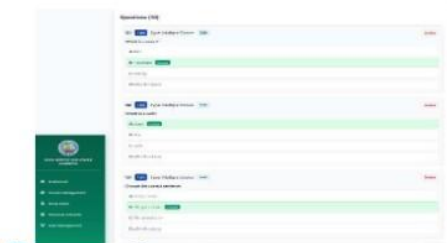
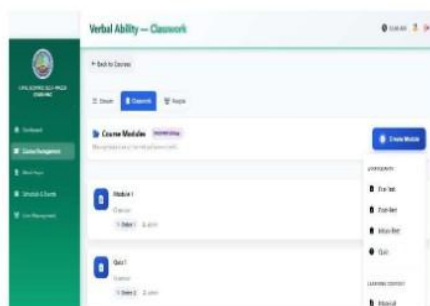
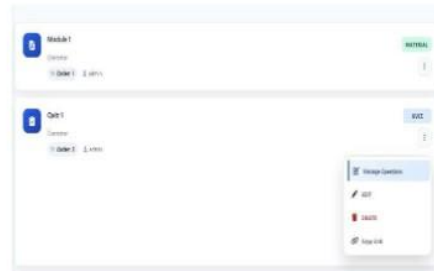
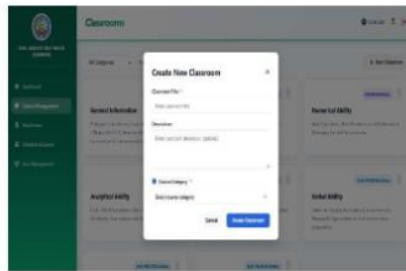
The Development

After gathering all essential data, the researchers proceeded to the second stage, which involved developing a web-based system using the Rapid Application Development (RAD) approach.

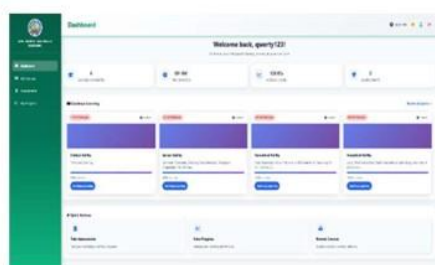
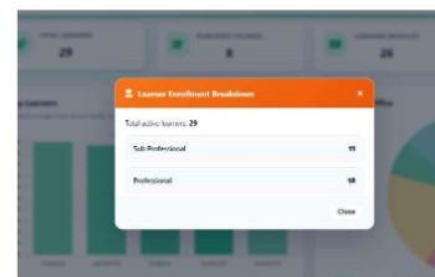
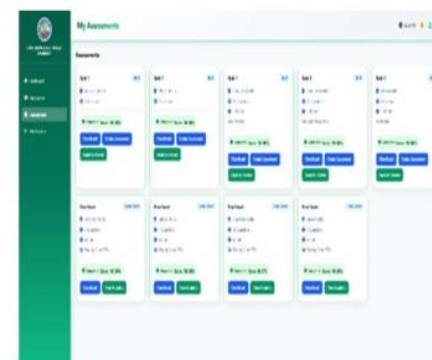
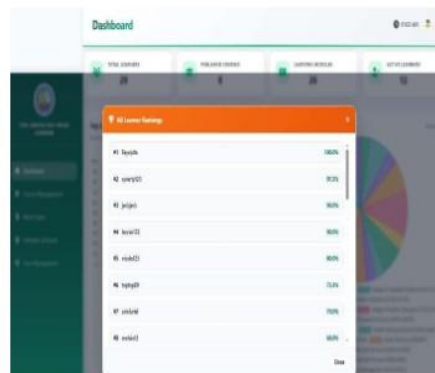
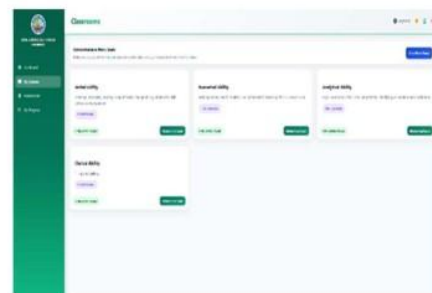
V. Result and Discussion

Research Objective No. 1: To design and develop a web-based **LEARNING MANAGEMENT SYSTEM FOR LSPU EMPLOYEES: CIVIL SERVICE EXAMINATION PREPARATION** that provides a centralized platform for delivering structured learning modules, enables users to take quizzes and examinations for self-assessment, facilitates efficient monitoring of learning progress, ensures accurate recording of users' performance and results, and ultimately enhances employees' readiness, engagement, and effectiveness in preparing for the Civil Service Examination at Laguna State Polytechnic University – Sta. Cruz Campus.





Learner Ability Averages					
Item	Elemental Ability	Technical Ability	Analytical Ability	Global Ability	Total Average
144161	80.0%	80%	8.0%	8.0%	80.0%
144162	80.0%	8.0%	8.0%	8.0%	8.0%
144163	8.0%	80.0%	8.0%	8.0%	8.0%
144164	8.0%	8.0%	8.0%	8.0%	8.0%
144165	8.0%	8.0%	8.0%	8.0%	8.0%
144166	8.0%	8.0%	8.0%	8.0%	8.0%
144167	8.0%	8.0%	8.0%	8.0%	8.0%
144168	8.0%	8.0%	8.0%	8.0%	8.0%
144169	8.0%	8.0%	8.0%	8.0%	8.0%
144170	8.0%	8.0%	8.0%	8.0%	8.0%
144171	8.0%	8.0%	8.0%	8.0%	8.0%
144172	8.0%	8.0%	8.0%	8.0%	8.0%
144173	8.0%	8.0%	8.0%	8.0%	8.0%
144174	8.0%	8.0%	8.0%	8.0%	8.0%
144175	8.0%	8.0%	8.0%	8.0%	8.0%
144176	8.0%	8.0%	8.0%	8.0%	8.0%
144177	8.0%	8.0%	8.0%	8.0%	8.0%
144178	8.0%	8.0%	8.0%	8.0%	8.0%
144179	8.0%	8.0%	8.0%	8.0%	8.0%
144180	8.0%	8.0%	8.0%	8.0%	8.0%
144181	8.0%	8.0%	8.0%	8.0%	8.0%
144182	8.0%	8.0%	8.0%	8.0%	8.0%
144183	8.0%	8.0%	8.0%	8.0%	8.0%
144184	8.0%	8.0%	8.0%	8.0%	8.0%
144185	8.0%	8.0%	8.0%	8.0%	8.0%
144186	8.0%	8.0%	8.0%	8.0%	8.0%
144187	8.0%	8.0%	8.0%	8.0%	8.0%
144188	8.0%	8.0%	8.0%	8.0%	8.0%
144189	8.0%	8.0%	8.0%	8.0%	8.0%
144190	8.0%	8.0%	8.0%	8.0%	8.0%
144191	8.0%	8.0%	8.0%	8.0%	8.0%
144192	8.0%	8.0%	8.0%	8.0%	8.0%
144193	8.0%	8.0%	8.0%	8.0%	8.0%
144194	8.0%	8.0%	8.0%	8.0%	8.0%
144195	8.0%	8.0%	8.0%	8.0%	8.0%
144196	8.0%	8.0%	8.0%	8.0%	8.0%
144197	8.0%	8.0%	8.0%	8.0%	8.0%
144198	8.0%	8.0%	8.0%	8.0%	8.0%
144199	8.0%	8.0%	8.0%	8.0%	8.0%
144200	8.0%	8.0%	8.0%	8.0%	8.0%





Research Objective No. 2: To design and develop a system that would help the personnel prepare effectively for the quiz, exam by improving their knowledge, skills, and confidence.

- Create Classroom
- Add Classwork
- Create Module
- Add Quiz/Examination
- Read Module
- Take Quiz/Examination
- View Analytics
- Take Mock Exam

Table 13. Black Box Testing Results for System Administrator

Functionality	First Trial			Second Trial		
	Pass	Fail	%	Pass	Fail	%
Dashboard	10	0	100%	10	0	100%
Sign up	10	0	100%	10	0	100%
Dashboard	10	0	100%	10	0	100%
Classroom	10	0	100%	10	0	100%
Classwork	10	0	100%	10	0	100%
User Management	9	1	90%	10	0	100%

The provided table illustrates the outcomes of black box testing conducted by 10 IT experts. It confirms that the primary functions of the system operated successfully. Specifically, all functions related to System Administrator Account achieved a flawless pass rate of 5 and 100% overall success rate. The performance of all functions collectively demonstrated effectiveness based on the overall percentage.

Table 14. Black Box Testing Results for Users Section

Functionality	First Trial			Second Trial		
	Pass	Fail	%	Pass	Fail	%
Login	10	0	100%	10	0	100%
Sign up	10	0	100%	10	0	100%
Dashboard	10	0	100%	10	0	100%
User Classroom	10	0	100%	10	0	100%
Stream	10	0	100%	10	0	100%
Classwork	10	0	100%	10	0	100%
Read Module	10	0	100%	10	0	100%
Take Exam/Quiz	9	1	90%	10	0	100%
User Assessment	10	0	100%	10	0	100%
Take Assessment	10	0	100%	10	0	100%
Retake Assessment	10	0	100%	10	0	100%
View Result	10	0	100%	10	0	100%
My Progress	9	1	90%	10	0	100%
User Analytics	10	0	100%	10	0	100%
Display Recent Quiz/Exam	10	0	100%	10	0	100%
Display Percentage of Each Course	10	0	100%	10	0	100%

The outcomes of the black box testing conducted by the ten IT experts serve as a crucial validation of the system's functionality and reliability. The flawless pass rate for all functions related to the employees' section, coupled with the 100% overall success rate in the second trial, underscores the system's robustness in supporting learning activities and user interaction. This high level of performance not only instills confidence in the system's ability to meet operational requirements but also reflects positively on the expertise and diligence of the development team. Furthermore, the comprehensive success across all functions indicates a thorough understanding of user needs and meticulous attention to detail during the development process.

Research Objectives No. 3: To assess and evaluate the acceptance of the user using Technology Acceptance Model in terms of:

- Quality Factors
- Perceive Ease to Use
- Perceived Usefulness
- Experience
- Attitude Towards to Use
- Behavioral Intention to Use

The researcher leaned on the Technology Acceptance Model (TAM) survey to gauge how well users took to the newtech and planned to use it, right in line with the study's aims. Once the questionnaire was drafted, they ran it by the research adviser and a stats expert for approval. With their green light, it was good to go. Feedback came via a Likert scale tracking satisfaction: Very Satisfied (Highly Acceptable), Satisfied (Acceptable), Neither (Undecided), Unsatisfied (Slightly Unacceptable), or Very Dissatisfied (Unacceptable). We handed out printed surveys in person to the system's key users for straightforward, real-time responses.

Table of Level of Acceptability

Identifiers	Mean	Interpretation
Quality Factors	4.48	Acceptable
Perceived Ease of Use	4.51	Acceptable
Perceived Usefulness	4.52	Acceptable
Attitude Towards Using	4.47	Acceptable
Behavioral Intention to Use	4.53	Acceptable
Experience	4.54	Acceptable
Total Mean	4.51	Acceptable

The evaluation also focused on user acceptance and satisfaction using several key factors: system quality, ease of use, perceived usefulness, attitude towards the system, behavioral intention to use, and overall user experience.

SUMMARY OF CONCLUSIONS AND RECOMMENDATIONS

This chapter discusses the primary findings on the level satisfaction with the developed web-based system. The conclusions were based on the study's goal, research



Page 12 of 14 - Integrity Submission
 questions, and results. The implications of these findings, as well as the recommendations that follow, are explained. The recommendations were based on the study's conclusions and purpose.

Summary

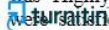
This study focused on the design and development of a Learning Management System for LSPU Employees: Civil Service Examination Preparation. The system was created to provide accessible, flexible, and organized review materials aligned with the competencies required for the exam.

The research examined the challenges faced by employees, such as limited time for review, lack of structured materials, and difficulty accessing reliable resources. To address these concerns, the LMS was developed with key features including user-friendly navigation, categorized review modules, practice quizzes, performance tracking. The system was evaluated based on usability, functionality, reliability, and overall user satisfaction. Feedback from respondents indicated that the LMS significantly improved their review experience by making learning more convenient and structured.

The researchers were able to obtain the necessary data via interviews, library research, web research, technological gap analysis, and brainstorming. The first survey was conducted to establish the problems that employees experience, as well as the present procedures for document tracking, document processing, and user management. The researchers used the Rapid Application Development (RAD) methodology when developing the web-based solution. With this methodology, the researchers can review and enhance the program without having to restart the development process from the beginning. To evaluate the performance and usefulness of the develop system, the researchers created questionnaire using Black Box testing. The develop system was tested virtually with the help of IT experts who served as a system tester. The tests show that all of the system's functionalities work properly.

During the system evaluation, the researchers develop and prepared a questionnaire based from the Technology Acceptance Model (TAM) and ISO/IEC 25010 Software Quality Metrics, which used a 5-point Likert Scale response to assess user satisfaction. Survey Questions were tested and sent to assess the system's usability, effectiveness, and user satisfaction. In this study, the researchers did a face-to-face evaluation to determine the user's level of satisfaction.

The data gathered from the 63 respondents were evaluated and interpreted. The interpreted data resulted in a response scale that was measured against each evaluation criteria, and the outcome was "Highly Acceptable," indicating that the users were satisfied with the developed **LEARNING**



Page 12 of 14 - Integrity Submission

Submission ID: tm:oid::29324:139841716

MANAGEMENT SYSTEM for LSPU EMPLOYEES: Civil Service Examination Preparation.

Conclusions

The feedback gathered after the testing and evaluation phase proved that the developed web-based system is working well and the respondents were satisfied and willing to use the developed system. In conclusion, the study was able to achieve its objective: (1) Develop a Learning Management System for LSPU Employees: Civil Service Examination Preparation; (2) To design and develop a system that would help the document handler and guest to track their documents; (3) To assess and evaluate the acceptance of the user using Technology Acceptance Model in terms of Quality Factors, Perceived Ease to Use, Perceived Usefulness, Experience, Attitude towards Using, and Behavioral Intention to Use towards the develop system. In the study, the researchers therefore conclude the following:

1. After gathering data through a comprehensive preliminary investigation, and interviews with employees of the National Irrigation Administration of Pila, Laguna, the researchers concluded that the current document tracking process needs to be improved in terms of efficiency and reliability, and that developing a web-based system can be a better solution to the difficulties that employees face when doing their specific tasks.
2. The Rapid Application Development (RAD) methodology used by the researchers is beneficial for developing the web-based system.
3. Using Black Box Testing, the researchers discovered that the developed web-based system met its specified functional requirements.

Recommendations

To further enhancement of "Learning Management System for LSPU Employees: Civil Service Examination Preparation", the following are recommended below:

1. To future researchers, the web-based system should be more responsive on the web but not responsive on mobile. Future researchers should make their work responsive to mobile devices so that mobile users can access their work more easily.

Literature Cited

Datta, A., et al. (2022) Modeling sub-GHz communication channels from internal to external body environments in ruminants, targeting advancements in smart livestock management. Published in IEEE Journals & Magazine (2023, April 1). Full text available at <https://ieeexplore.ieee.org/document/9915440>.

Jindal, A., et al. (2023). This work presents a method for creating feedback-linearizable discrete models of continuous-time systems via retraction mapping. Refer to IEEE Journals & Magazine, 2023. See <https://ieeexplore.ieee.org/document/10153969/> for details.

Submission ID: tm:oid::29324:139841716



A. M. NOMA, ET AL. (2022) Design of intelligent and secure hospital appointment scheduling system. (2022, April 17). IEEE Conference Publication IEEE Xplore. <https://ieeexplore.ieee.org/document/9803093>





APPENDIX C

Curriculum Vitae



Patricia Mae O. Etron

Computer Engineer

My Contact

✉ etronmaepatricia@gmail.com

☎ 09923319312

📍 Brgy. Calios, Sta. Cruz, Laguna

👤 Patricia Etron

📅 August 3, 2002

Skills

- IT Service Management (ITSM): Hands-on experience with ServiceNow platform, including Incident, Change, Problem, Request, and Knowledge Management modules.
- Familiar with REST integrations, basic HTML, and system troubleshooting
- ServiceNow administration: User/group management, roles, forms, and catalog items Possess fundamental knowledge in different language such as Javascript, Java and Python
- Holds knowledge about Arduino Uno
- Capable of managing documents and data using Microsoft Word, Excel, PowerPoint, MS Outlook and MS Teams.
- Time Management
- Communication Skills
- Can work under pressure

Career Objective

My career objective is to leverage my technical skills, innovative mindset, and passion for emerging technologies to contribute to the dynamic field of computer engineering. I aim to pursue a challenging and rewarding position that allows me to apply my knowledge in areas such as software development, hardware design, and system optimization

Educational Background

Tertiary

Laguna State Polytechnic University | Bachelor of Science in Computer Engineering

2020 – 2024

Secondary

Laguna Senior High School | Science Technology Engineering and Mathematics

2014 – 2020

Primary

Santo Angel Sur Elementary School

2008-2014

Work Experience

- Guest Experience Leader in McDonalds for 2 years
- Key Account Officer at Learning Is Fun
- Associate Software Engineer at Accenture
- ServiceNow System Administrator at Qiagen Manila

Certification

- Certified System Administrator (ServiceNow)

Character References:

Abigail Viar	Guidance Councilor	09077641233
Ellen Joy Dulin	Manager	09363246285
Renz Manuel	College Instructor	09092487271

Patrick Dan Alan Arisga

Laguna, Philippines | 09765102332 | patrickarisga20@gmail.com | www.linkedin.com/in/patrick-arisga

EDUCATION

Laguna University **Santa Cruz, Laguna**
Computer Science **Undergraduate**

WORK EXPERIENCE

Yoke Studios **London, UK**
Full Stack Web Developer (Contractual) **Mar 2021 - Jun 2023**

- Developed and implemented the front end and backend of a highly responsive and user-friendly marketplace webpage, resulting in a 40% increase in user engagement.
- Collaborated with cross-functional teams to optimize webpage loading time by 50%, enhancing overall user experience and reducing bounce rate.
- Implemented advanced security measures, including encryption protocols and secure authentication, resulting in zero reported security breaches over a period of 2 years.

Aquatic Holdings LLC / BookALifeguard.com **California, USA**
Full Stack Web Developer (Contractual) **Aug 2020 - Mar 2021**

- Developed and integrated a seamless appointment scheduling feature into the interface of Calendly, resulting in a 40% increase in user engagement and a 50% reduction in manual appointment setup.
- Collaborated with UX designers and product managers to refine the user experience of the appointment setting feature, resulting in an average rating of 4.8 out of 5 stars from users.
- Implemented advanced analytics tracking within the appointment scheduling feature, providing valuable insights on user behavior and enabling data-driven optimizations that increased conversion rates by 15%.

The Audition Platform Ltd **London, UK**
Full Stack Web Developer (Contractual) **Feb 2023 - Jul 2023**

- Developed and deployed an innovative matching algorithm that increased successful user-client matches by 30%, resulting in improved client satisfaction.
- Implemented a user-friendly interface for clients to easily input their preferences, leading to a 20% increase in the number of users matched with compatible clients.
- Optimized the algorithm's efficiency, reducing processing time by 50% and enabling faster and more accurate results for users.

GDSLlink Global **Manila, Philippines**
QA Automation Engineer **Oct 2023 - Aug 2024**

- Developed and implemented a Pentaho script to generate comparison sheets between test cases and actual results, ensuring bureau responses matched expectations and reducing backlog issues by 99%.
- Developed a Pentaho script to automate data extraction from databases into readable text files, reducing manual effort by 70% and significantly increasing client efficiency.

SKILLS & INTERESTS

Skills	Microsoft Office (Excel, Word, PowerPoint), HTML, CSS, JavaScript, MySQL, Postman, Selenium
Interests	Cooking, Badminton, Anime, American Sitcoms, Gaming



Padua, Arteo Don Gerrard

Calamba City, Laguna

Email ID: adgvpadua@yahoo.com

Contact no.: +63927 707 4082

Profile

Total experience: 13 Years 2 months

DXC experience: 6 years 8 months

Current Job Title: Associate Manager Software Engineering

Time in Current Role: 2 years 1 month

Skills

Top Skills (Technical/Nontechnical skills)

- Software Testing Tools & Best Practices (13 Years)
- Defect & Bug Discovery (13 Years)
- Issue Identification & Documentation (13 Years)
- Defect Logging, Tracking & Monitoring (13 Years)
- Atlassian JIRA (8 Years)
- POLISY/400 and Group/400 (6 Years)
- Agile Testing Methodology (7 Years and 6 Months)
- Waterfall Testing Methodology (6 years)
- Microsoft SQL Server, Oracle PL/SQL and Greenplum Database (1 Year)
- Redmine (1 Year)
- SOAPUI (6 Months)

Key Roles Performed

- Manual Test Engineer - 11+ years
- Automation Test Engineer - 1+ years

Industry Experience

- Non-life insurance (6+ years)
- Banking (1+ years)
- Healthcare Claims (2+ years)
- Retail (1+ years)



Delivery

- Applies innovative methods, processes and tools to enable cost-effective Defect Prevention activities
- Ensures compliance to operational level agreements with internal and external entities as applicable
- Performs other related functions that may were assigned from time to time

Education

STI College - Bacoor City, Cavite

Bachelor of Science in Information Technology



MATHEW BAUTISTA



Contact



137 M. Acosta Street, Pasay
City



+63 9693930876



mcb022492@gmail.com



Skills

IT Support



Administrative Task



Technical Troubleshooting



Character References

- Mr. Dante Gumaua Assessor (Excellent Star Maritime Center Inc.) Contact #: 0906 461 2478
- Mr. Randolph Barbosa Colleague (Excellent Star Maritime Center Inc.) Contact #: 0955 882 1899



About Me

I am currently looking for a new career opportunity. I am a passionate worker with an impeccable work ethic. A graduate with four years of experience in technology and willing to acquire new knowledge and skills.



Education

2009-2014

Pasig Catholic College

Bachelor of Science in Information
Technology



Work Experience

Metropolitan Bank and Trust
Company Endorsed by: Staff
Alliance, Inc.

July 2014 - May 2015

Peripheral Operator

- To compile all the daily reports for each branch.
- Collaborate with other colleagues to ensure the printers are all working to be used for the printing of daily reports.
- Inspection of all the printed copies to ensure it's completed and readable.

Excellent Star Maritime Center Inc.
Room 401 4th Floor Travellers Life
Bldg. Kalaw Ave, Ermita, Manila,
December 2016 - December 2019

IT Support

- Responsible for the company's data access and management of software and hardware assets.
- Check that all Computer hardware used on the Practical Simulator Assessment is ready to use during the examination.
- Assist the assessors in the preparation of briefing materials, during the briefing of the examinee.
- Encoding of the name of the examinee in Pharos Server Master Computer.
- Ensure that CCTV cameras are regularly checked and maintained.
- Clean and check all computers and each accessory for orderly use.
- Perform other duties as may be directed by the superior.



WESLEY DYRON C. GIMUTAO

IT AUTOMATION ENGINEER

ABOUT ME

Enthusiastic IT Automation Engineer skilled in web application technology, with hands-on experience in agile project management and content management systems. Proficient in the Bootstrap framework and experienced with data visualization tools like Power BI and Tableau. A strong communicator and collaborator with excellent organizational skills, eager to learn and grow.

EDUCATION

BS Information Systems

National University Laguna
2020 - 2024

ITMAWD

STI COLLEGE CALAMBA
2018 - 2020

SKILLS

- Web Development
- Time Management
- Content Management System
- Front End Coding
- Data Analysis / Data Visualization
- Computer Literacy
- Project Management Tools
- Strong Communication
- Coremedia
- Willingness to Learn

PORTFOLIO

[Web Development Portfolio](#)

CERTIFICATES

[Cybersecurity workshop](#)

+639611257382

weasleydyron.gimutao@gmail.com

LinkedIn

Calamba Laguna, 4027

EXPERIENCE

IT Automation Engineer

August - Present

Slix Phils Inc.

- Digitalized the manual checklist of the machines used productions. Paperless checklist that uses web application technology : html, css, javascript, jquery.php, mysql
- Monitored Dashboard that shows the productivity level of each operator

Content Loader Intern

Jan - April 2024

Essilor Shared Services Philippines, Inc. (ESSPI)

- Mastered Coremedia CMS for content management
- Practiced Agile Scrum methodology using Jira
- Collaborated with teams to deploy live sites, accepting feedback and suggestions for improvement.

Student Assistant

Mar 2023 - Sep 2023

National University Laguna

- Organized and maintained filing system to keep documentation easily accessible to fulfill requests for information, and Helped with administrative support by managing incoming calls, coordinating files, and sorting mail
- Communicated effectively with faculty and staff and accepted critiques and suggestions for areas of improvement.

John Lester Domingo

Bachelor of Science in Information Technology

Dedicated Customer Service Representative dedicated to providing quality care for ultimate customer satisfaction. Proven ability to establish and maintain excellent communication and relationships with clients. Dedicated to identifying customer needs and delivering effective solutions to all problems. Excellent time management skills combined with a superior knowledge of the customer service industry. Bilingual, hardworking, and ready to join my next team.



✉ mr.johnlester.domingo@gmail.com

📍 Santa Cruz, Laguna

☎ 09274979176

🌐 linkedin.com/in/john-lester-domingo-319943281

EDUCATION

College

Laguna State Polytechnic University Sta. Cruz

08/2018 - 08/2022

Santa Cruz, Laguna

Courses

- Bachelor of Science in Information Technology

Senior High School

AMA Computer College

08/2016 - 06/2018

Santa Cruz, Laguna

Courses

- ICT (Information in Computer Technologies)

WORK EXPERIENCE

Data Encoder/ SPES

Provincial Public Employment Service Office

06/2016 - 06/2022

Santa Cruz, Laguna

Special Program for Employment of Students. Only worked 20 days every summer for the past 6 years while studying

Achievements/Tasks

- Developed a system for encoding and decoding data from multiple databases, allowing for easier data sharing
- Developed a system for encoding and decoding data in multiple formats, including text, audio, video, and image

Contact: Mark Tuazon

Secretariat

Alagang Tingog Center

02/2024 - 04/2024

Santa Cruz, Laguna

Achievements/Tasks

- Greeted customers in a friendly and professional manner
- Handled customer complaints calmly and professionally, resolving customer issues in a timely manner
- Created reports and documents as requested by the office manager

Contact: Nero Emcs

SKILLS

Ability to Work Under Pressure

Microsoft Excel

Computer Skills

Customer Service

Communication Skills

Leadership

Microsoft Office

CERTIFICATES

Certificate of Participation (04/2022 - 04/2022)

PSITE NCR Webinar

Certificate of Attendance (04/2022 - 04/2022)

Startup Conference: Shaping up the Future of youths through Startups

Certificate of Completion (04/2022 - 04/2022)

Analyze Data with Python - Learn Basic Statistics - Programming for Intermediate User- Visualize Data- Build Python Web Apps- Programming for beginners

LANGUAGES

English

Full Professional Proficiency

Tagalog

Native or Bilingual Proficiency



RICHARD D. RAMOS

Junior Web Developer

CONTACT

-  Santa Cruz, Laguna
-  ramosrichard310@gmail.com
-  +63 9978906269

PERSONAL INFORMATION

Age	24
Birth Date	July 22, 1999
Birth Place	Santa Cruz, Laguna
Civil Status	Single
Religion	Iglesia Filipina Independiente

TECHNICAL SKILLS

- Java / Spring
- SQL / PL/SQL
- HTML
- CSS / Bootstrap
- JavaScript / JQuery
- PHP
- MySQL
- MS Word
- MS PowerPoint
- React JS
- Oracle APEX
- Jasper Report

EDUCATION

TERTIARY

Laguna State Polytechnic University – Santa Cruz Main Campus

- Bachelor of Science in Information Technology - Major in Web & Mobile Application Development
- S.Y. 2019 - 2023

SECONDARY

Pedro Guevara Memorial National High School

- Santa Cruz, Laguna
- S.Y. 2011 - 2015

PRIMARY

Santisima Cruz Elementary School

- Santa Cruz, Laguna
- S.Y. 2005 - 2011

CAPSTONE PROJECT

Animal Shelter Aid System

- Developed a web system for an Animal Shelter.
- Assigned as the main programmer for the team.
- Converted wireframe design from Adobe XD to an actual web design using HTML as structure, CSS / Bootstrap as style sheet, and JavaScript for other functionalities.
- Created the Flow Chart for the web system's process flow.
- Used PHP as the main server-side programming language, and MySQL as database.
- Used PayPal API for donations, with PayPal sandbox during testing.
- Integrated email notification.
- Integrated YouTube's platform for streaming.

SOFT SKILLS

- Time Management
- Problem Solving
- Communication
- Work Ethic
- Adaptability
- Observation

LANGUAGE

- FILIPINO
- ENGLISH

INTERNSHIP**NAR Power System Specialists Corporation**

- Back-end developer
- Worked on the server-side programming of the website for the company. Using PHP as the primary programming language, and MySQL as database.
- February 27, 2023 – June 19, 2023
- 600 Hours

WORK EXPERIENCE**Samsung Electro Mechanics Philippines Corporation**

- Machine Operator
- Assigned to operate and monitor the machines in the semiconductor taping department.
- 2019

Computer Professionals Inc.

- Junior Web Developer
- Worked as a front-end developer for my previous project using React JS.
- Working as a web developer in my current project using Oracle APEX
- 2023 - present


GRETCHEN R. BESANDE

BARANGAY BIÑAN, PAGSANJAN, LAGUNA

OBJECTIVE

To acquire experience that would contribute in my personal learning and development, as well as enhancing my knowledge and skills through observation and participation in actual work, and adapt to a challenging working environment.

CONTACT

MOBILE NUMBER: 0997-496-9133

EMAIL: chen.besande@gmail.com

SKILLS

- HTML, Java and C++ programming
- Visual Studio .Net
- Data and information security
- Software Development Lifecycle (SDLC)
- Application Development

EDUCATION

Pedro Guevara Memorial National High School
Santa Cruz, Laguna
June 2002 – March 2006

ACTS COMPUTER COLLEGE
Santa Cruz, Laguna
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
JULIE 2011 – MAY 2015

TESDA
Cagandahan Skills Development, Inc.
Paete, Laguna
Food and Beverages Services
June 05, 2017 – July 28, 2017

EXPERIENCE

PHINMA PROPERTIES
29 EDSA, Mandaluyong City
Philippines 1559
Property Consultant
January 15, 2016 – April 15, 2016

BIGA WATER REFILLING STATION
Living Water
#192 Santa Anastacia, Sto Tomas, Batangas
Secretary
May 20, 2016 – May 28, 2018

FUJITSU DIE-TECH CORPORATION
113 East Science Ave.,
Special Export Processing Zone
Laguna Technopark
Biñan, Laguna (4024) Philippines
Warehouse Personnel
November 5, 2018 – April 5, 2019

FREELANCE SOFTWARE DEVELOPER
Upwork Philippines
June 10, 2019 – present

- Attended regular project technical team meetings.
- Provided senior project staff with full communication of project progress.
- Assisted with development of project code.
- Developed code adhering to established processes and standards.
- Coordinated with deployment staff by providing necessary documentation.
- Created and maintained project documentation, in accordance with established standard.



REYNALDO A. YASOÑA

CONTACT

0991-483-9977

reynaldo_yasona@outlook.com

Pagsanjan, Laguna

SKILLS

- **Programming Languages:** Java, Python, C++, Visual Basic, C#, JavaScript
- **Web Development:** HTML, CSS, PHP, React JS, AJAX, JQuery
- **API Integrations**
- **Database Management:** MySQL, DB2
- **Robotics:** ROS (Robot Operating System), Arduino
- **Mobile App Development:** Android Studio, Java
- **Software Development:** ASP.NET, Visual Studio
- **Macro Programming**
- **Adaptability to New Technologies**
- **Attention to Detail**
- **Problem-solving Skills**

CAREER OBJECTIVE

Dynamic and driven programmer with a passion for innovation and problem-solving. Seeking to leverage my diverse skill set in programming languages such as Java, Python, C++, and JavaScript, along with experience in web and mobile app development, database management, and robotics. Eager to contribute to a forward-thinking team where I can apply my technical expertise to develop cutting-edge solutions and drive business success. Committed to continuous learning and staying updated with emerging technologies to deliver impactful results.

EDUCATION

ACTS COMPUTER COLLEGE

Santa Cruz, Laguna
Bachelor of Science in Information Technology
S.Y. 2011 - 2015

Lumban National High School

Lumban, Laguna
S.Y. 2007 - 2011

Wawa Elementary School

Lumban, Laguna
S.Y. 2001 - 2007

PERSONAL DATA

SEX:	Male
DATE OF BIRTH:	May 21, 1995
PLACE OF BIRTH:	Lumban, Laguna
NATIONALITY:	Filipino
CIVIL STATUS:	Single
RELIGION:	Roman Catholic
HEIGHT:	5'10
WEIGHT:	71 kgs
FATHER'S NAME:	Ronaldo Yasoña Sr.
MOTHER'S NAME:	Buenaventura Yasoña
LANGUAGES:	Filipino, English

WORK EXPERIENCE**MICROVENTURE'S PHILIPPINES (ONE PUHUNAN)**

Position: Account Officer
 Brgy. Longos Kalayaan, Laguna
 March 17, 2016 – January 23, 2017
 Mobile Number: 09989680305

FUJI XEROX (DENSO OFFICE)

Position: DocuCare Specialist
 Top Serve Service Solutions Inc.
 9694 A&N Bldg, Kamagong st. cor.,
 St. Paul rd. Brgy. San Antonio,
 Makati City Phil
 May 7, 2018 – April 30, 2019

DENSO PHILIPPINES CORPORATION

Position: Full Stack Developer / Lead
 Programmer
 109 Unity Avenue, Special Economic Zone,
 CIP 1, Canlubang, Calamba City, Laguna
 Philippines 4037
 May 06, 2019 – March 11, 2023

NITYO INFOTECH

Open System Developer (DB2)
 2nd Floor Vicente Madrigal Building,
 6793 Ayala Avenue, Makati
 July 3, 2023 – Present

Duties and Responsibilities:

- Creating servers and databases for functionality.
- Developing front end website architecture.
- Designing user interactions on web pages.
- Developing back-end website applications.
- Ensuring responsiveness of applications.
- Seeing through a project from conception to finished product.
- Give support to the other software developer.
- Give support to the end users.
- Conduct training and create a user's manual for the end users.
- Database management and monitoring and DB2 Database Developer and Migration.
- Staying abreast of developments in web applications and programming languages.

CHARACTER REFERENCES**Ronald Sancon**

Assistant Manager
 DENSO Philippines Corp.
 109 Unity Avenue, Special Economic Zone,
 CIP 1, Canlubang, Calamba City, Laguna
 Philippines 4037
 09209131270

MARVIE ALVARADO

Regional Manager
 Microventure's Philippines
 Brgy. Pagsawitan
 Sta. Cruz, Laguna
 0998-9680-305



JULIUS ROY D. ASUNCION

CONTACT

-  +639652935330
-  juliusroyauncion@gmail.com
-  Sitio 6, Bagumbayan the, Sta. Cruz, Laguna

EDUCATION

TERTIARY
LAGUNA STATE POLYTECHNIC
UNIVERSITY
STA. CRUZ MAIN CAMPUS

- Bachelor of Science in Information Technology Major in Service Management Program

UNDERGRADUATE

SENIOR HIGH

- Sta Cruz Integrated National High School

2019 - 2021

SKILLS

- Hardworking
- Trustworthy
- Able to adapt in certain situation
- Time Management
- Leadership
- Effective Communication
- Critical Thinking

LANGUAGE

- Tagalog
- English

PROFILE SUMMARY

A hardworking and trustworthy professional with strong adaptability and the ability to thrive in dynamic environments. Adept at time management, ensuring timely and efficient completion of tasks while maintaining high-quality standards. Equipped with leadership skills to guide teams toward achieving objectives, along with effective communication to foster collaboration. Known for critical thinking and problem-solving, consistently delivering well-thought-out solutions to complex challenges. Dedicated to contributing to organizational success through reliability and a proactive approach to every situation.

OBJECTIVES

- To Improve my personal skill, abilities, and knowledge while working for a company that values hard work and trusts me to take on responsibilities and challenges.

SEMINAR

- Guarding the Gates: Defending Against Cyber Attacks and Malware in Today's Digital world 2024
- 13th EdukCircle International Convention On Engineering and Computer Technology 2023

PERSONAL INFORMATION

Age	: 23
Date of Birth	: July 23 2001
Place of Birth	: Bagumbayan, Santa Cruz, Laguna
Civil Status	: Single
Citizenship	: Filipino
Height	: 5'6"
Weight	: 65 kg.
Religion	: Catholic

JULIUS ROY D. ASUNCION
APPLICANT



FRANCIS ANSELME V. BANGKIL

CONTACT

☎ 09629733678
✉ francisbangkil28@gmail.com
📍 2210 A. Teodoro St. Brgy Poblacion 1 Sta. Cruz Laguna

PROFILE SUMMARY

A young motivated professional with passion and determination for achieving goals through utilizing the knowledge and skills acquired in previous training.

EDUCATION

TERTIARY
LAGUNA STATE POLYTECHNIC
UNIVERSITY
STA. CRUZ MAIN CAMPUS
• Bachelor of Science in Information
Technology Major in Service
Management Program

UNDERGRADUATE

SENIOR HIGH
• Union College of Laguna

2018 - 2020

SKILLS

- Hardworking
- Trustworthy
- Able to adapt in certain situation
- Time Management
- Effective Communication
- Critical Thinking

SEMINAR

- 13th EdukCircle International Convention On Engineering and Computer Technology 2023
- Device Configuration and Management 2026

PERSONAL INFORMATION

Age : 24
Date of Birth : November 21, 2001
Place of Birth : Pasay City
Civil Status : Single
Citizenship : Filipino
Height : 5'4
Weight : 53 kg.
Religion : Born Again

REFERENCE

FRANCIS ANSELME V. BANGKIL

APPLICANT

JON CHRISCHAN A. ALOBA

APPLICANT

 Sitio 2 Bagumabayan Santa Cruz Laguna
 0992-413-8234
 chrischanaloba944@gmail.com



OBJECTIVE

To succeed in an environment of growth and excellence and earn a job which provides me job satisfaction and self development and help me achieve personal as well as organizational goals.

EDUCATION

Pedro Guevara Memoria
National Highschool
JUNIOR HIGH SCHOOL
 • SY. 2014-2019
LAGUNA SENIOR HIGH SCHOOL
SENIOR HIGH SCHOOL
 • ACCOUNTANCY BUSINESS
 MANAGEMENT
 • SY. 2019-2021
LAGUNA STATES
POLYTECHNIC UNIVERSITY
 • BACHELOR OF SCIENCE IN
 INFORMATION TECHNOLOGY
 • 2021-2026

SKILLS

- Empathy
- Problem Solving
- Strategic planning and scheduling skills
- Time-management skills
- Verbal and written communication skills

WORK EXPERIENCE

Laundry Shop (2022-2023)

- I work at a laundry shop. I start work at 8:00 AM and sometimes go home at 5:00 PM or 7:00 PM. My job is to load the clothes into the washing machine, then add fabric conditioner after washing. After 30 minutes, I transfer the wet clothes to the dryer to dry them. After 40 minutes, the clothes are dry and ready to be folded and packed in plastic. In the afternoon, I also deliver laundry to different places.

Palengke Groceries (2019-2020)

- I work at the public market. I start working at 4:00 AM and usually go home at 7:00 PM. My job involves many different responsibilities, and most of the time I stay busy throughout the whole day. One of my tasks is packing bags early in the morning. I also arrange and organize the display of different products to make sure everything looks clean and presentable for customers. Aside from that, I repack sugar and other products into smaller portions for selling. Part of my work also includes carrying heavy loads, especially sacks of sugar that weigh around 50 kilos. Sometimes, I ask my co-workers to help lift the sacks onto my head so I can carry them properly. I also assist customers and sell products while doing my other tasks because serving customers is our top priority. Working at the market is very tiring because we are not allowed to sit down during working hours. We are required to stand the whole day while working and assisting customers. Even though the job is physically exhausting, I continue to work hard and do my responsibilities properly every day.

LANGUAGE

- Filipino
- English

PERSONAL PROFILE

- Birthdate : December 25 2001
- Birth place : Polilio Quezon
- Gender : Male
- Civil Status : Single
- CitezenShip : Filipino
- Religion : Roman Catholic

Jon Chrischan A. Aloba

I hereby certify that the information above is true and correct to the best of my knowledge